Biochemistry



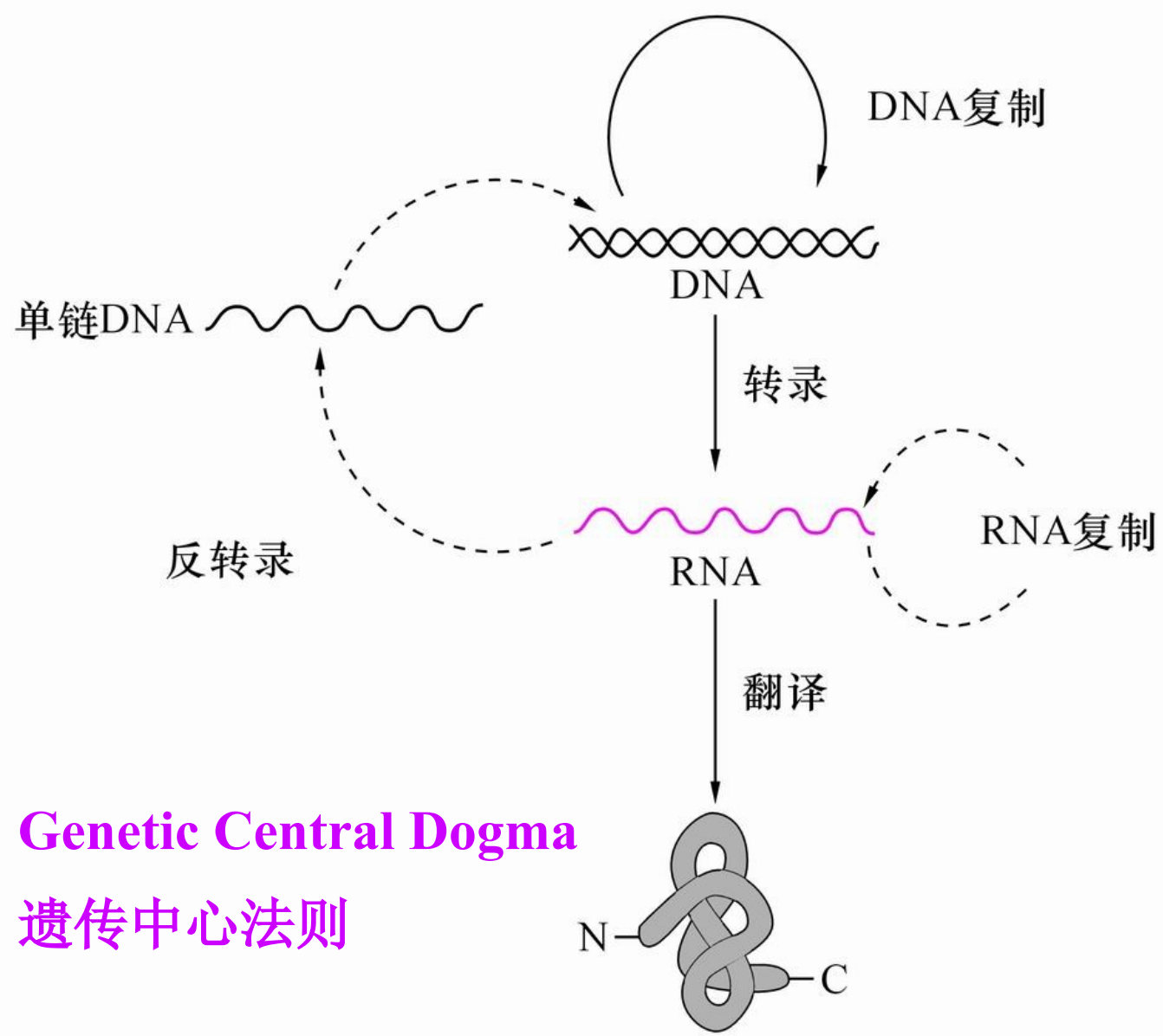
# Section G
# RNA synthesis and processing

**Guodong Yang**

**Biochemistry and Molecular Biology, College of Life Sciences**

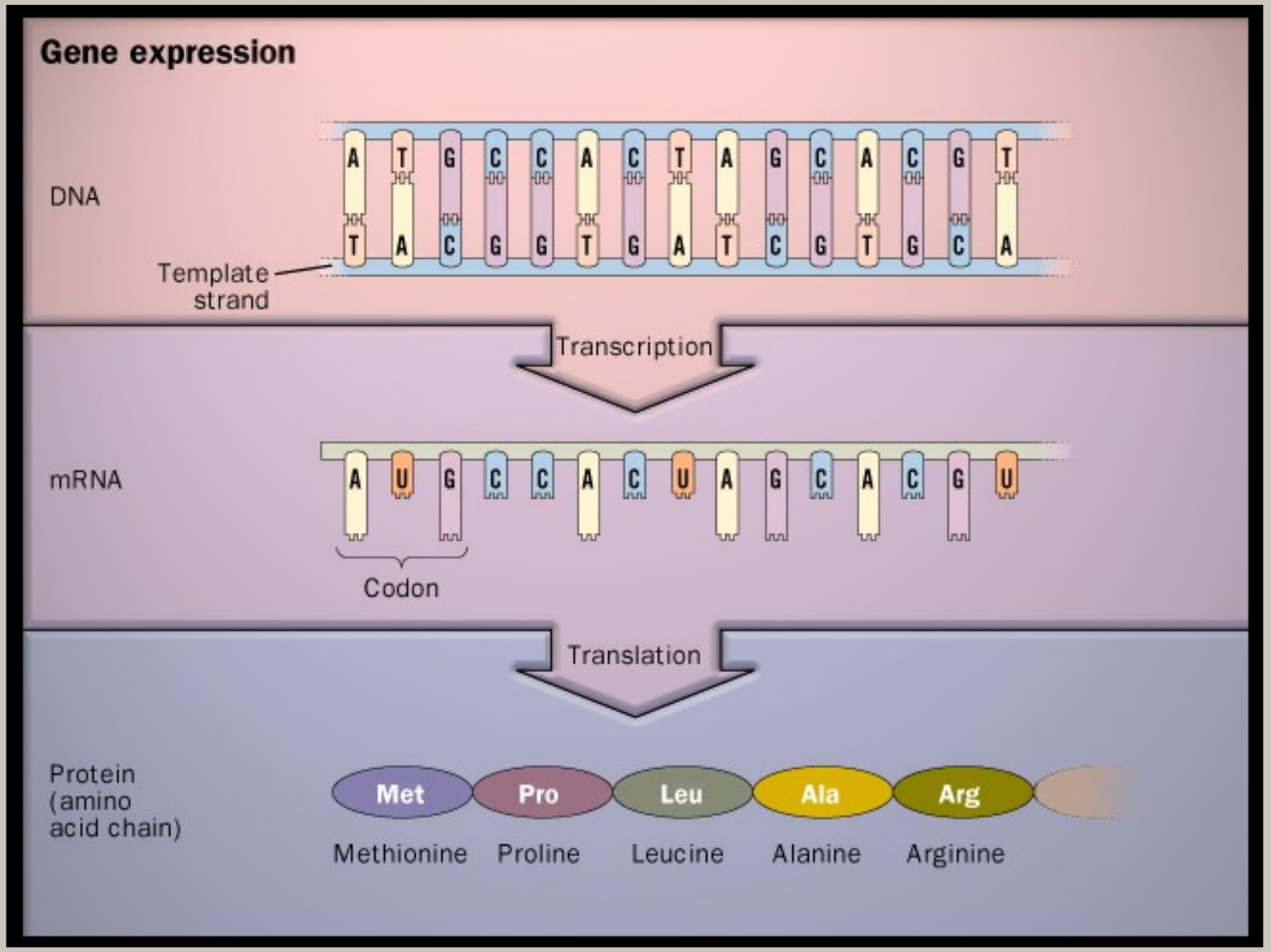
gdyang@sdau.edu.cn

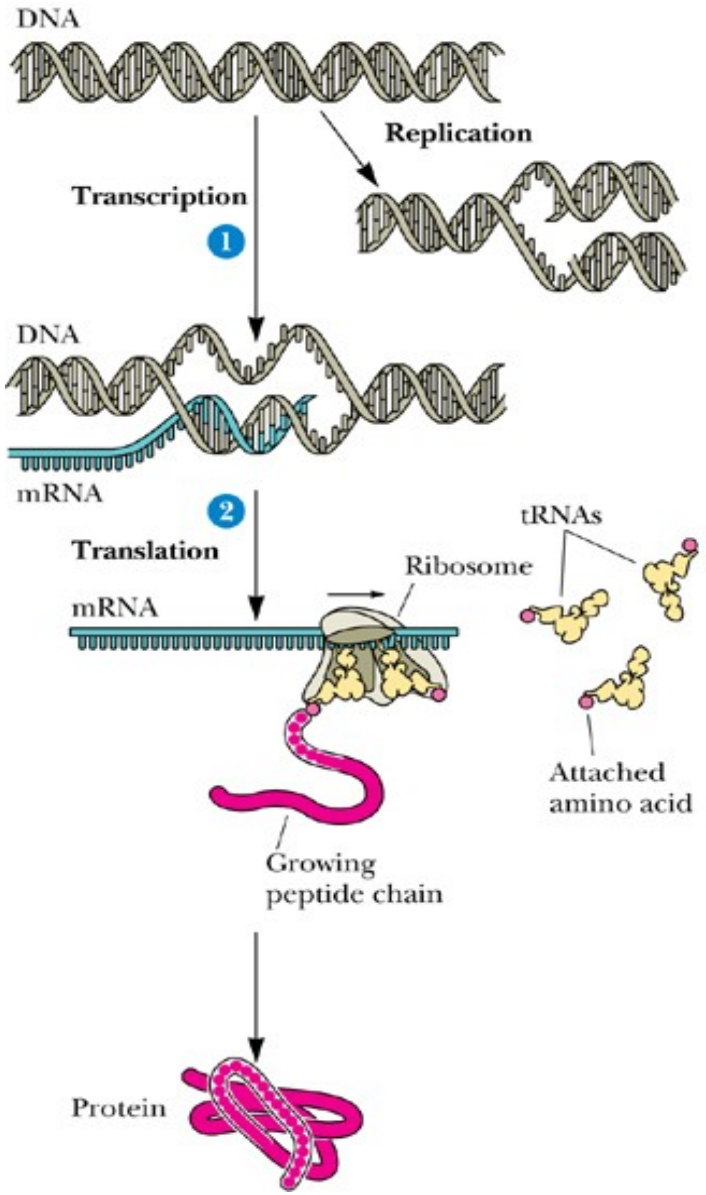
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**Genetic Central Dogma**

**遗传中心法则**


Gene expression
DNA
Template strand
Transcription
mRNA
Codon
Translation
Protein (amino acid chain)
Met
Pro
Leu
Ala
Arg
Methionine
Proline
Leucine
Alanine
Arginine




**Replication**
DNA replication yields two DNA molecules identical to the original one, ensuring transmission of genetic information to daughter cells with exceptional fidelity.

**Transcription**
The sequence of bases in DNA is recorded as a sequence of complementary bases in a single-stranded mRNA molecule.

**Translation**
Three-base codons on the mRNA corresponding to specific amino acids direct the sequence of building a protein. These codons are recognized by tRNAs (transfer RNAs) carrying the appropriate amino acids. Ribosomes are the "machinery" for protein synthesis.

G1 RNA structure: tRNA cloverleaf

G2 RNA Transcription in Prokaryotes

G3 The Lac operon

G5,G6 Transcription in Eukaryotes

G7 Processing of eukaryotic pre-mRNA

G8,G9 Ribosomal RNA, tRNA





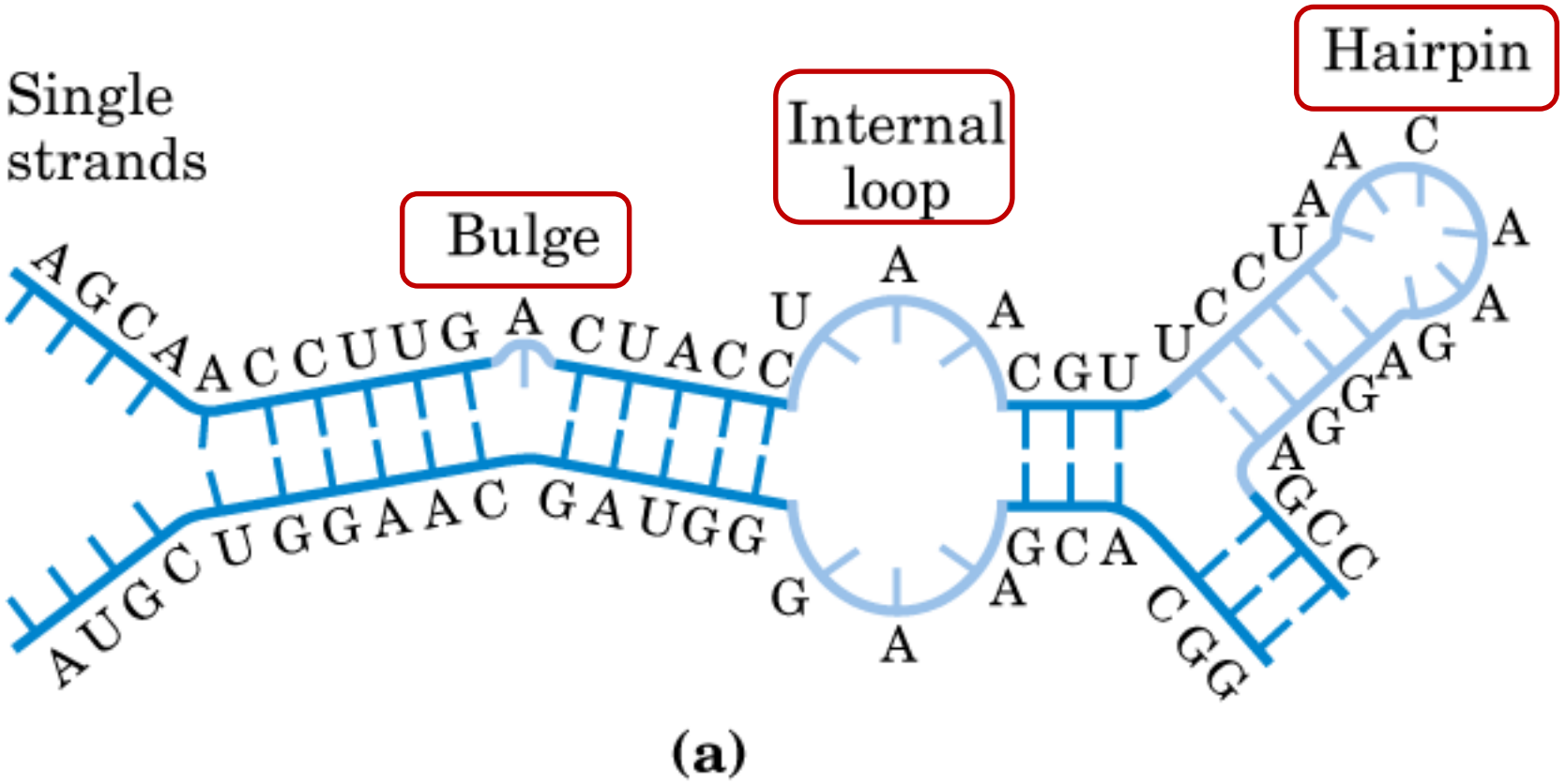
# G1 RNA structure

## RNA molecules do not form simple, regular secondary structure but many form complex and unique three dimensional structures. (G1)

1. Single strand RNA tends to take right-handed helical conformation.
2. Self-complementary sequences on a RNA molecule lead to specific structures.
3. The standard base-pairing rules are followed (i.e., A with U, G with C).
4. Intrastrand base pairing makes structures including internal loops, and hairpins (helical).

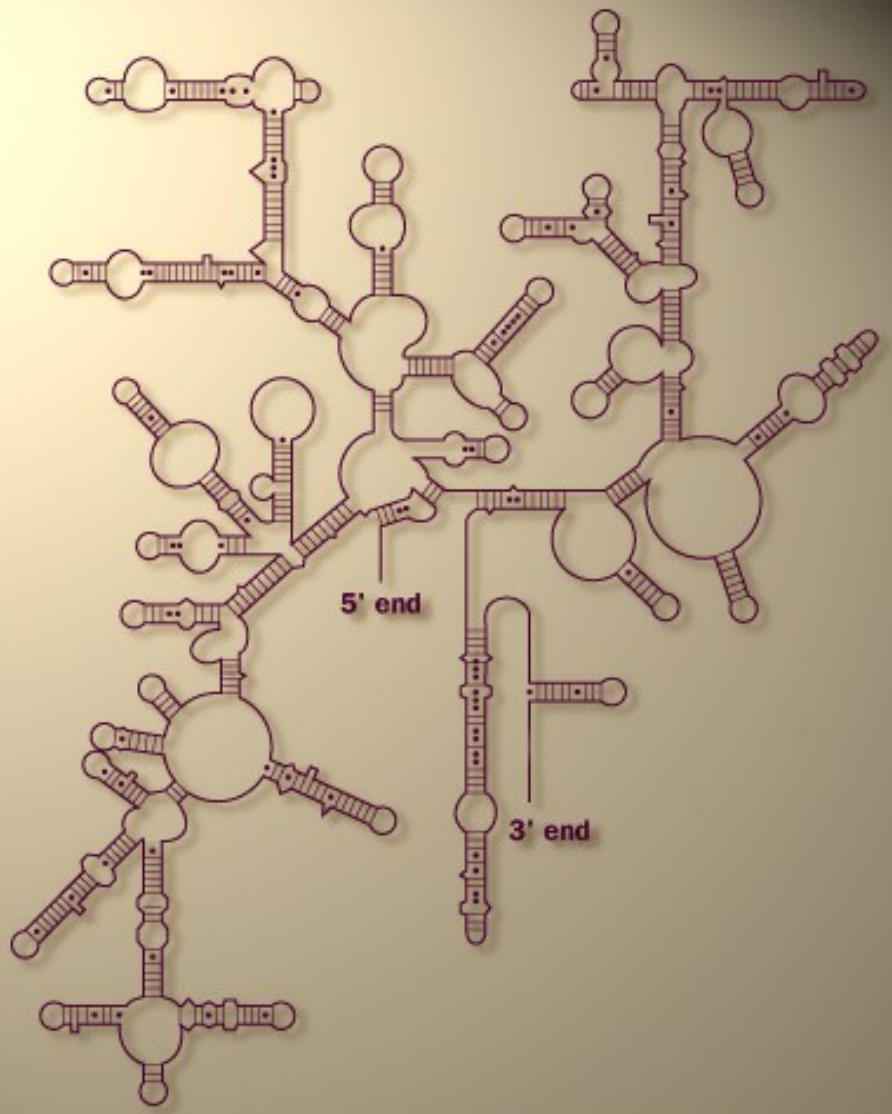


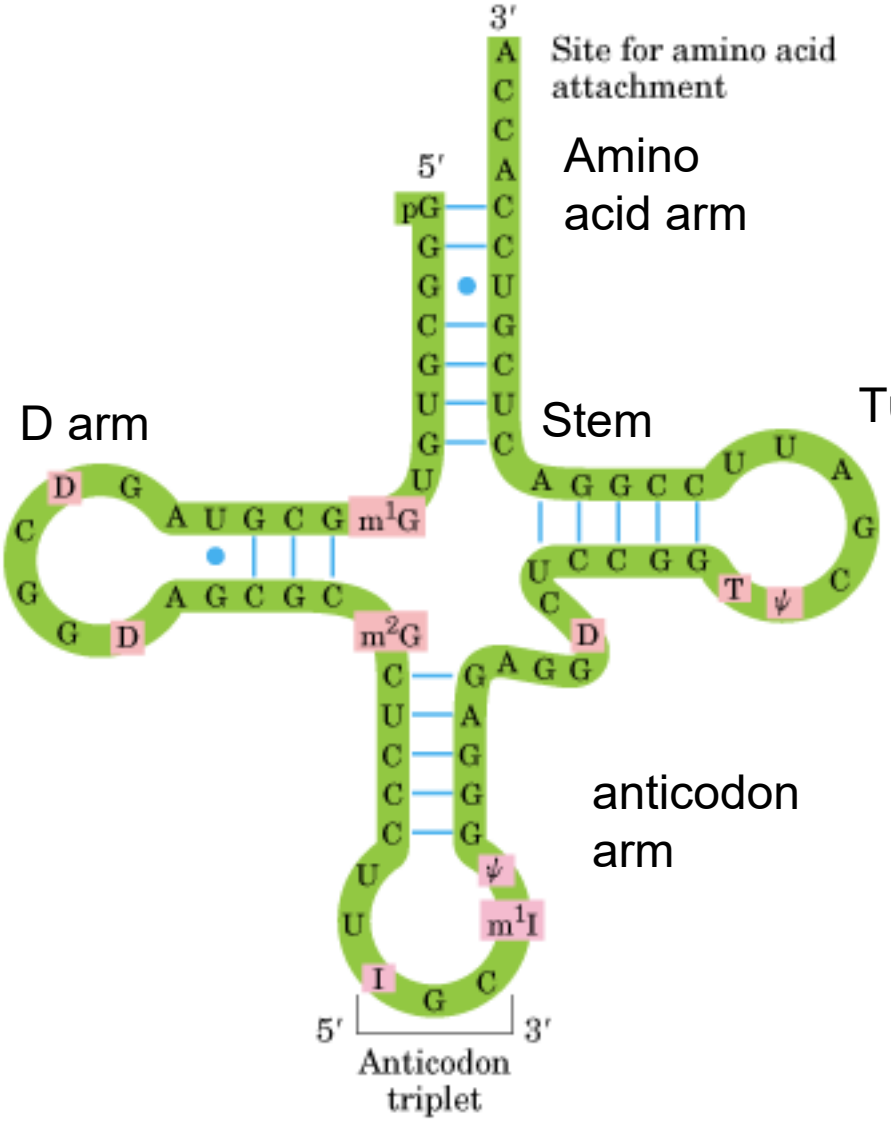



Single strands
Bulge
Internal loop
Hairpin
AGCAACCUUG A CUACC
AUGCUGGAAC GAUGG
U A A
G A
CGU
GCA
A
CGG
UCCU
A C
A A
A
G G A G
AGCC

**(a)**

## Ribosomal RNA








cloverleaf in secondary

# RNA Transcription in Prokaryotes (G2)





# 1. DNA and RNA syntheses are similar in some aspects but different in others

**Similar in fundamental chemical mechanism:**

both are guided by a **template**;

both have the same polarity in strand extension (**5` to 3`**);

both use triphosphate nucleotides (**dNTP or NTP**).





## Different aspects:

- Needs **no primers**;
- Only involves a **short segment** of a large DNA molecule;
- Uses only **one** of the two complementary DNA strands as the template strand;
- Has **no proofreading** activities;
- Be subject to **great variation**.





# 2. The multimeric RNA polymerase in *E.coli* has multiple functions

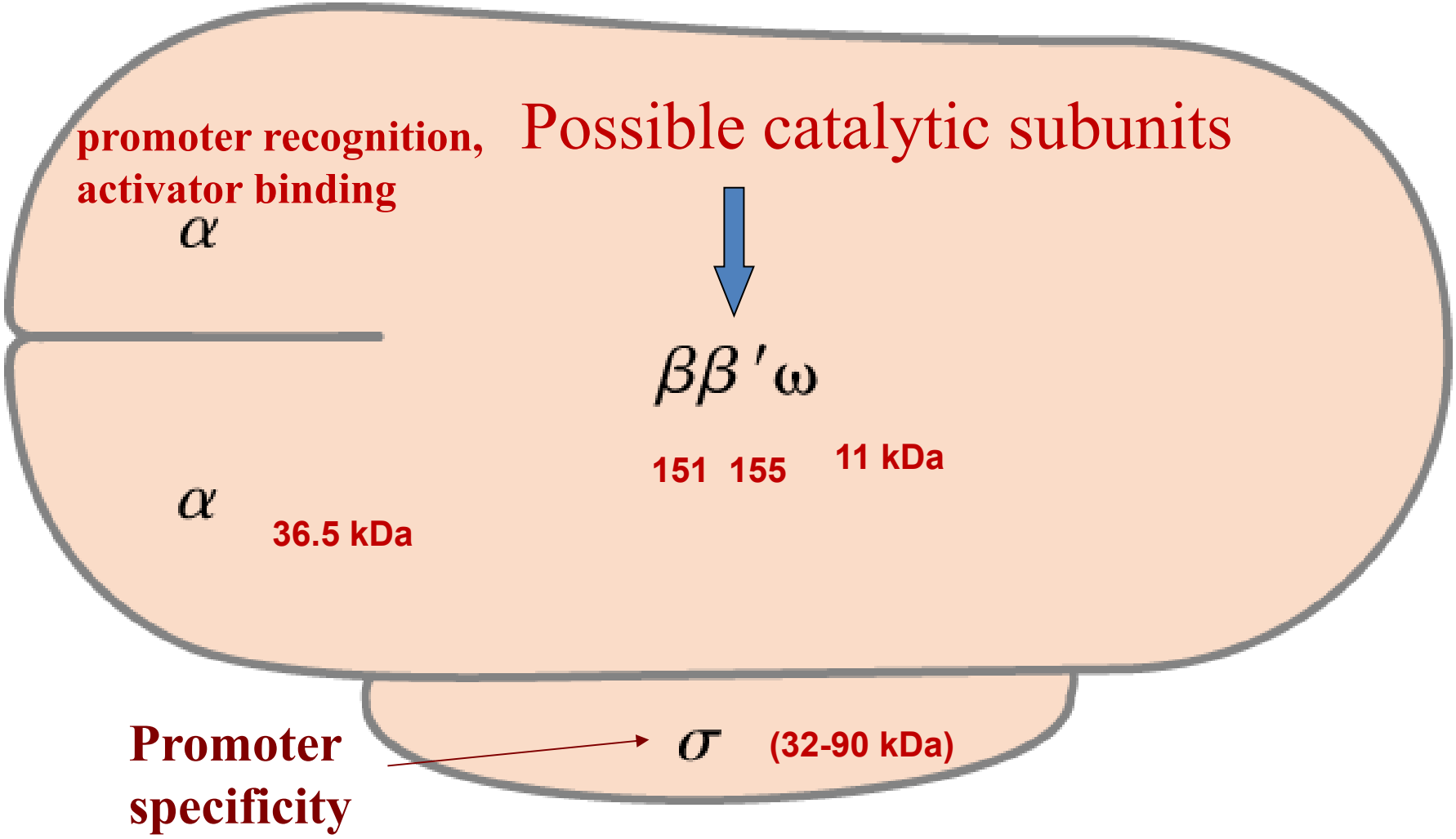
The holoenzyme consists of five types (五种，六个) of subunits ($\alpha_2\beta\beta'\omega\ \sigma$)and its is used to synthesize all the RNA molecules *in E. coli*.

The multiple functions include:

searches for initiation sites on the DNA molecule and unwinds a short stretch of DNA (**initiation**);

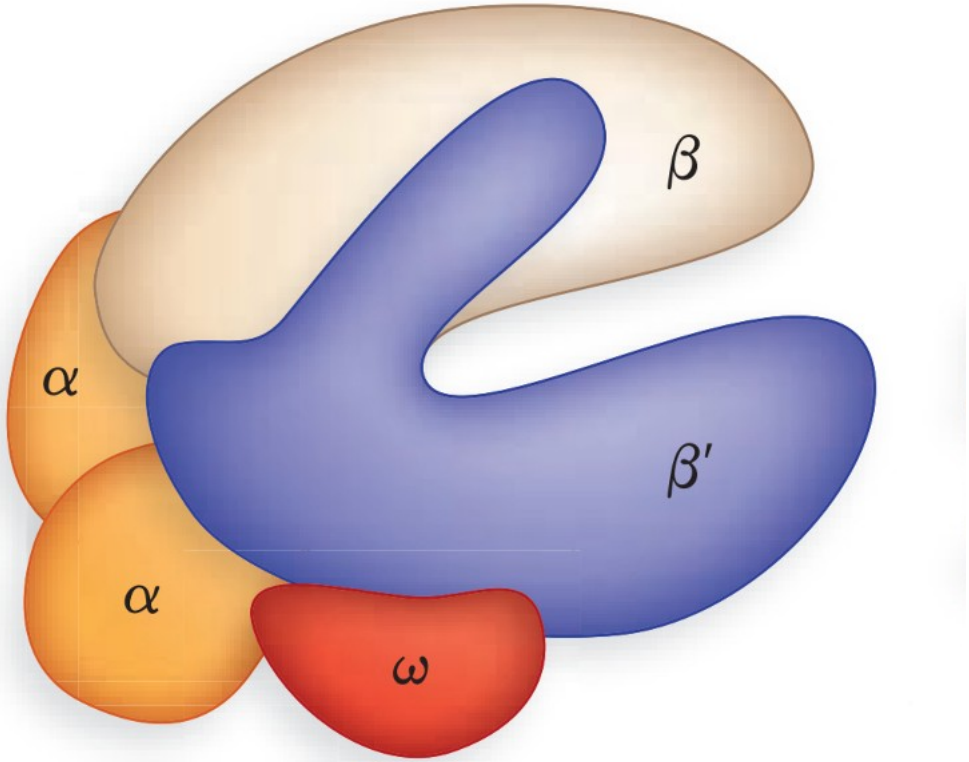
selects the **correct** NTP **and** catalyzes the formation of phosphodiester bonds (**elongation**);

detects termination signals for RNA synthesis (**termination**).

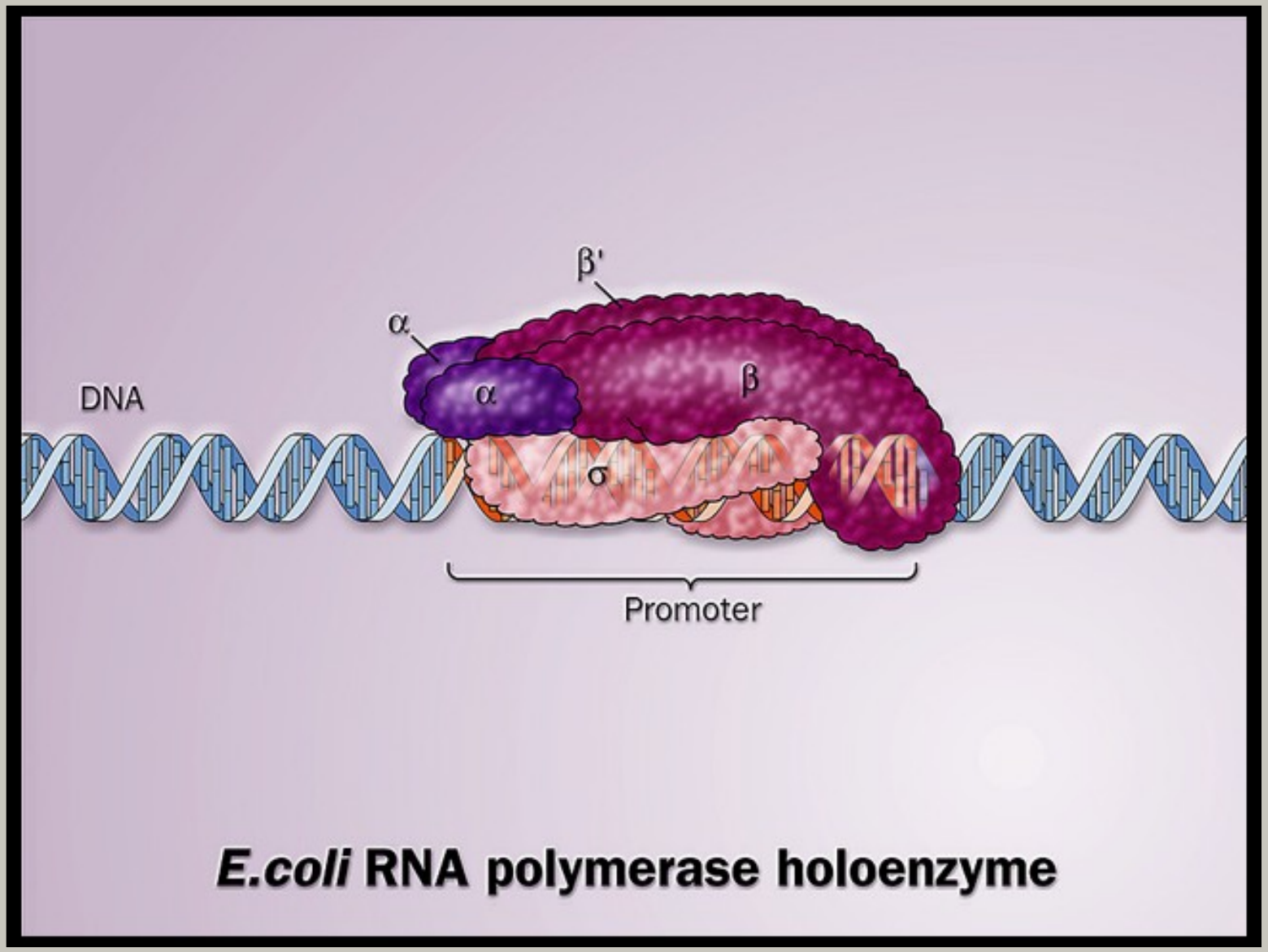



The *E. coli* RNA polymerase holoenzyme consists of six subunits: $\alpha_2\beta\beta'\omega\ \sigma$.

# core enzyme




five core subunits $(\alpha_2\beta\beta'\omega;\ M_r\ 390{,}000)$




***E.coli* RNA polymerase holoenzyme**

## Subunits of RNA Polymerase from *E. coli:*

| Subunit | Number | Mass (kd) | Gene | Role |
|---|---|---|---|---|
| $\alpha$ | 2 | 37 | *rpoA* | Binds regulatory sequences |
| $\beta$ | 1 | 151 | *rpoB* | Forms phosphodiester bonds |
| $\beta'$ | 1 | 155 | *rpoC* | Binds DNA template |
| $\sigma^{70}$ | 1 | 70 | *rpoD* | Recognizes promoter, initiates synthesis |

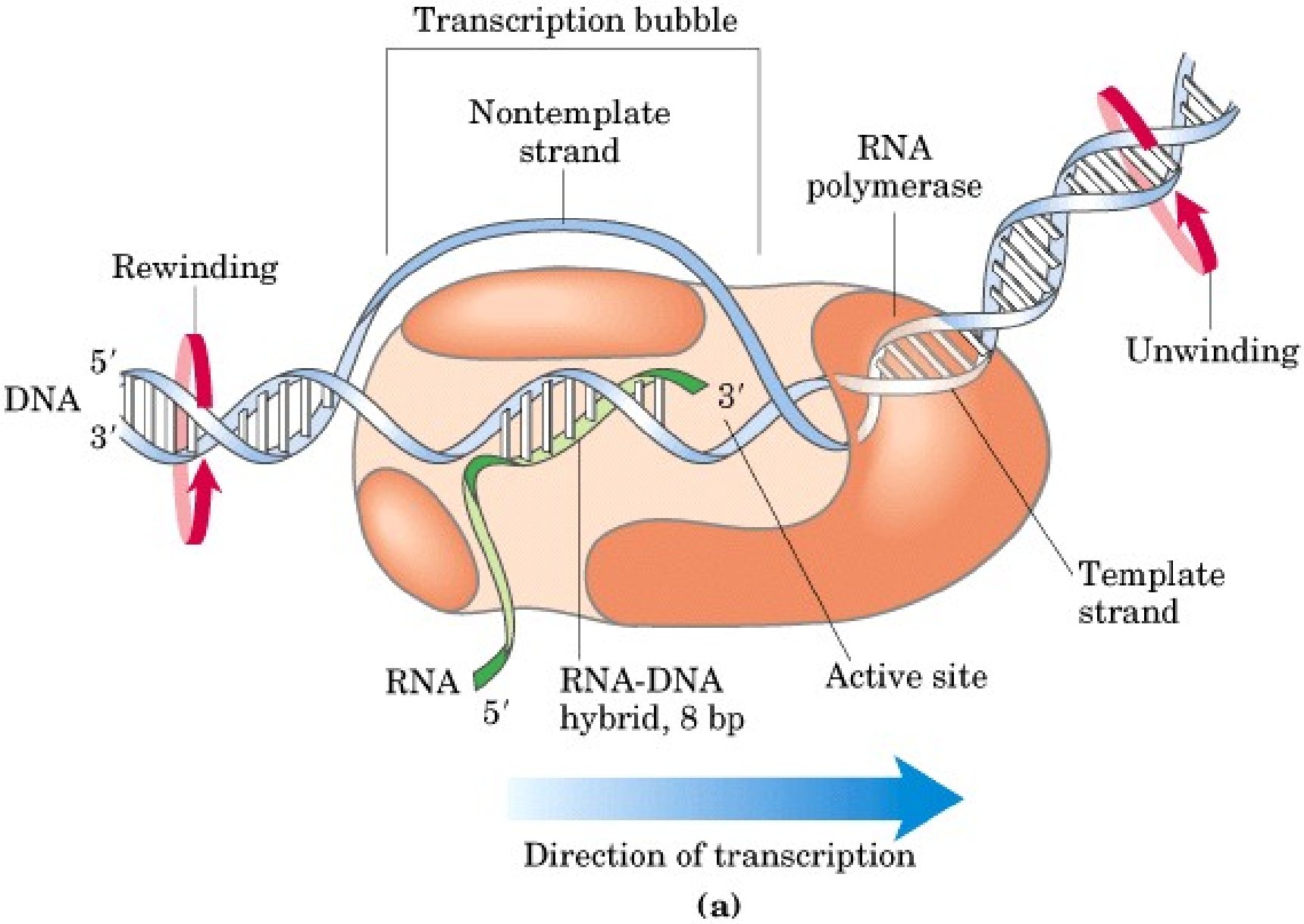


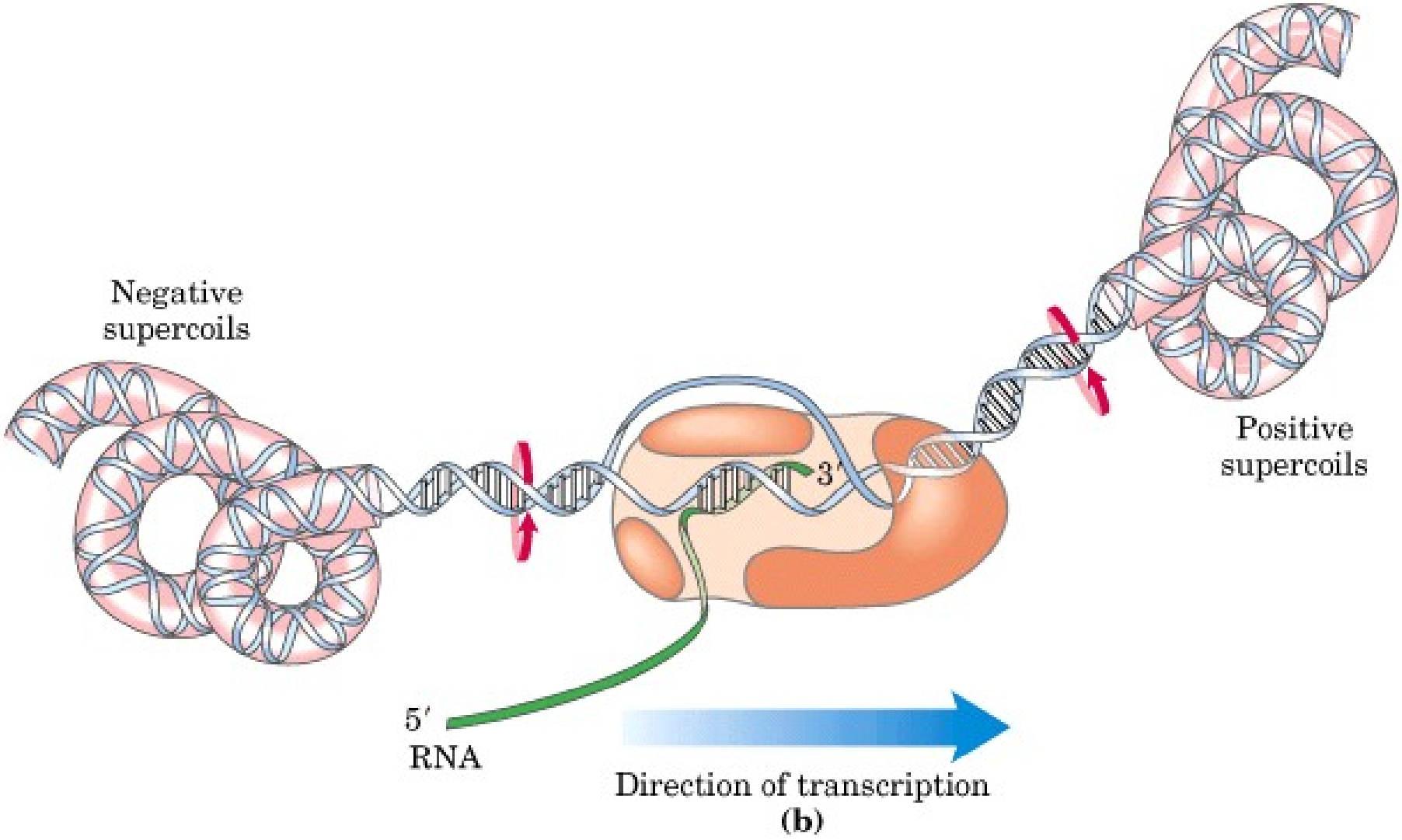


# 3. RNA synthesis occurs in a “moving” transcription bubble on the DNA template






Transcription bubble
Nontemplate strand
RNA polymerase
Rewinding
Unwinding
5′
DNA
3′
3′
Template strand
RNA
5′
RNA-DNA hybrid, 8 bp
Active site
Direction of transcription
(a)

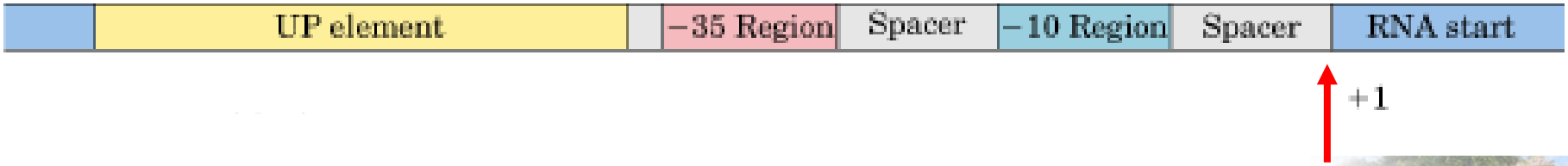

Negative
supercoils
3′
Positive
supercoils
5′
RNA
Direction of transcription
(b)

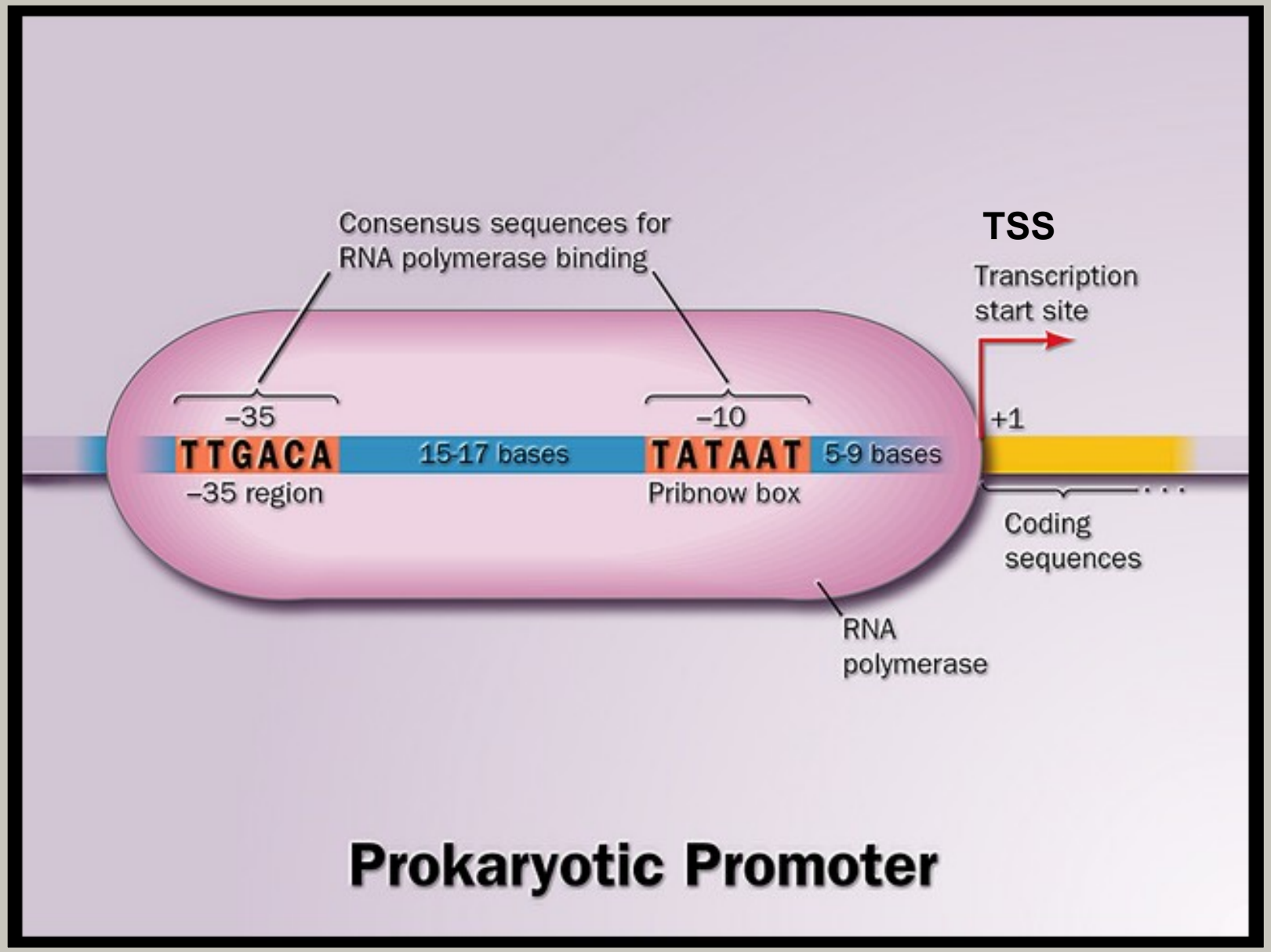
# 4. RNA polymerase recognizes specific promoter sequences on DNA to initiate transcription

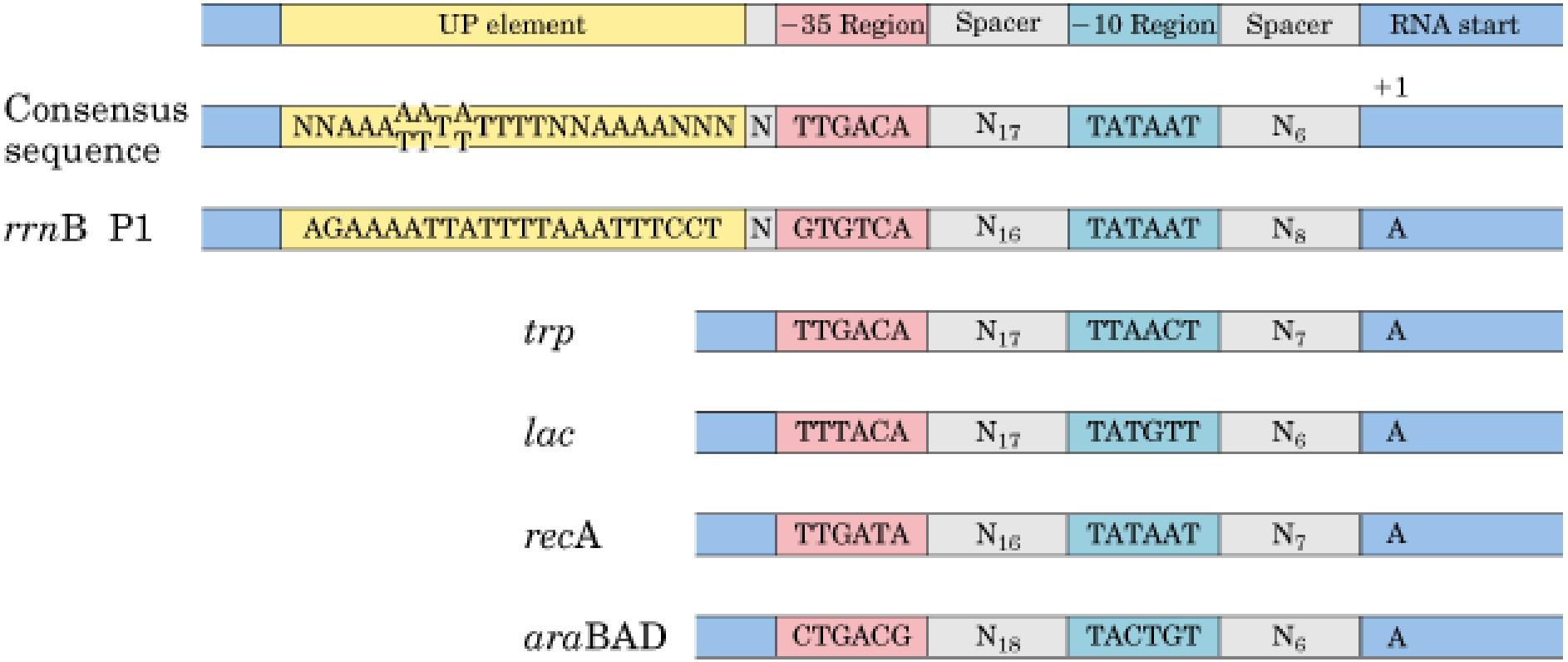
**Promoter** sequences are located adjacent to genes.

Promoters have two **consensus sequences** centered at –10 and –35 positions (the first residue of the RNA is given +1).

**TSS**
**Transcription start site**

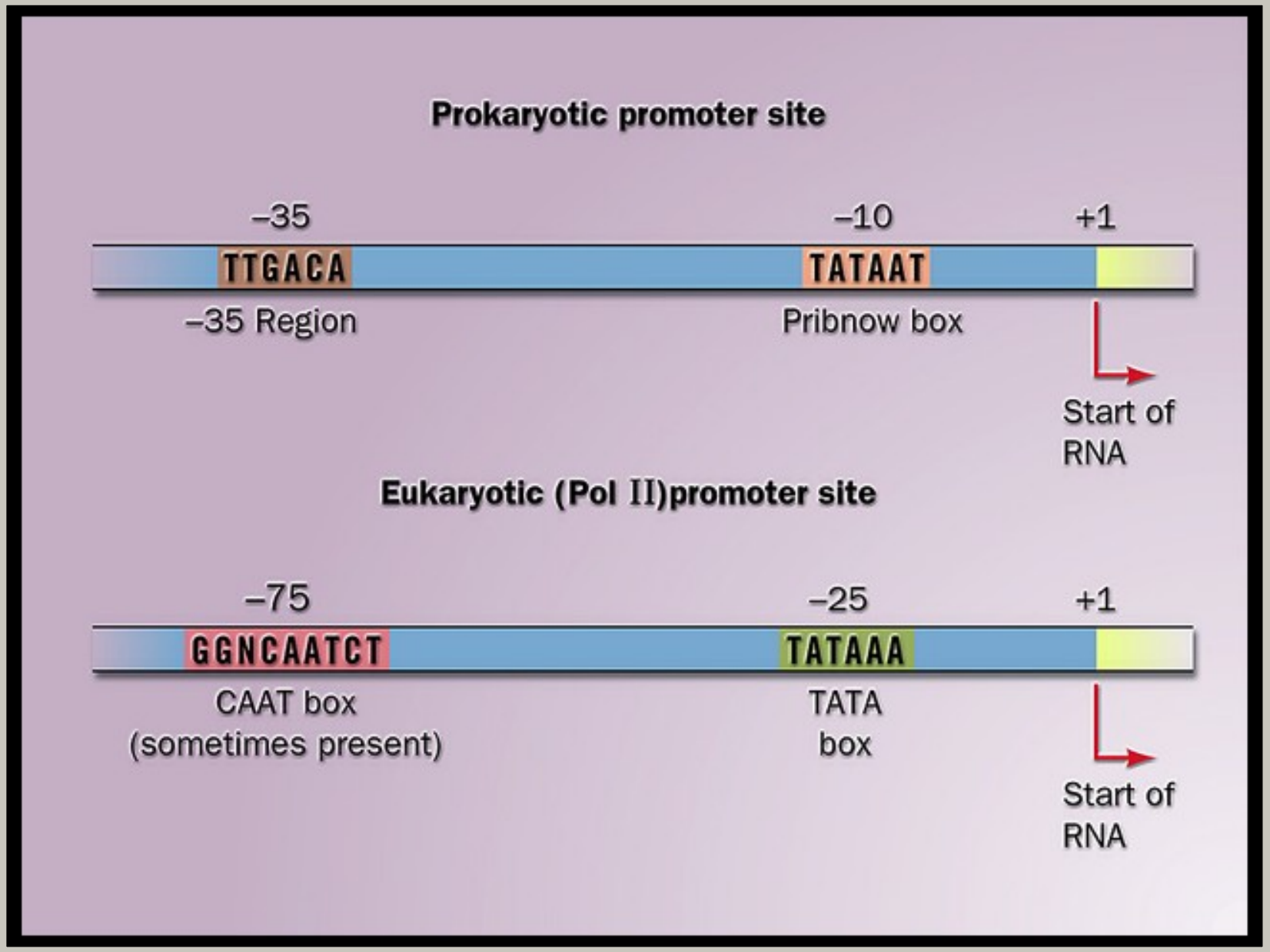



Consensus sequences for
RNA polymerase binding
TSS
Transcription
start site
–35
TTGACA
–35 region
15-17 bases
–10
TATAAT
Pribnow box
5-9 bases
+1
Coding
sequences
RNA
polymerase
Prokaryotic Promoter




the –10 sequence, TATAAT (the Pribnow box)

the –35 consensus region, TTGACA


Prokaryotic promoter site
–35
–10
+1
TTGACA
TATAAT
–35 Region
Pribnow box
Start of RNA
Eukaryotic (Pol II)promoter site
–75
–25
+1
GGNCAATCT
TATAAA
CAAT box
(sometimes present)
TATA box
Start of RNA

# 5. The σ subunits enable the *E.coli* RNA polymerase to recognize specific promoter sites

The RNA polymerase without the σ subunit (i.e., the $\alpha_2\beta\beta'\omega$, **core enzyme**) is unable to start transcription at a promoter.

Promoter vary up to 1000-fold in their efficiency of initiation which depends on the exact sequence of the key promoter elements as well as flanking sequences.





# Three stages of transcription

**Initiation**

**Elongation**

**Termination**





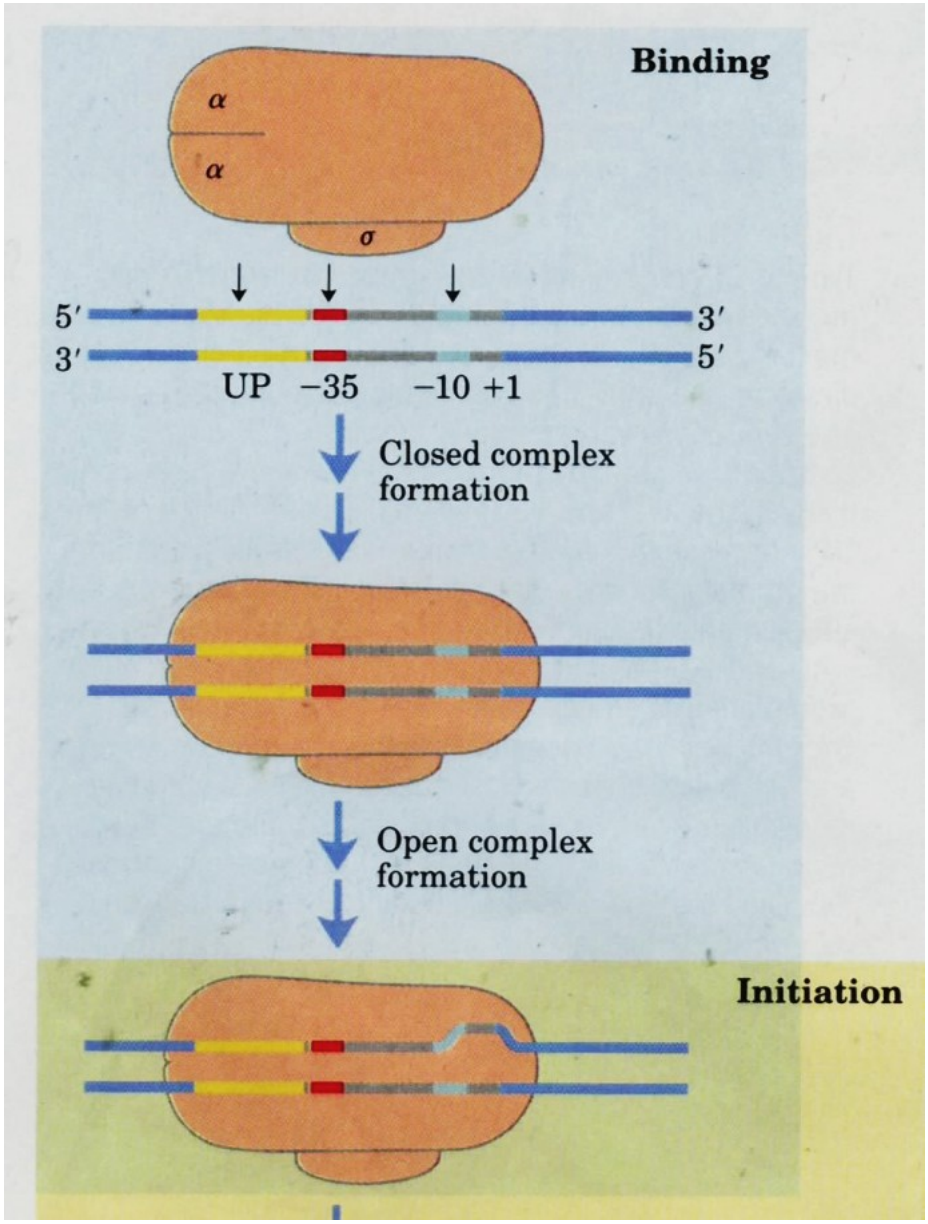
# 6. RNA polymerase unwinds the template DNA then initiate RNA synthesis - Initiation

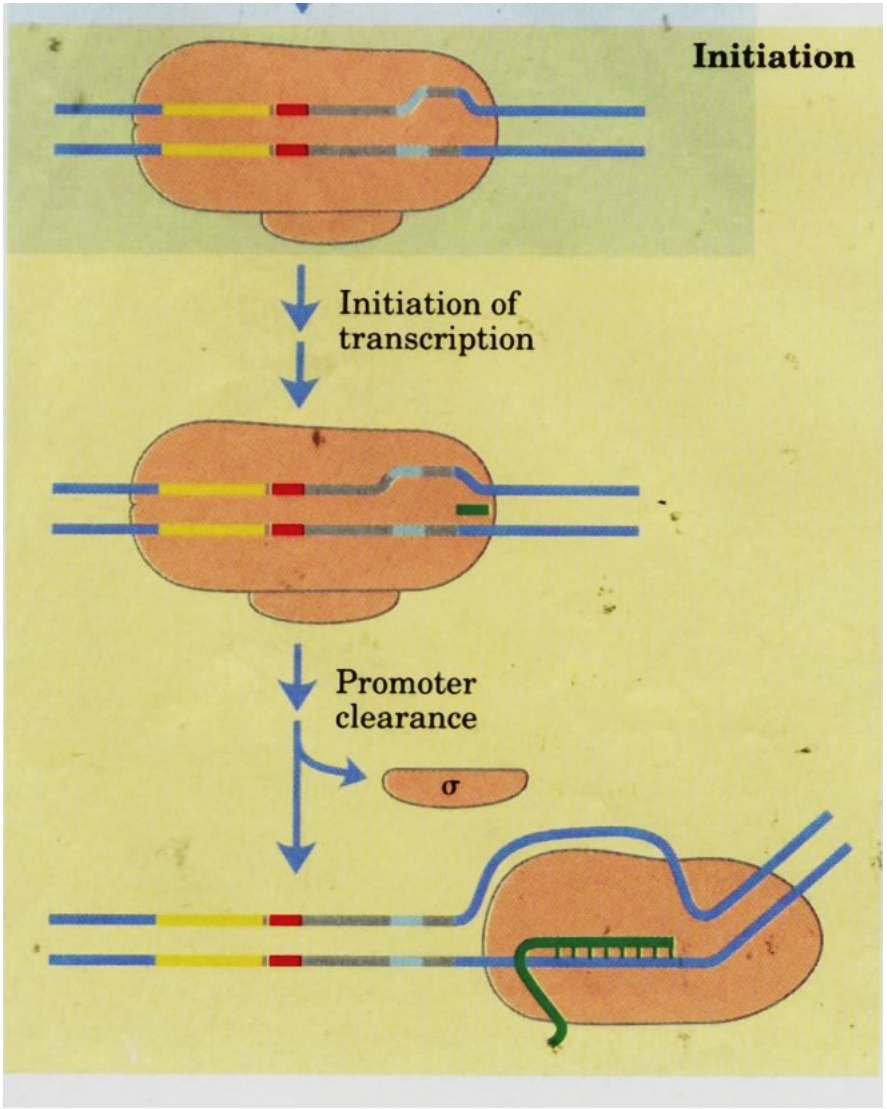
The enzyme slides to a promoter region and forms a more tightly bound "**closed complex**". Then the polymerase-promoter complex has to be converted to an "**open complex**", in which a 12-15 bp covering the region from the AT-rich –10 site to +3 site is unwound.

The essential transition from a "closed" to an "open" complex sets the stage for RNA synthesis, after which the core polymerase moves away from the promoter.






Binding
α
α
σ
5′
3′
3′
5′
UP
−35
−10 +1
Closed complex formation
Open complex formation
Initiation

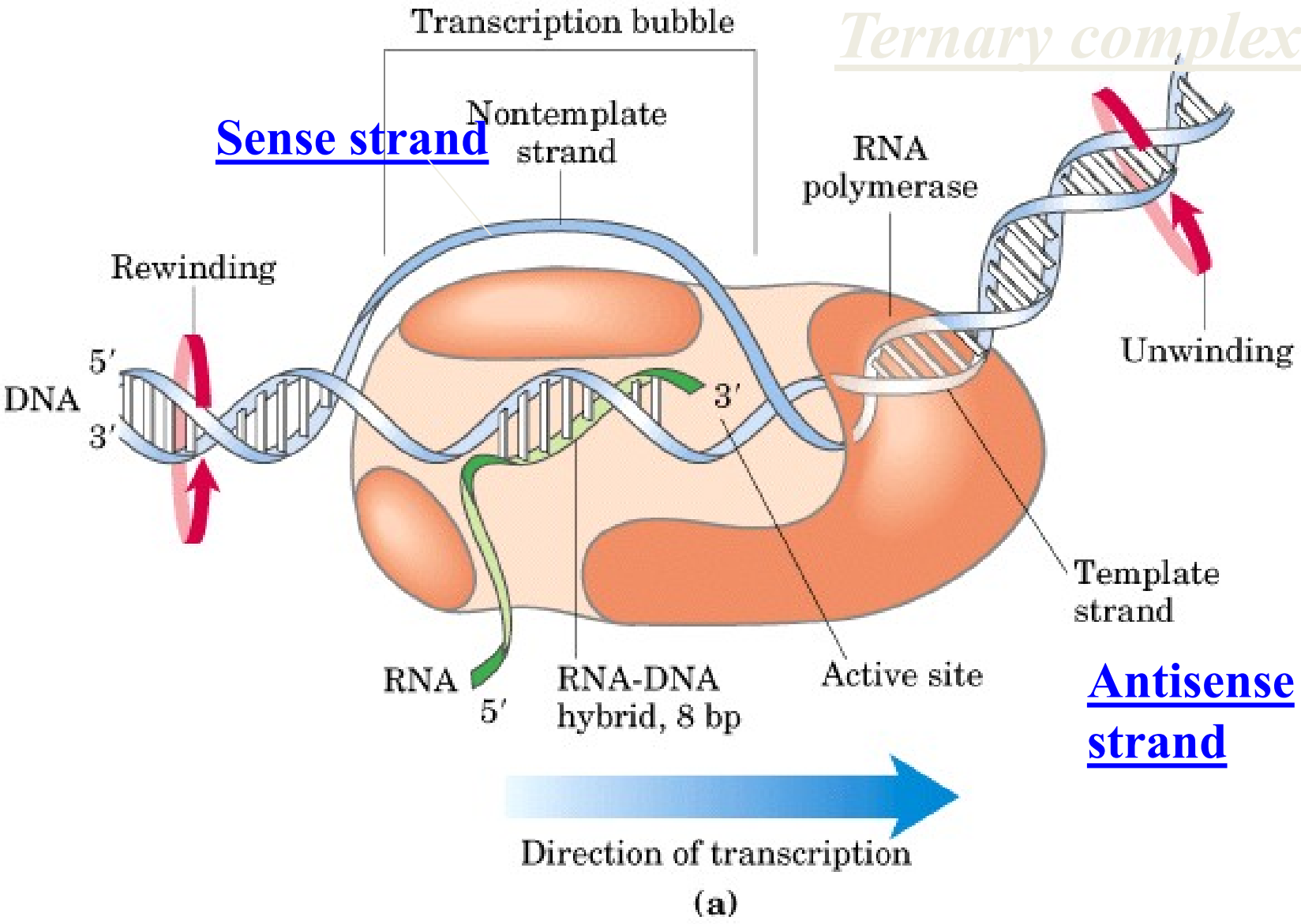

Initiation
Initiation of transcription
Promoter clearance
σ

# 7. After initiation, the factor σ is released to leave the core enzyme which continues elongation

The core enzyme ($\alpha_2\beta\beta'\omega$) continues elongation of the RNA transcript. The RNA synthesizes RNA in the 5'——3' direction, using the four ribonucleoside 5' triphosphates (ATP, GTP, CTP, UTP) as precursors.








Ternary complex

**Sense strand**

| | |
|---|---|
| (5′) CGCTATAGCGTTT (3′) | DNA nontemplate (coding) strand |
| (3′) GCGATATCGCAAA (5′) | DNA template strand |
| (5′) CGCUAUAGCGUUU (3′) | RNA transcript |

**Antisense strand**





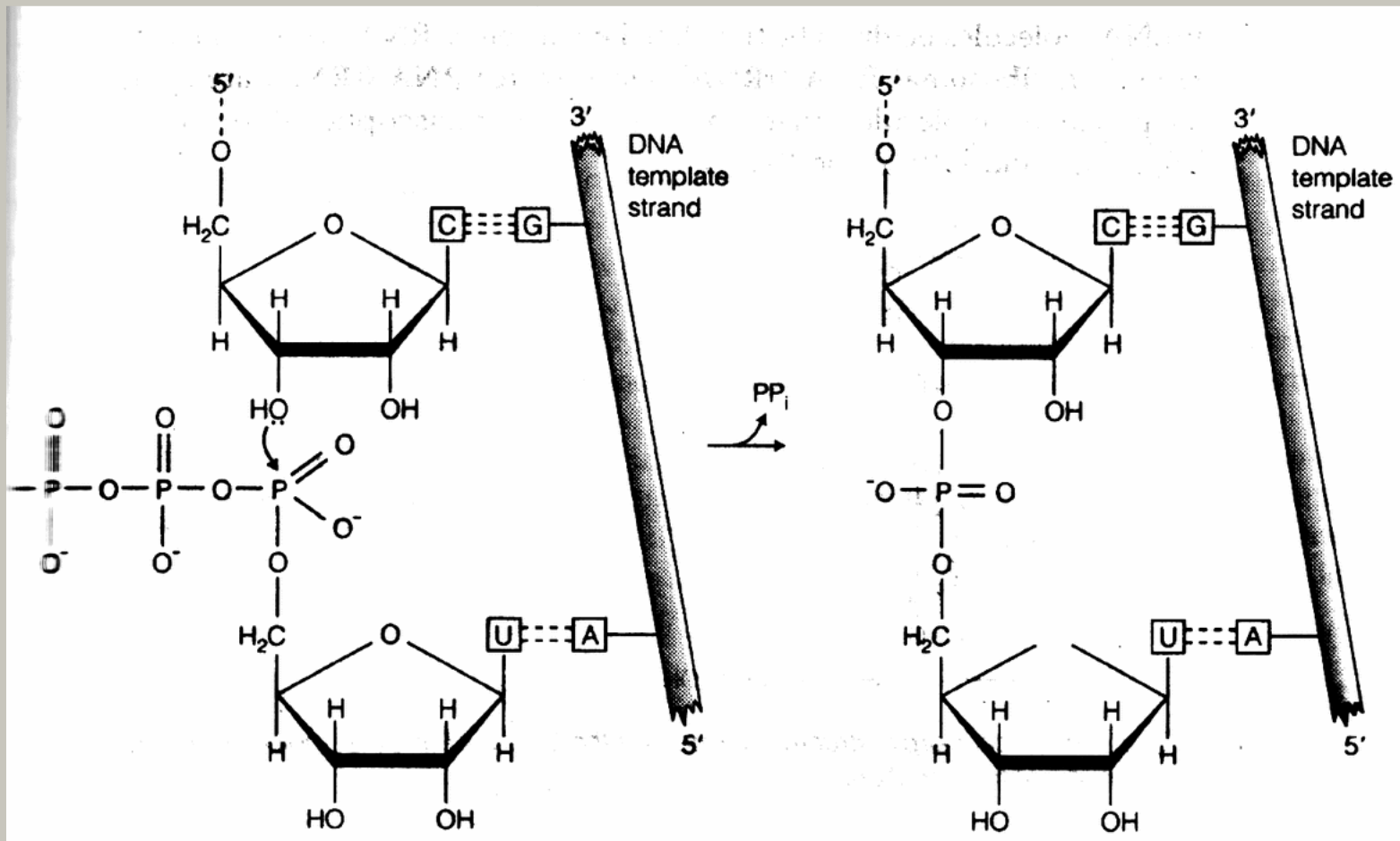



Fig. 2. *Transcription by RNA polymerase. In each step the incoming ribonucleotide selected is that which can base-pair with the next base of the DNA template strand. In the diagram, the incoming nucleotide is rUTP to base-pair with the A residue of the template DNA. A 3′5′ phosphodiester bond is formed, extending the RNA chain by one nucleotide, and pyrophosphate is released. Overall the RNA molecule grows in a 5′ to 3′ direction.*

# 8. *E.coli* RNA polymerase stops synthesizing RNA at specific terminator DNA sequences

## -Termination

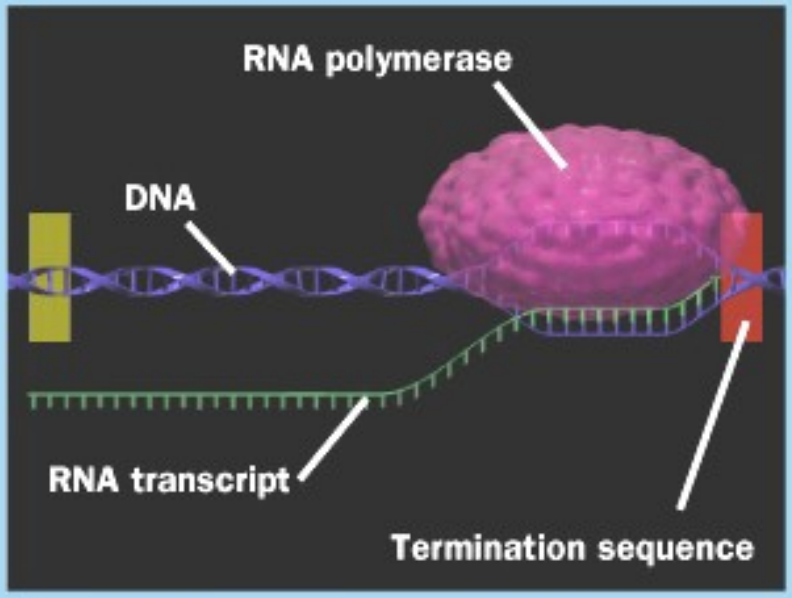
A common termination signal is a hairpin structure formed by a palindromic GC-rich region, followed by an AT-rich sequence.

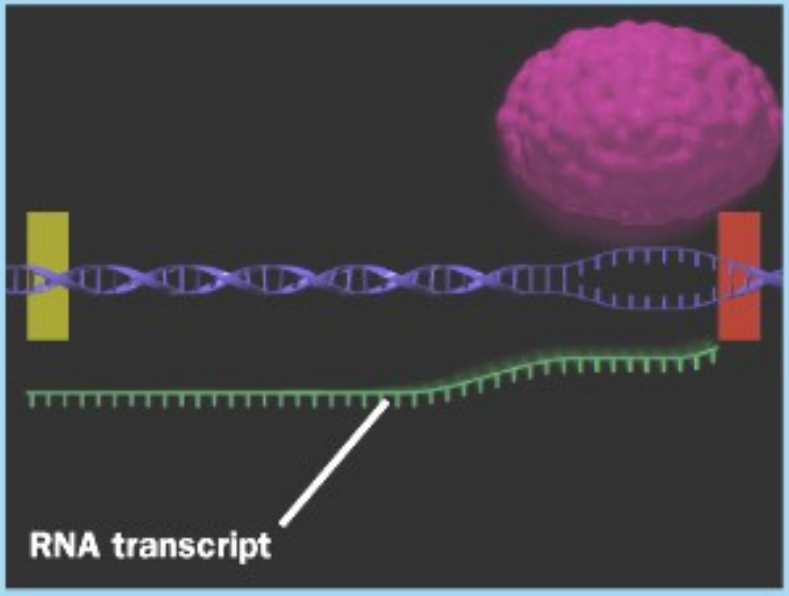
Other signals are also used which require the assistance of ρ protein for effective termination.





# TERMINATION of TRANSCRIPTION




When RNA polymerase reaches the termination sequence on the DNA template strand...

...the RNA transcript and RNA polymerase are released.

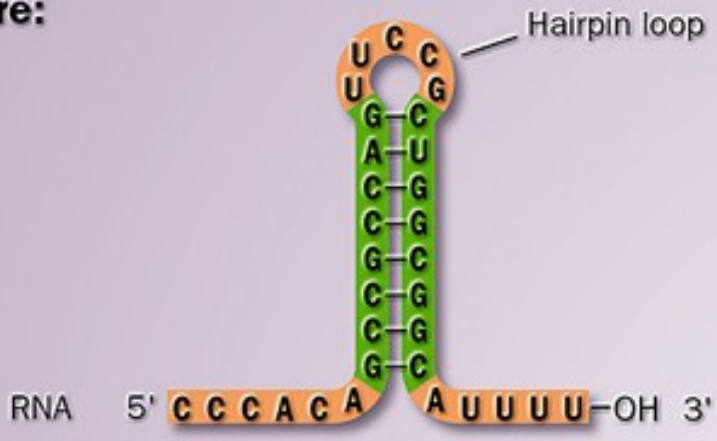
Palindrome sequences
回文序列

**Primary structure:**

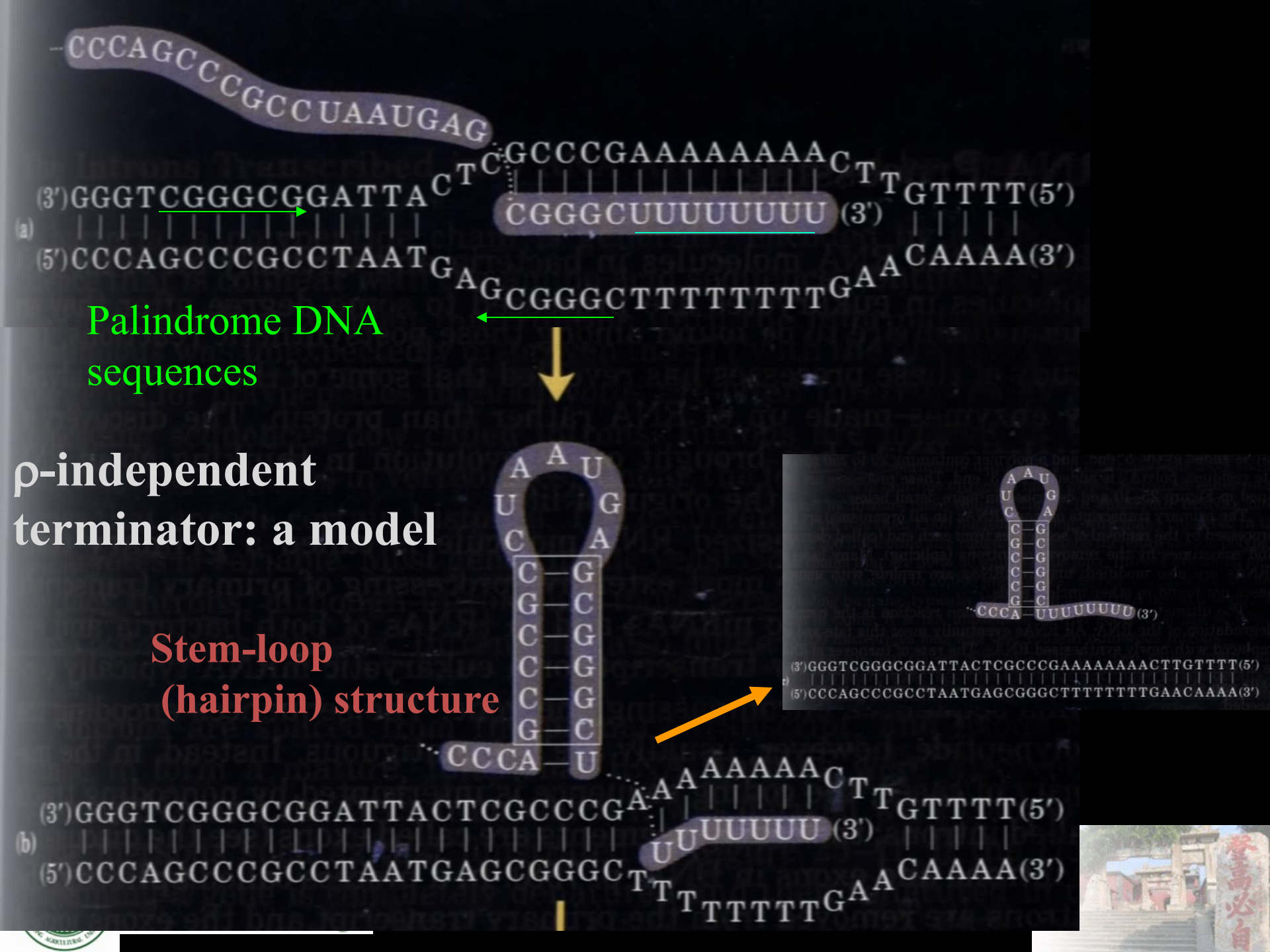
RNA 5' C C C A C A G C C G C C A G U U C C G C U G G C G G C A U U U U –OH 3'

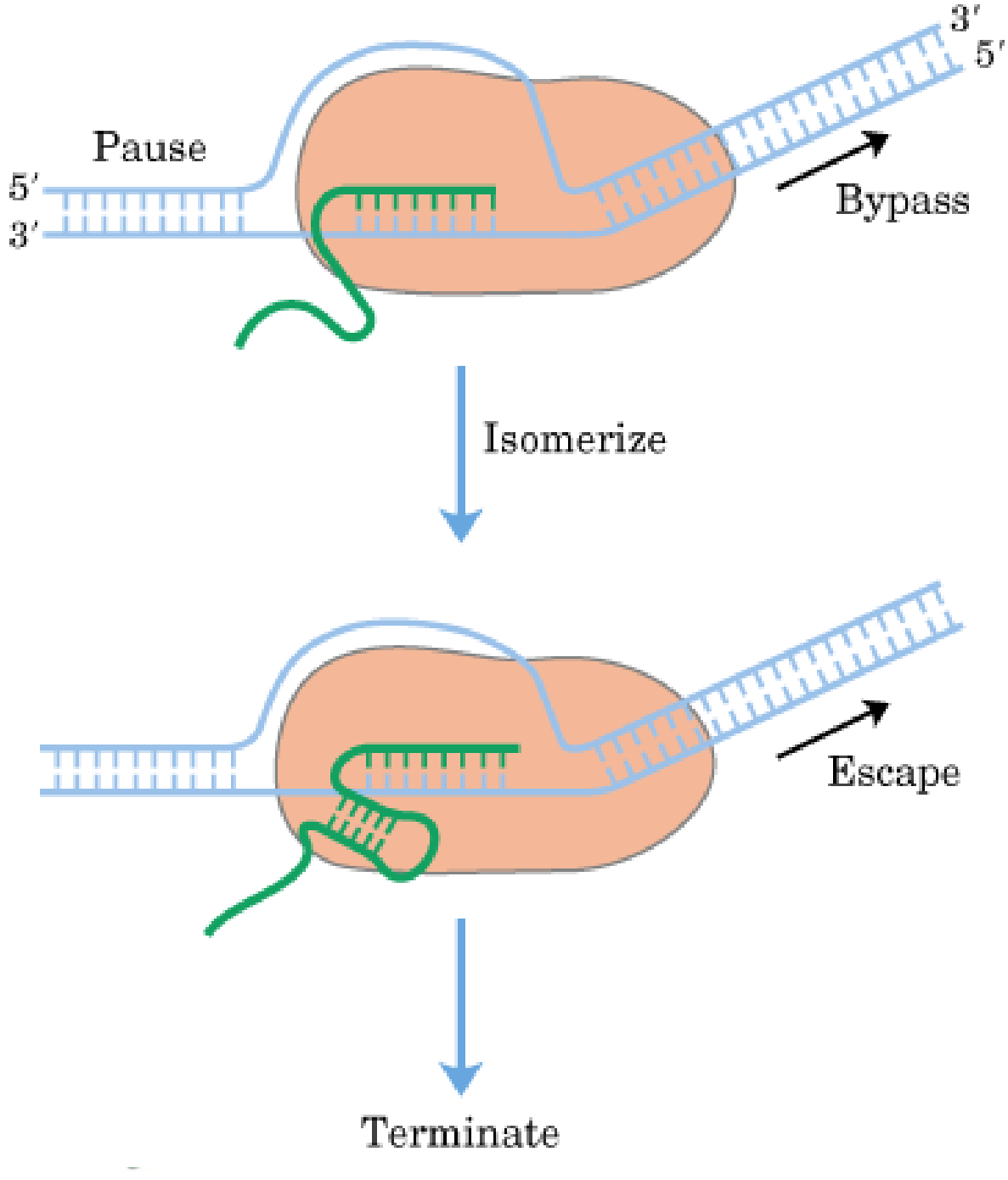
3′–C G G C G G U C  G A C C G C C G –5′

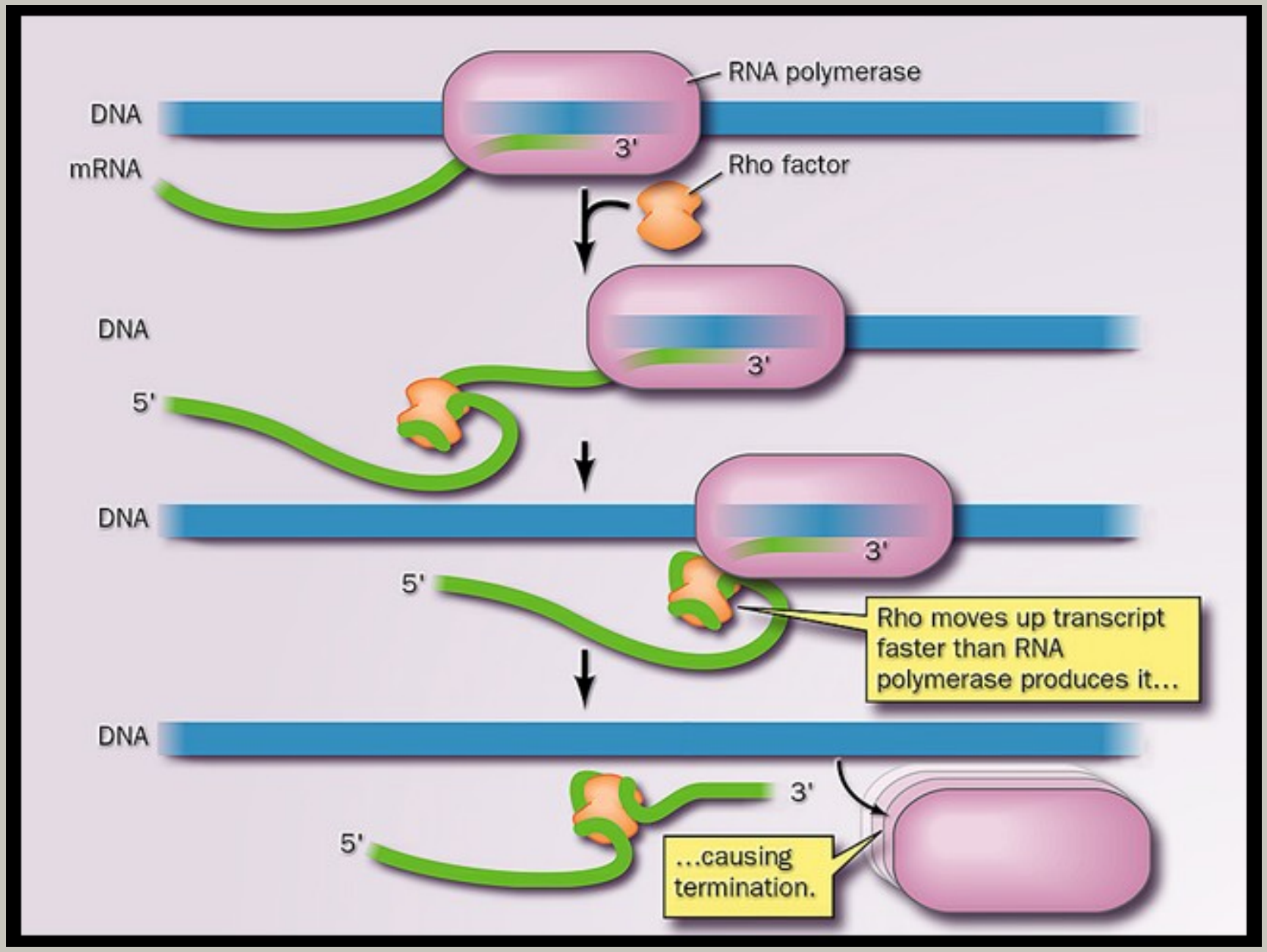
**Secondary structure:**




# Termination Sequence


Palindrome DNA sequences
ρ-independent terminator: a model
Stem-loop (hairpin) structure


3′
5′
Pause
5′
3′
Bypass
Isomerize
Escape
Terminate


RNA polymerase
DNA
mRNA
3'
Rho factor
DNA
3'
5'
DNA
3'
5'
Rho moves up transcript faster than RNA polymerase produces it…
DNA
3'
5'
…causing termination.

# Model for an ρ-dependent terminator

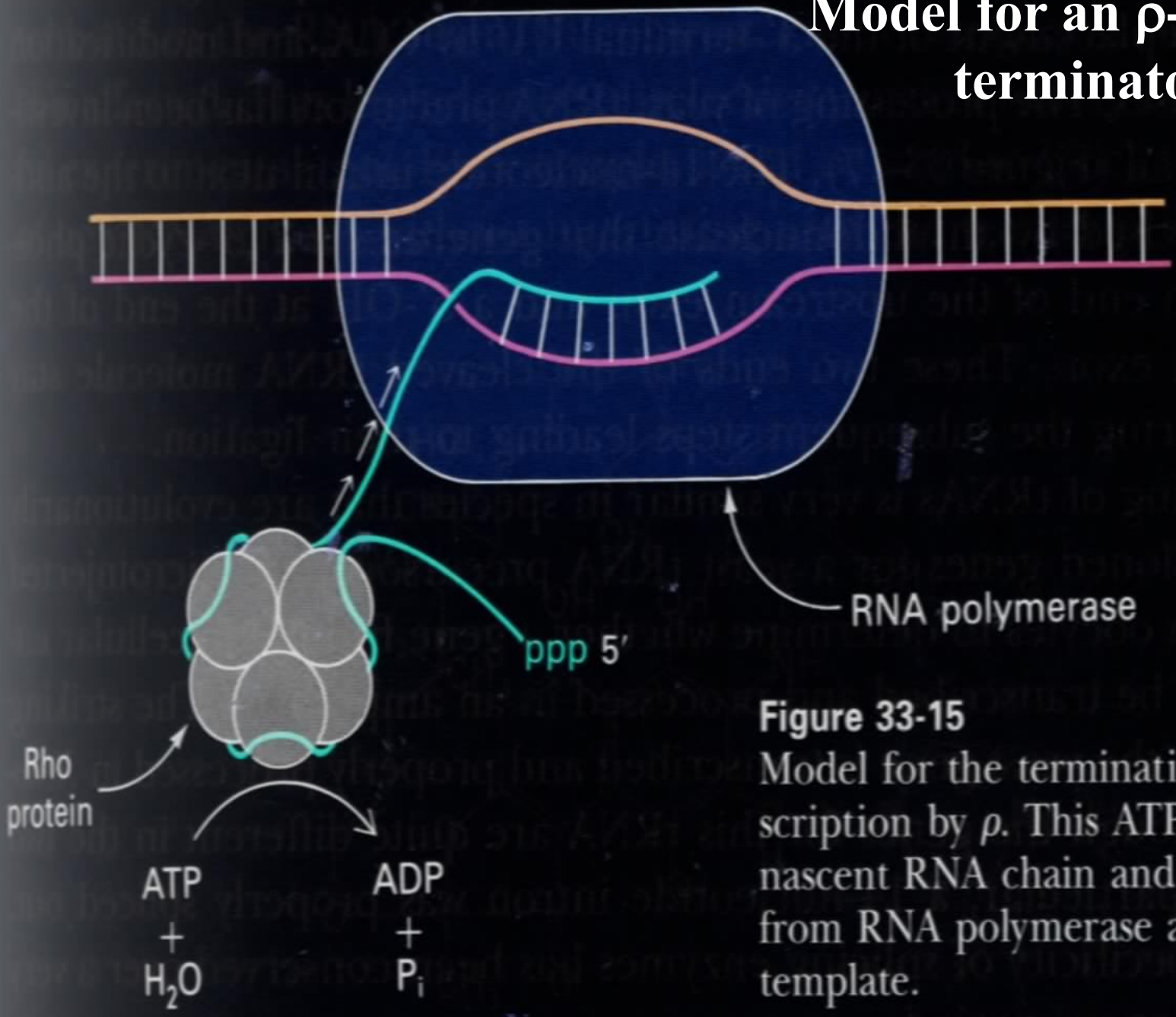
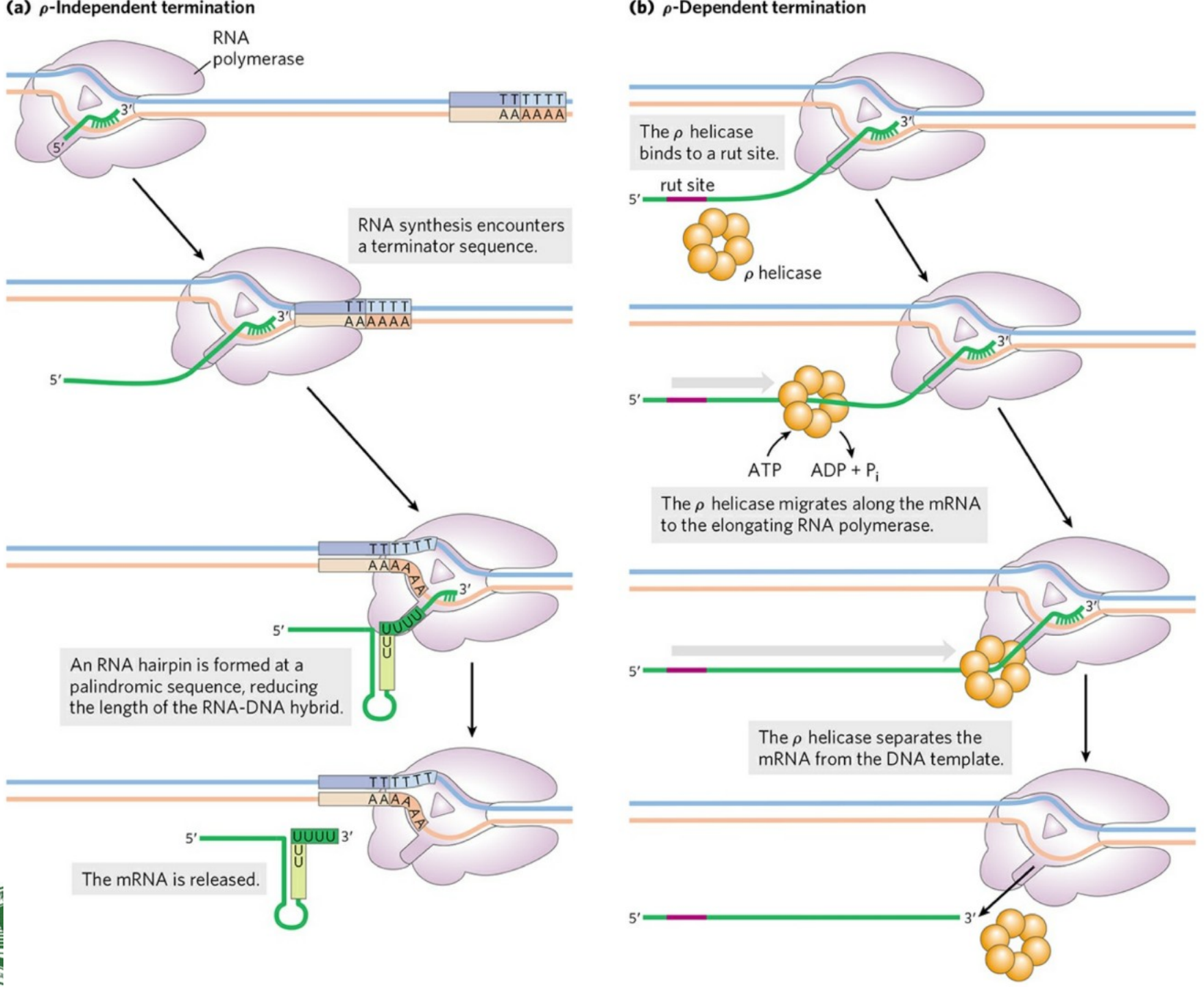
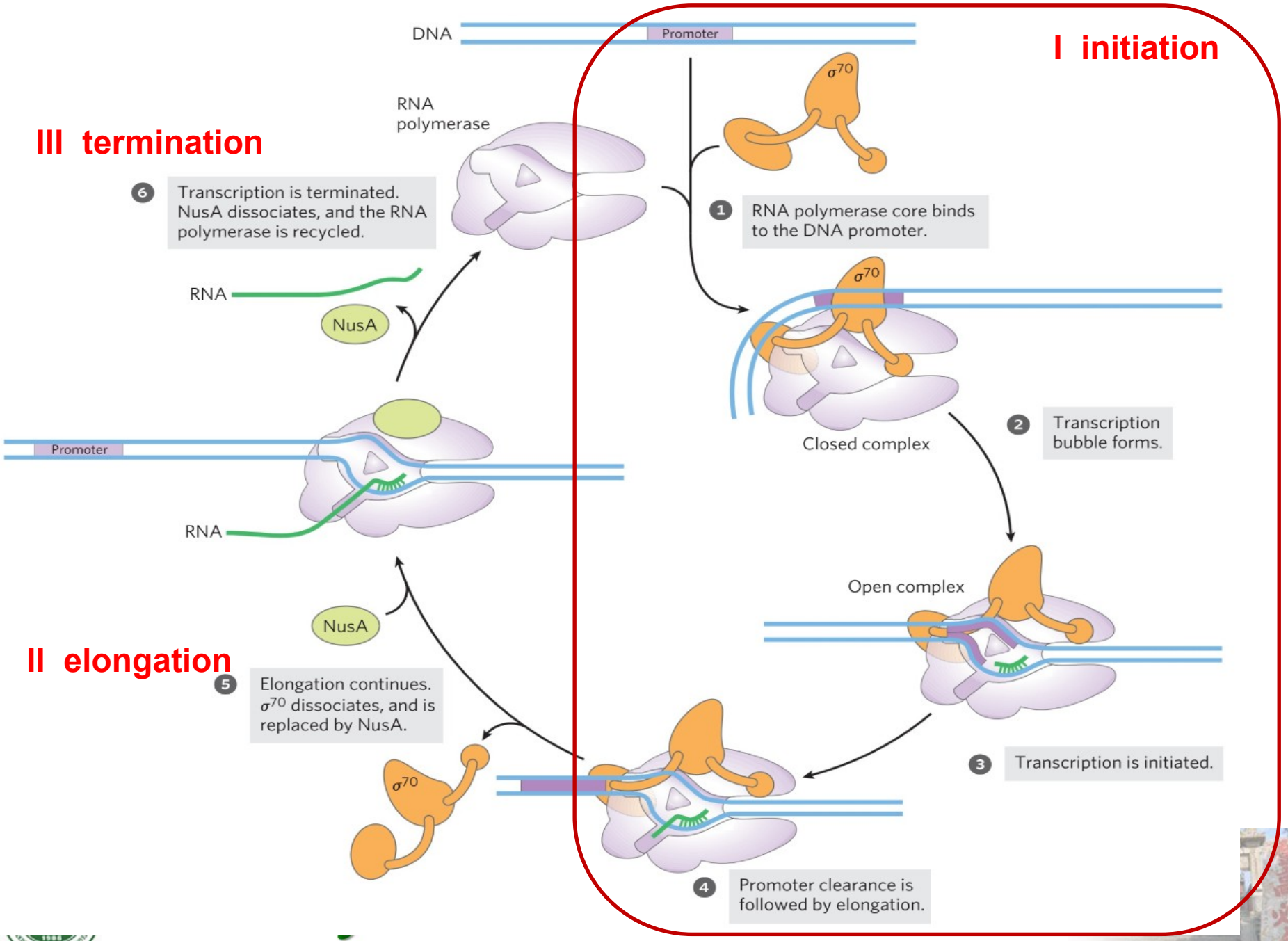



**Figure 33-15**
Model for the termination of transcription by *ρ*. This ATPase binds the nascent RNA chain and pulls it away from RNA polymerase and the DNA template.


(a) ρ-Independent termination
RNA polymerase
3′
5′
TTTTTT
AAAAAA
RNA synthesis encounters a terminator sequence.
An RNA hairpin is formed at a palindromic sequence, reducing the length of the RNA-DNA hybrid.
UUUU
CC
The mRNA is released.
(b) ρ-Dependent termination
The ρ helicase binds to a rut site.
rut site
ρ helicase
ATP
ADP + Pi
The ρ helicase migrates along the mRNA to the elongating RNA polymerase.
The ρ helicase separates the mRNA from the DNA template.


I initiation
II elongation
III termination
DNA
Promoter
RNA polymerase
σ70
1 RNA polymerase core binds to the DNA promoter.
Closed complex
2 Transcription bubble forms.
Open complex
3 Transcription is initiated.
4 Promoter clearance is followed by elongation.
5 Elongation continues. σ70 dissociates, and is replaced by NusA.
NusA
RNA
6 Transcription is terminated. NusA dissociates, and the RNA polymerase is recycled.

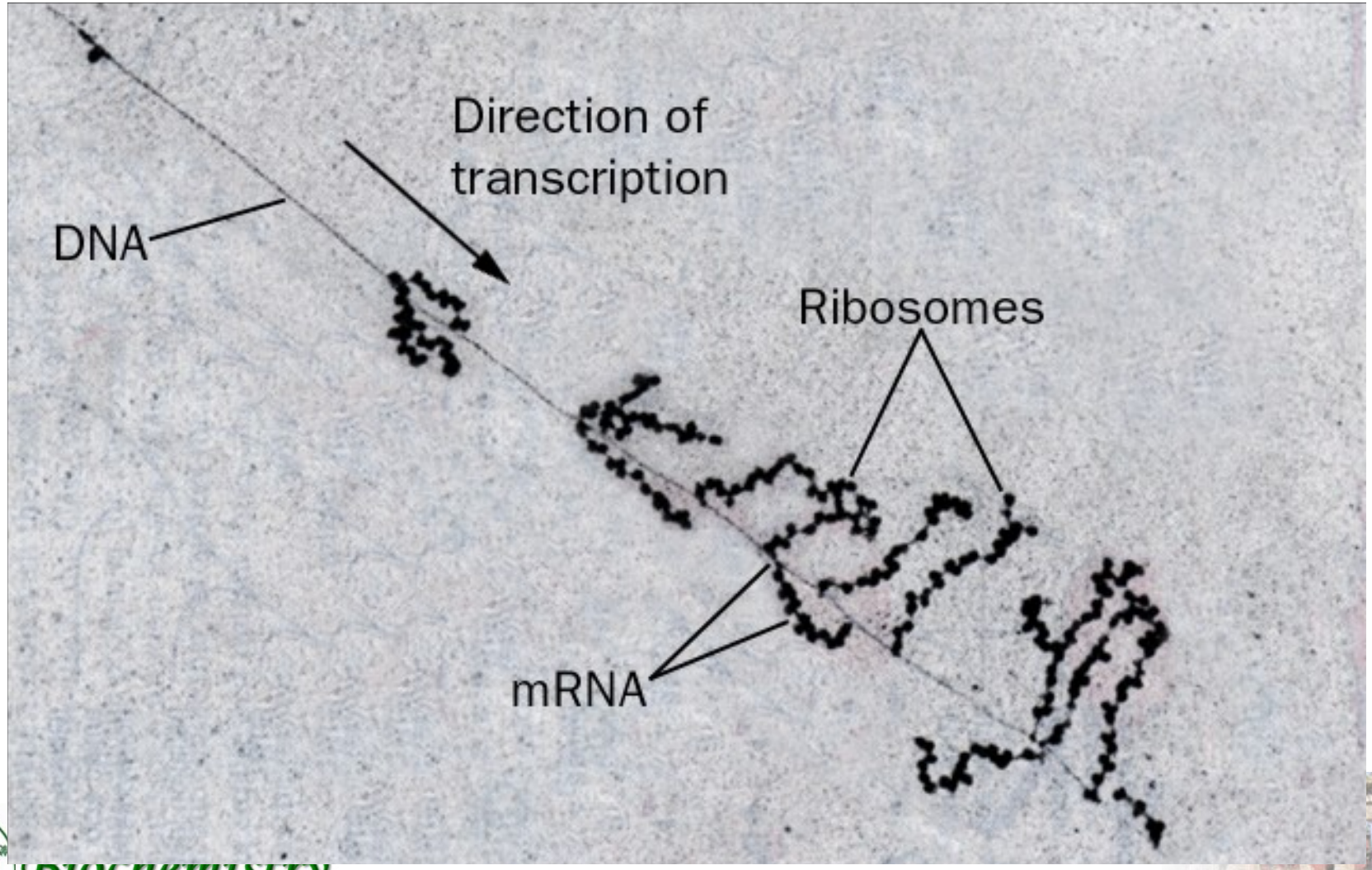
# 9. RNA processing

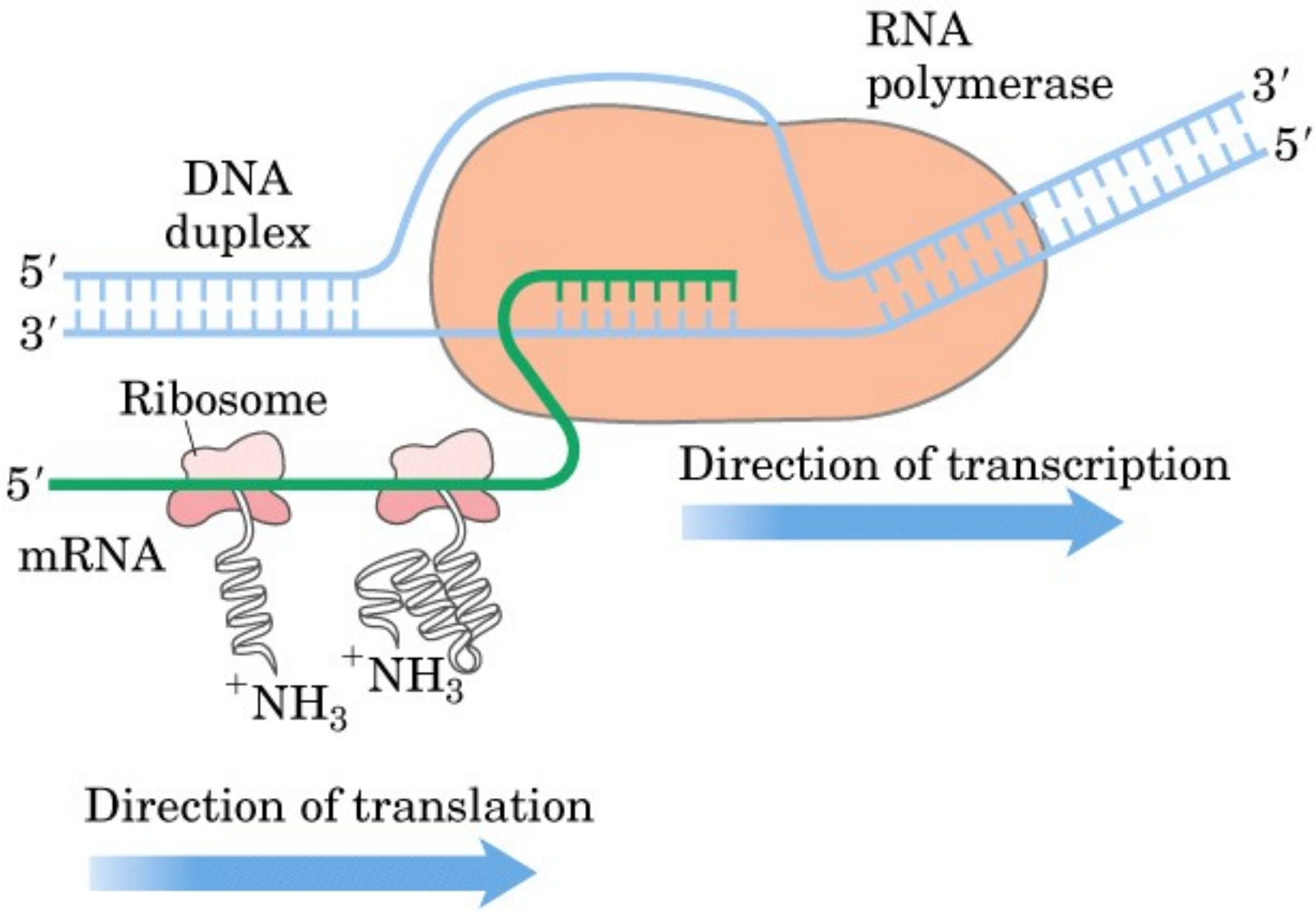
In prokaryotes, RNA transcribed from protein-coding genes (messenger RNA, mRNA), requires little or no modification prior to translation However, mRNA in eukaryotes, ribosomal RNA (rRNA) and transfer RNA (tRNA) are synthesized as precursor molecules that do require post-transcriptional processing（转录后加工） ) (G8, G9, G10)





# This E. coli gene is being simultaneously transcribed and translated




RNA
polymerase
3′
5′
DNA
duplex
5′
3′
Ribosome
5′
mRNA
Direction of transcription
+NH3
+NH3
Direction of translation

**顺反子（ cistron ）：**

即结构基因，决定一条多肽链合成的功能单位。

**单顺反子（ monocistron ）：**

真核基因转录产物为单顺反子，即一个基因编码一条多肽链或 RNA 链，每个基因转录有各自的调节元件。

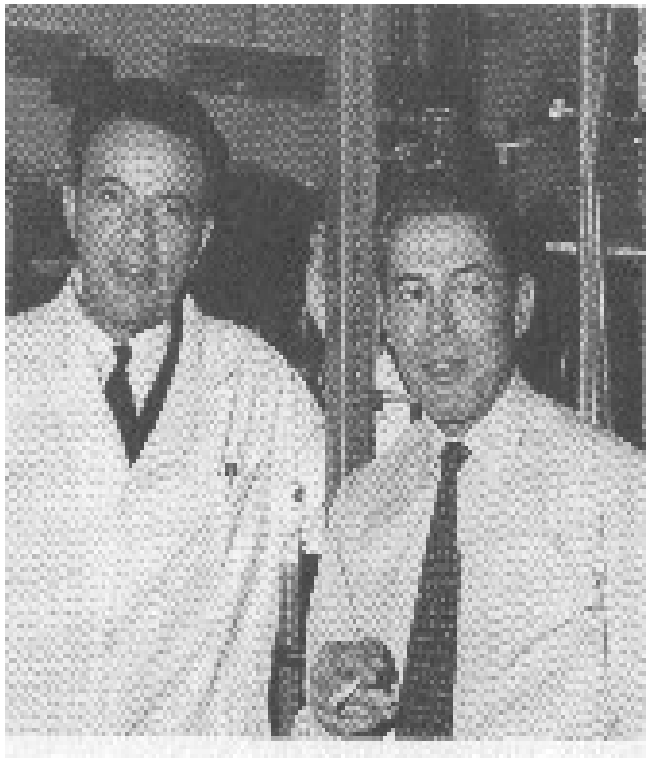
**多顺反子（ polycistron ）：**

在原核细胞中，通常是几种不同的 mRNA 连在一起，相互之间由一段短的不编码蛋白质的间隔序列所隔开，这种 mRNA 叫做多顺反子mRNA。





# G3 The *Lac* Operon



## 操纵子学说

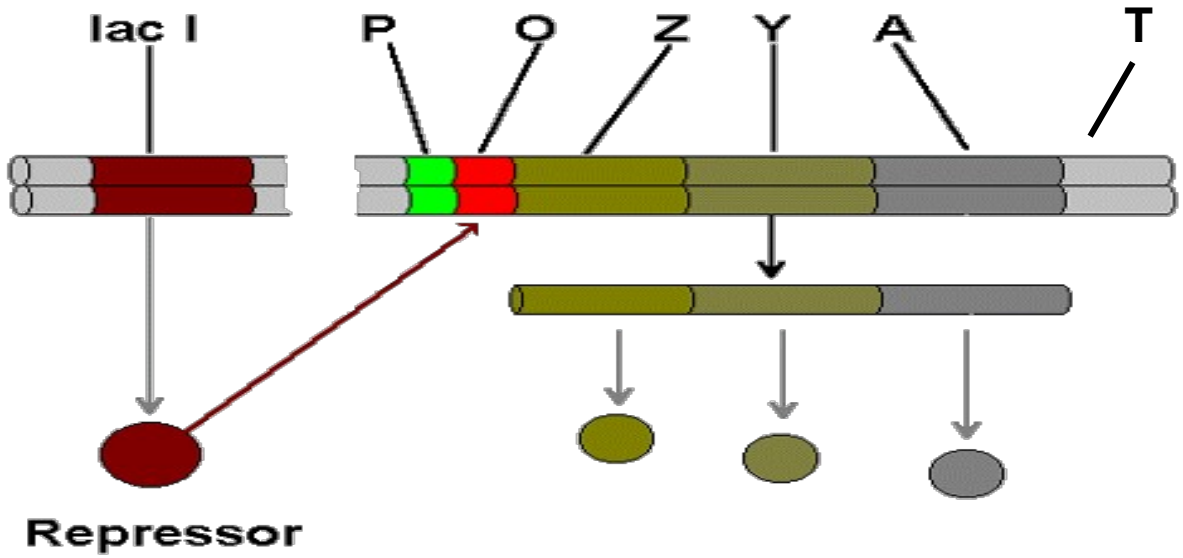
### 1、操纵子模型的提出

1961 年， Monod 和 Jacob 提出乳糖操纵子模型

获 1965 年诺贝尔生理学和医学奖

# 2、操纵子（operon）的定义

操纵子：是基因表达的协调单位，由启动子、操纵基因及其所控制的一组功能上相关的结构基因所组成。操纵基因受调节基因产物的控制。



## ■操纵子模型（ lac operon ）

操纵子的基本组成元件包括：

启 动 子（ promoter， P ）

操纵基因（ operator， O ）

结构基因（ structural gene(s) ）

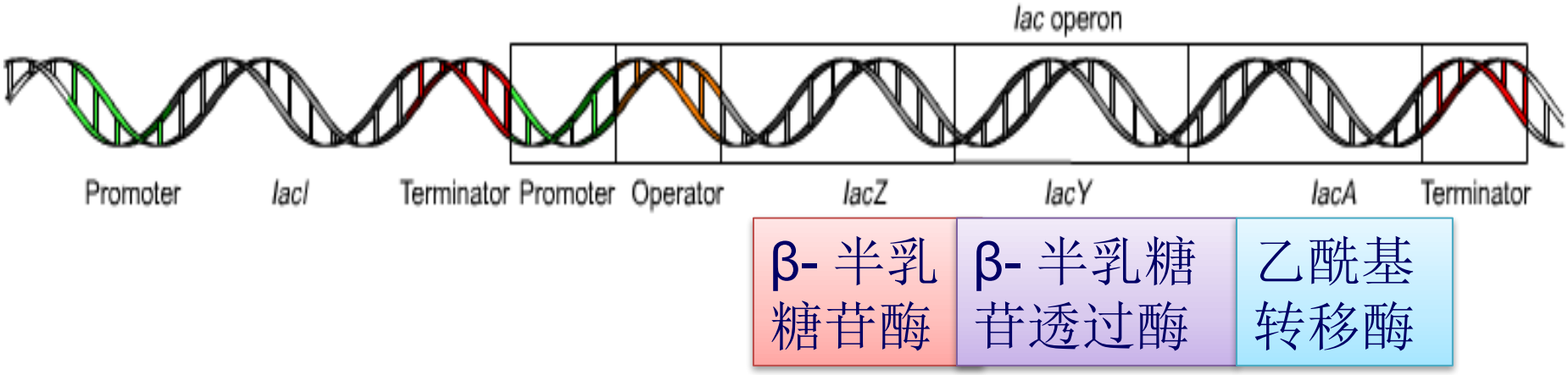
终 止 子 （ terminator ）

调节基因（ regulatory gene ）



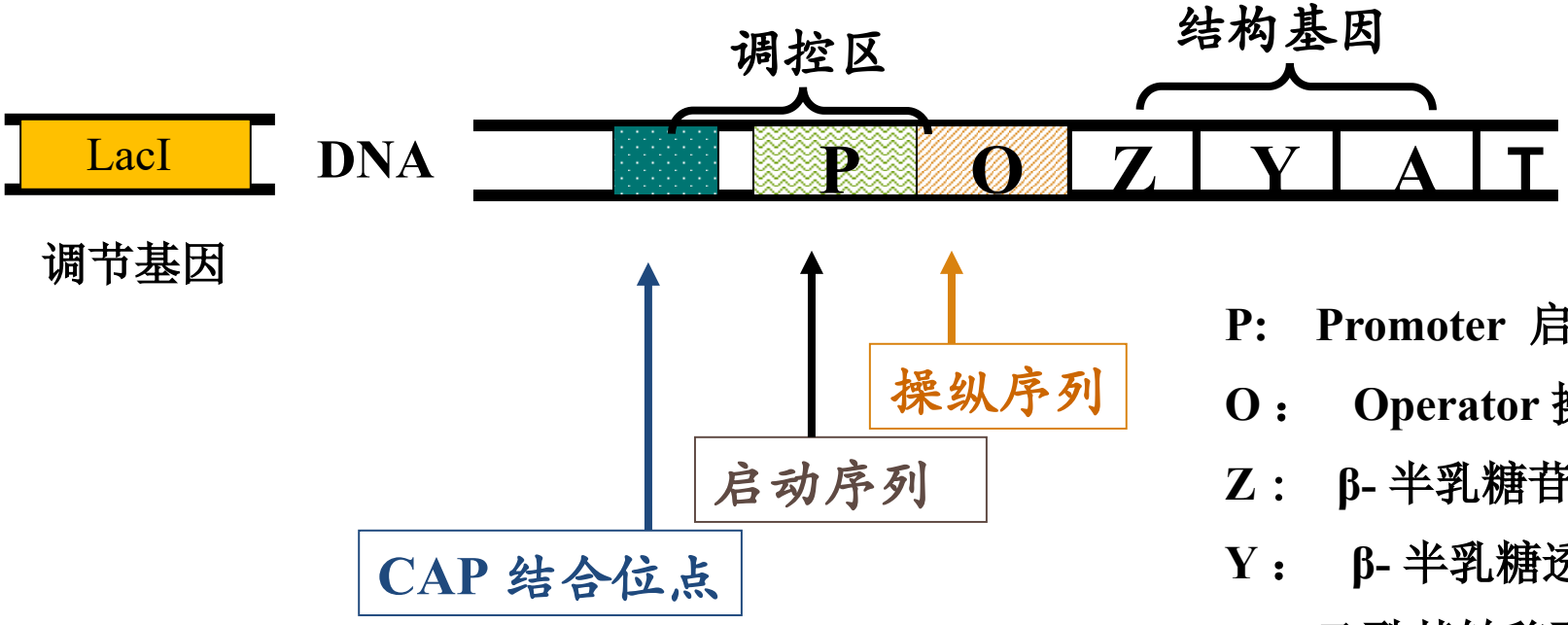


# Lac operon（乳糖操纵子）




*β*-galactosidase (*Z*),
galactoside permease (*Y*),
thiogalactoside transacetylase (*A*)

# 乳糖操纵子 - 正、负调控模型




P: Promoter 启动子

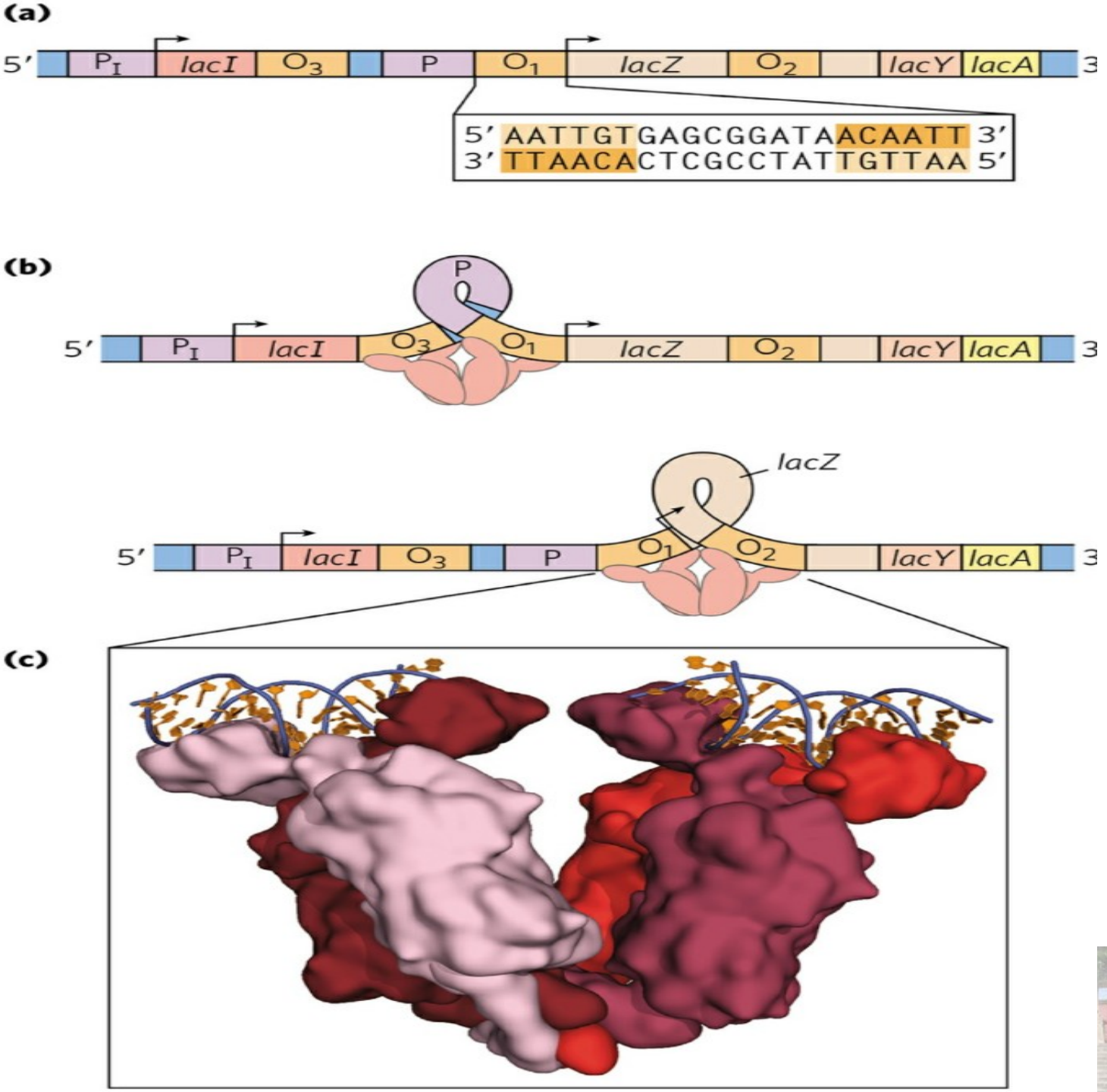
O： Operator 操纵基因

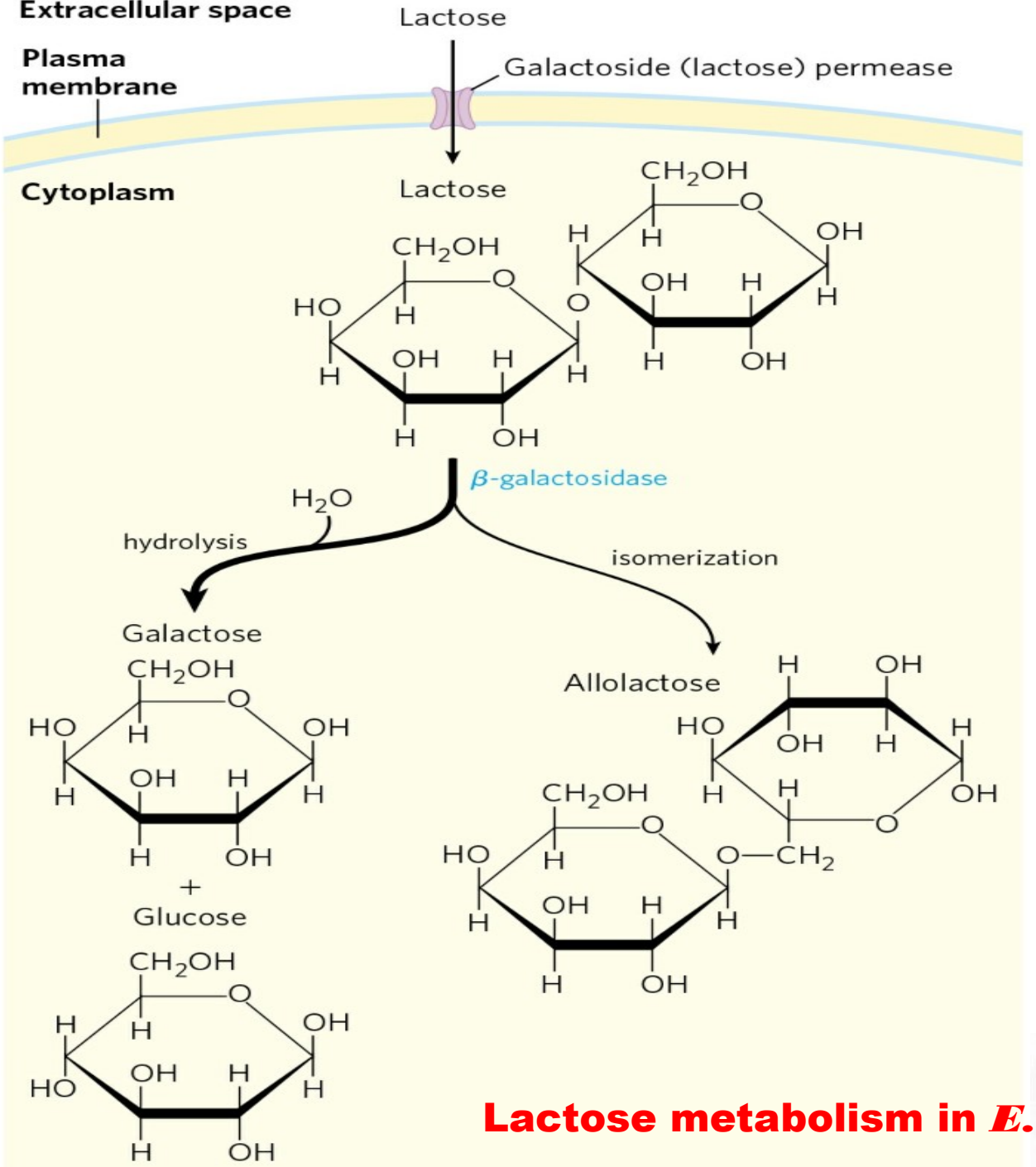
Z： β- 半乳糖苷酶

Y： β- 半乳糖透过酶

A： 乙酰基转移酶

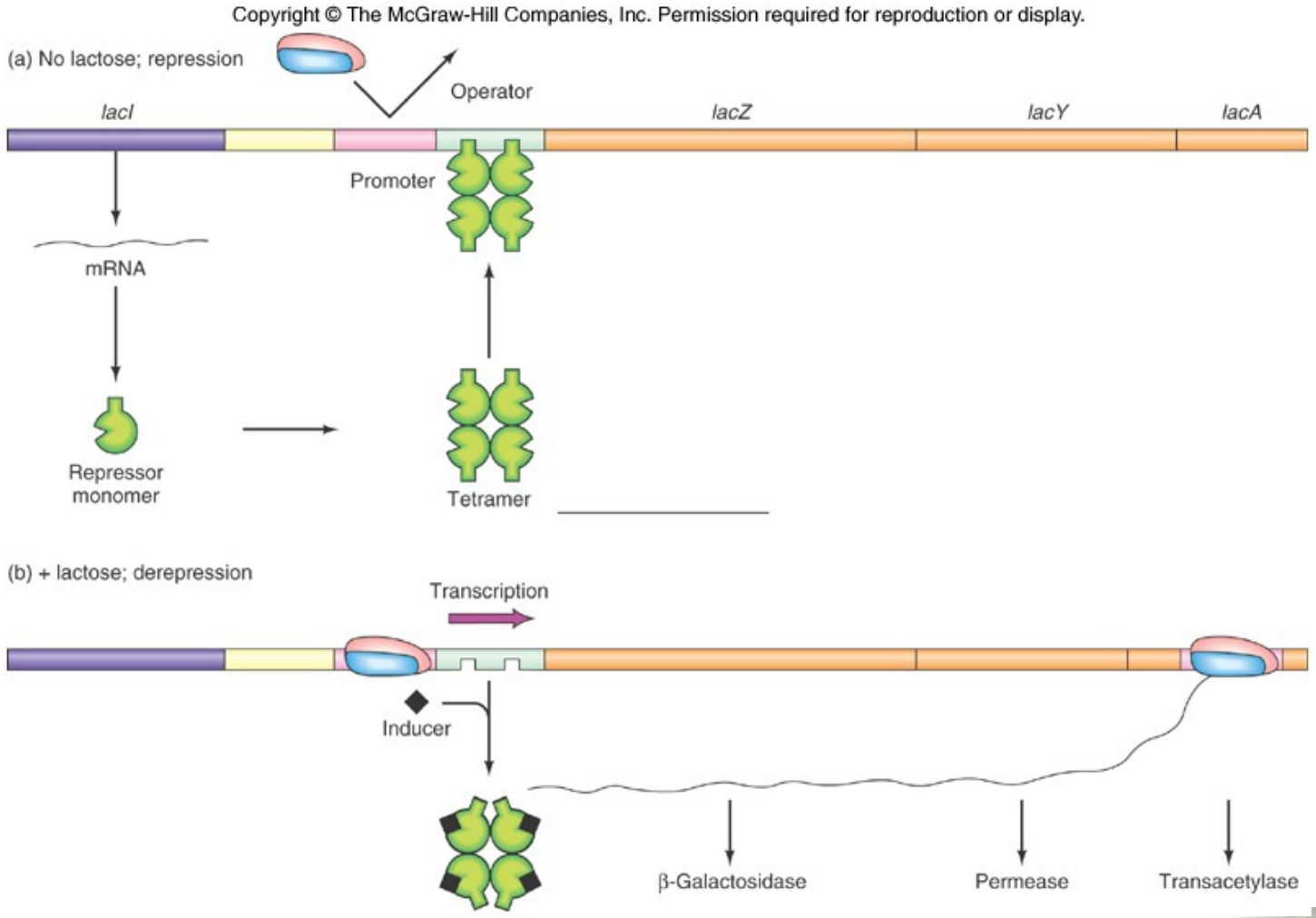
T： Terminator 终止子

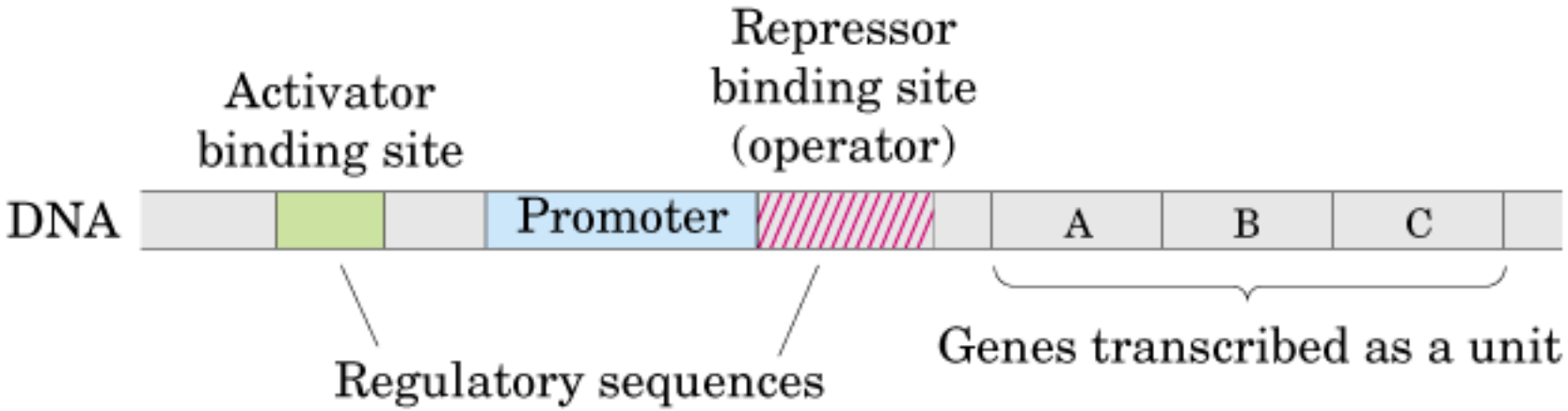

(a)
5′
$P_I$
lacI
$O_3$
P
$O_1$
lacZ
$O_2$
lacY
lacA
3
5′ AATTGTGAGCGGATAACAATT 3′
3′ TTAACACTCGCCTATTGTTAA 5′
(b)
P
5′
$P_I$
lacI
$O_3$
$O_1$
lacZ
$O_2$
lacY
lacA
3
lacZ
5′
$P_I$
lacI
$O_3$
P
$O_1$
$O_2$
lacY
lacA
3
(c)




**Lactose metabolism in *E. coli*.**

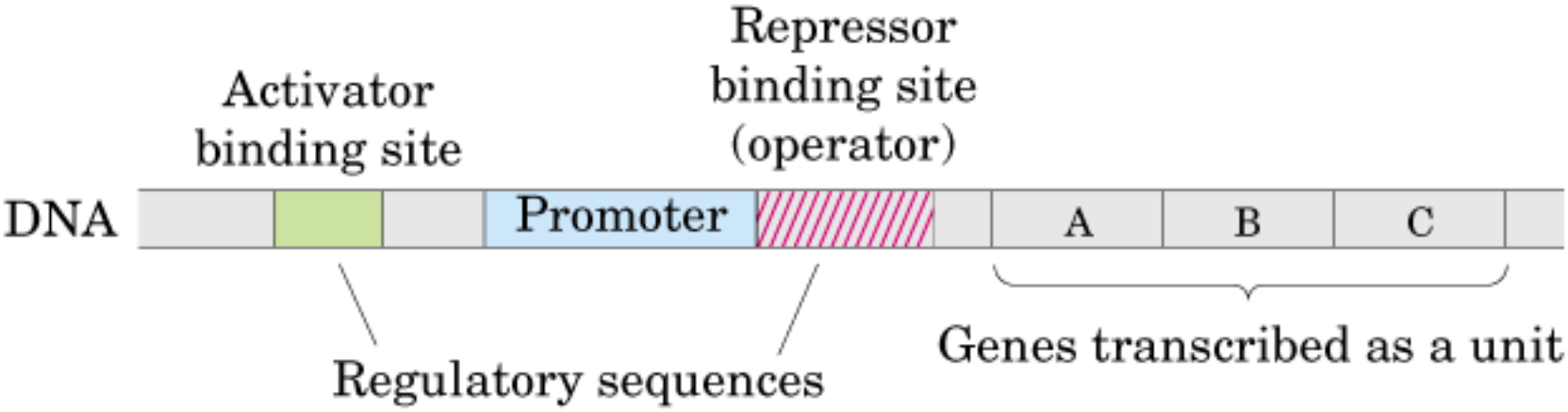
# Negative Control of the *lac* Operon




Activator
binding site
Repressor
binding site
(operator)
DNA
Promoter
A
B
C
Regulatory sequences
Genes transcribed as a unit

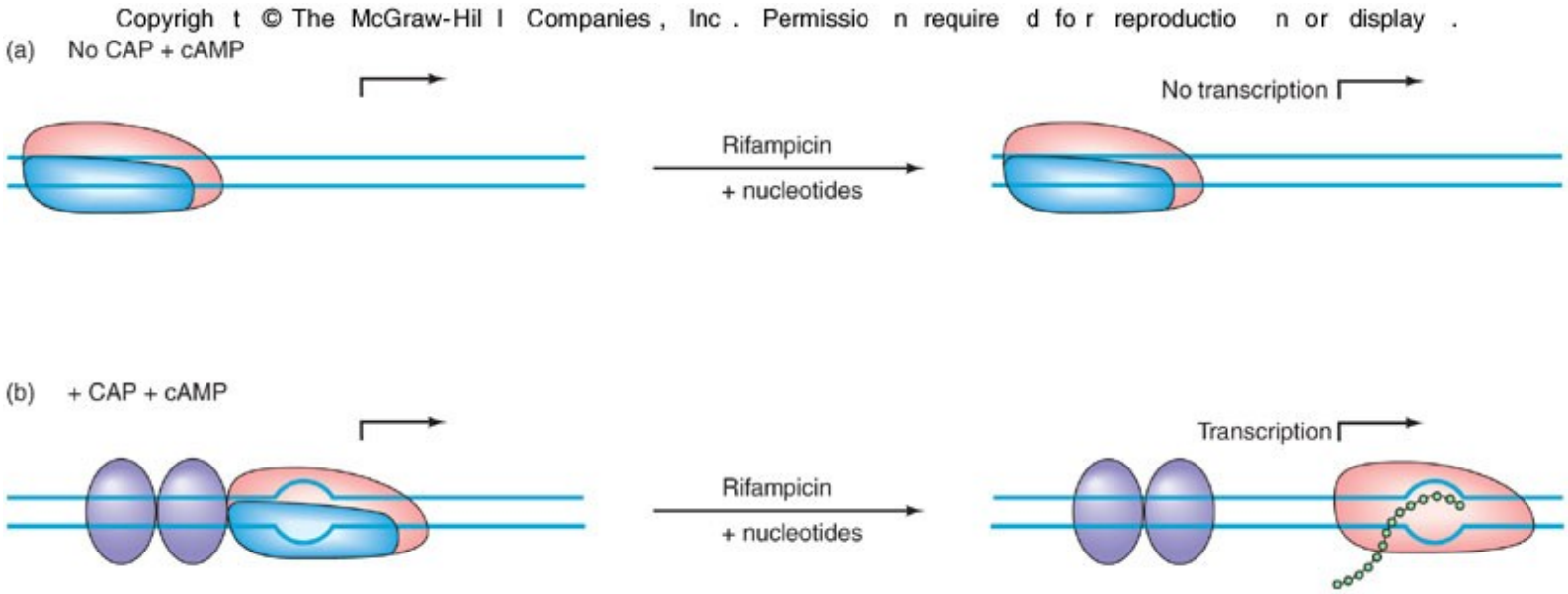
# cAMP 与代谢物激活蛋白－正调控模型

**代谢物激活蛋白（ CAP/CRP ） (catabolite activator protein/cAMP receptor protein)**



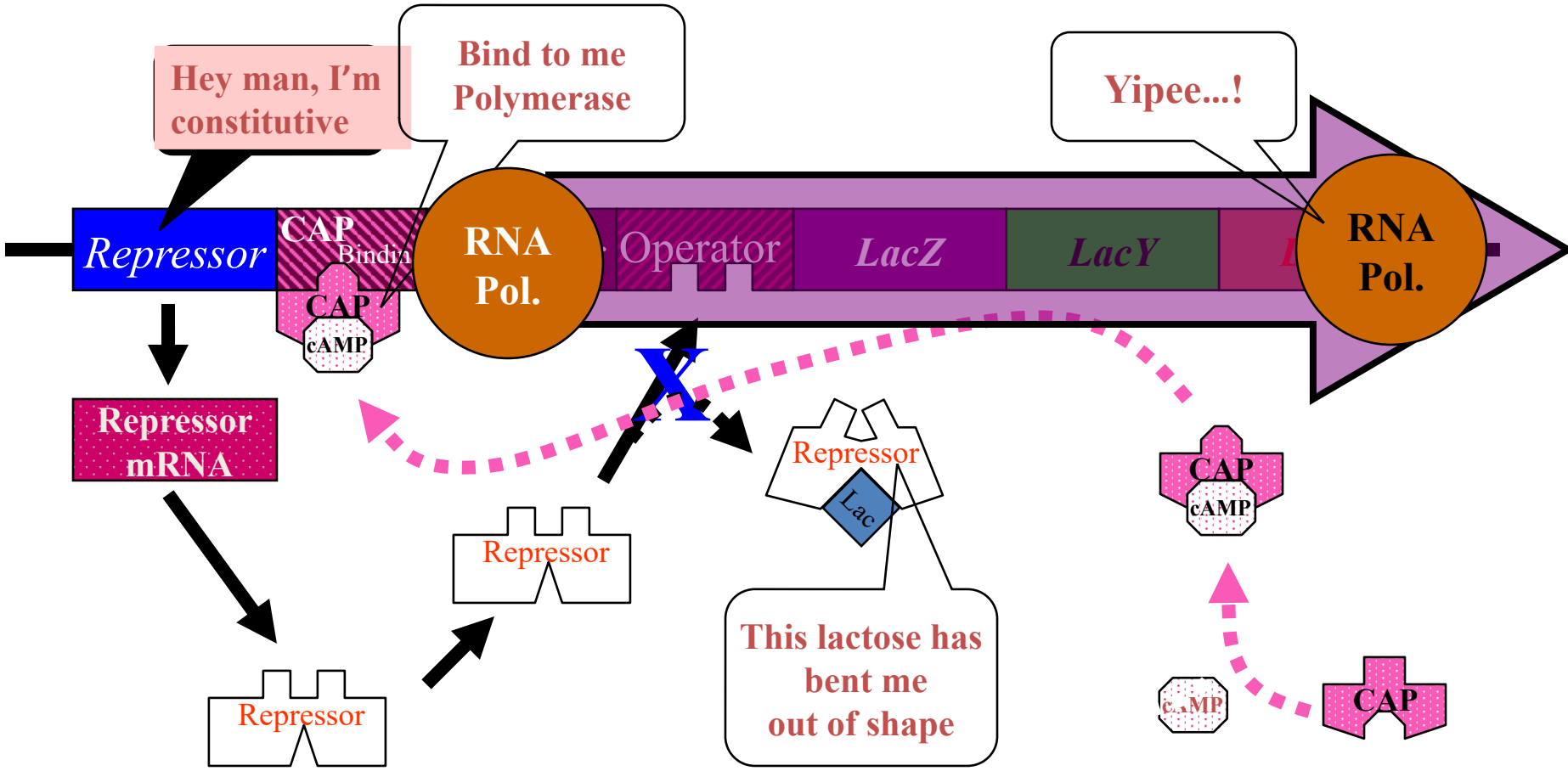
# CAP Plus cAMP Action

The open promoter complex does not form even if RNA polymerase has bound the DNA unless the CAP-cAMP complex is also bound



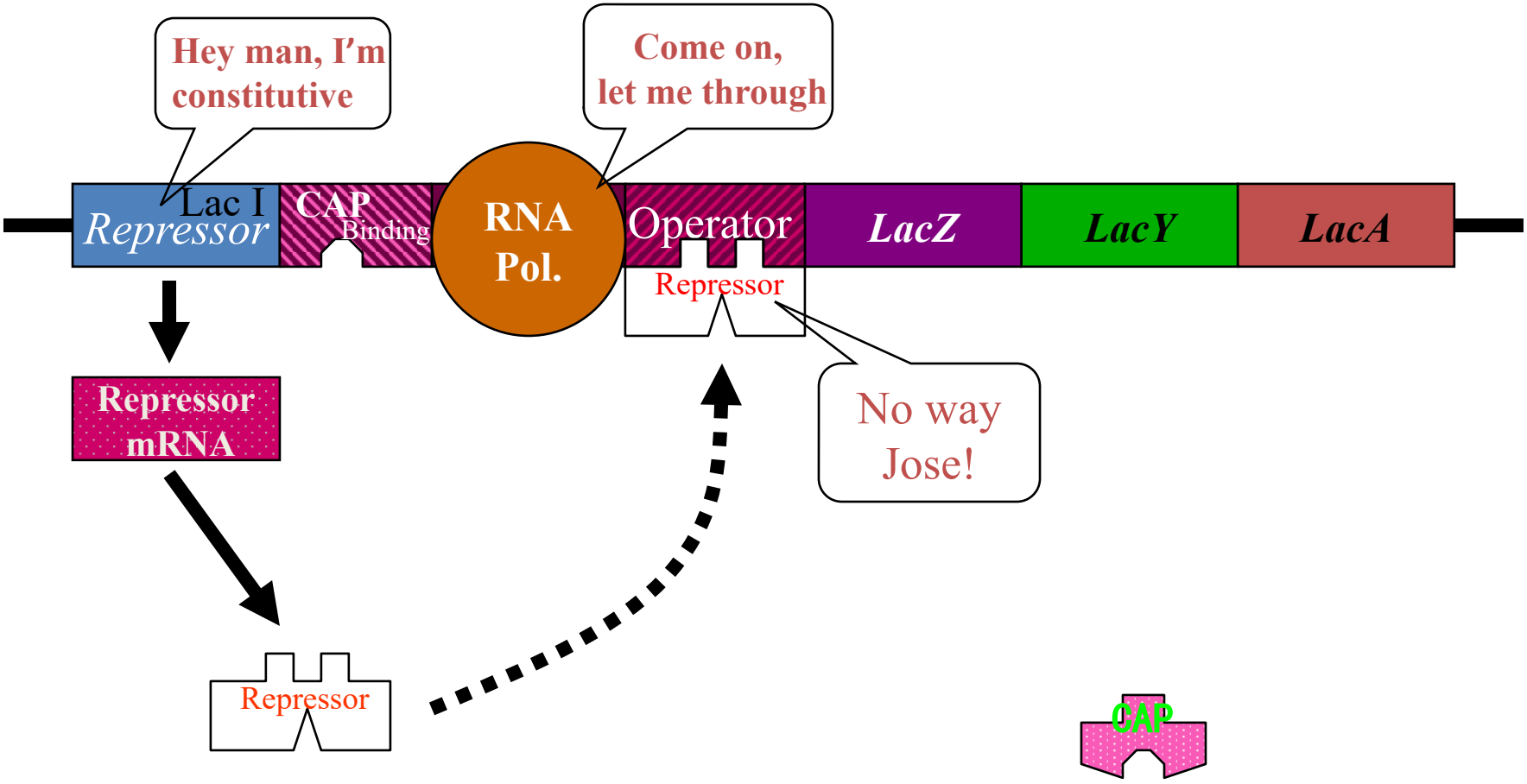
# The *Lac* Operon:

## When Lactose Is Present But Not Glucose



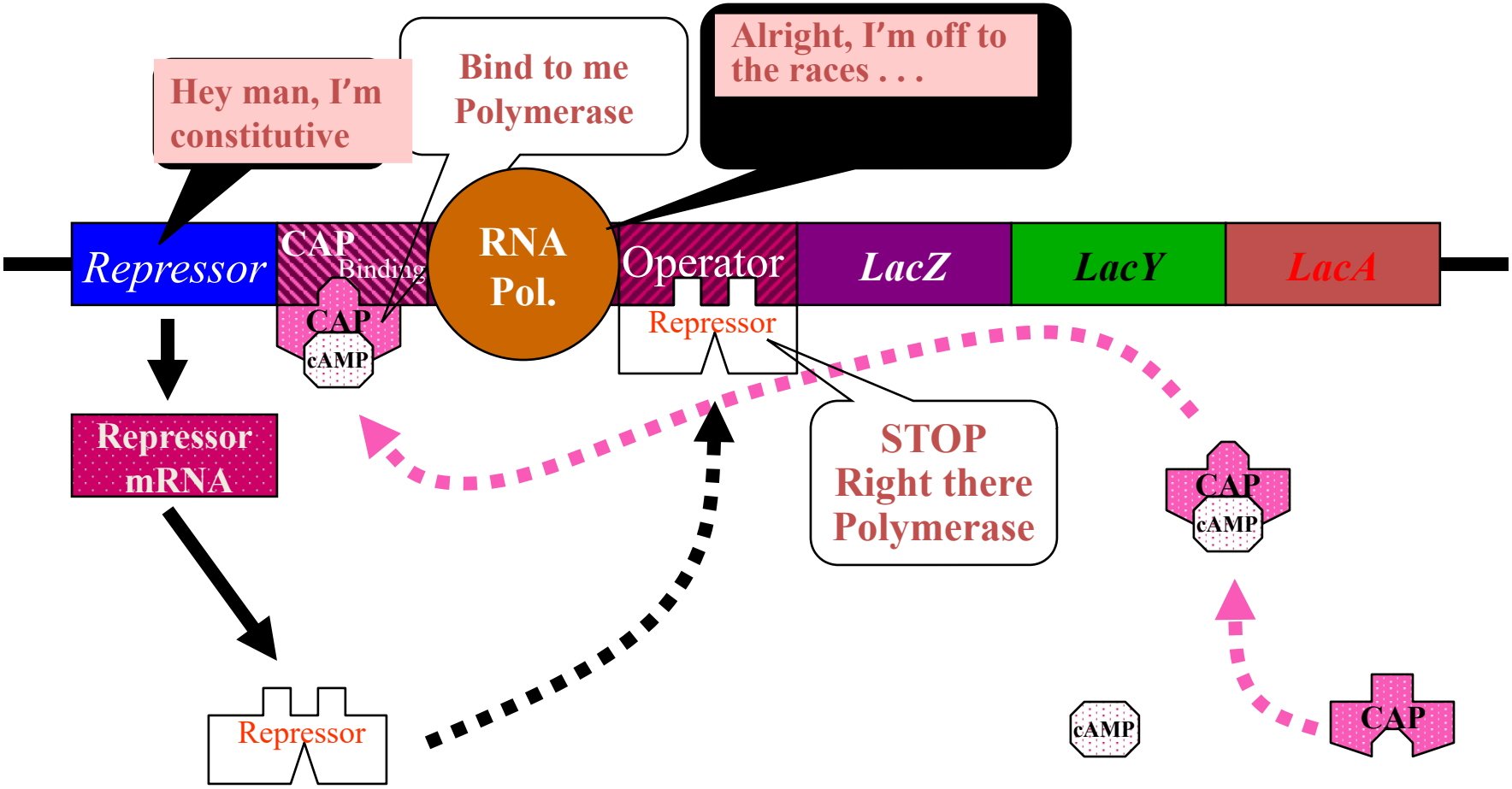
# The *Lac* Operon:

## When Glucose Is Present But Not Lactose

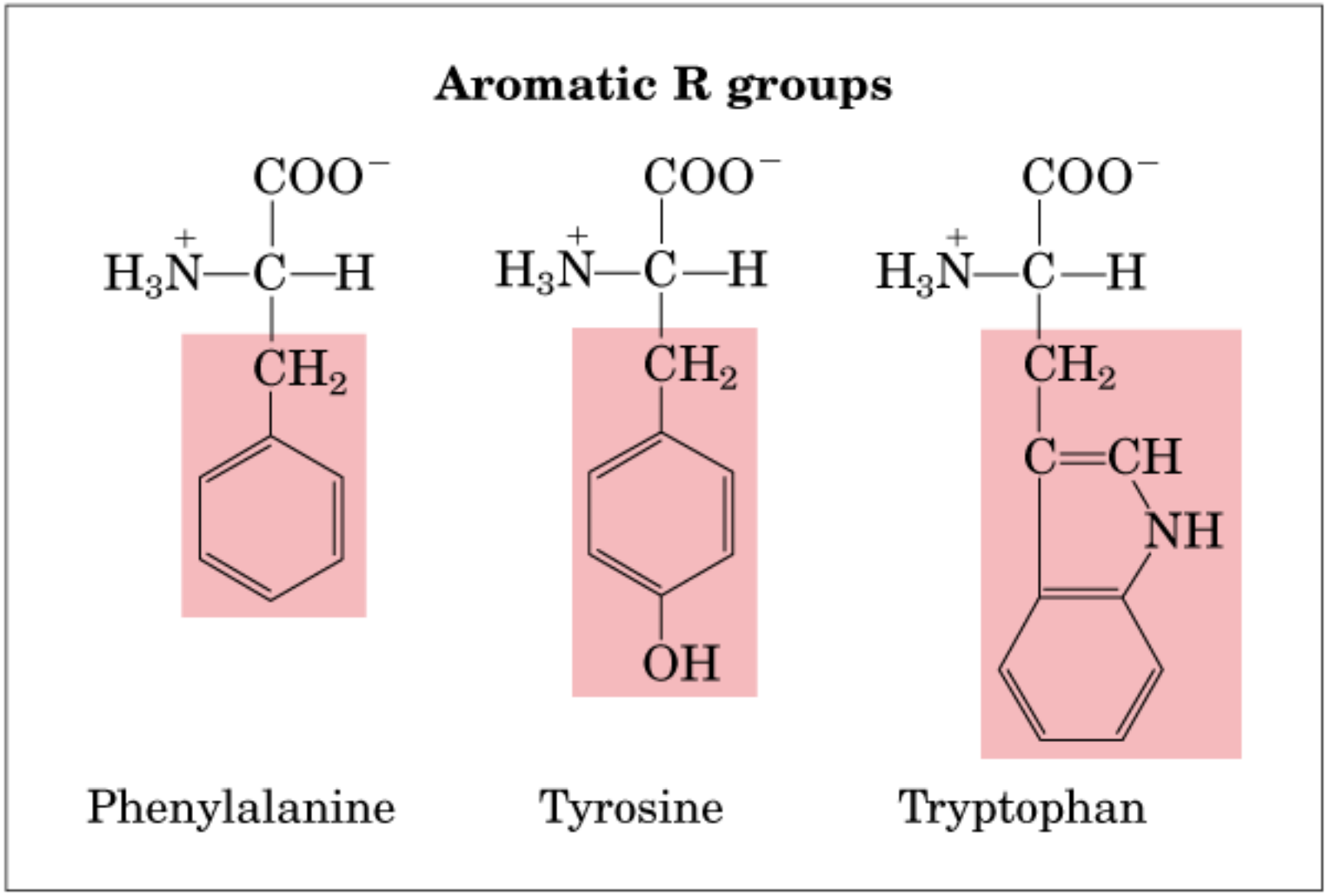


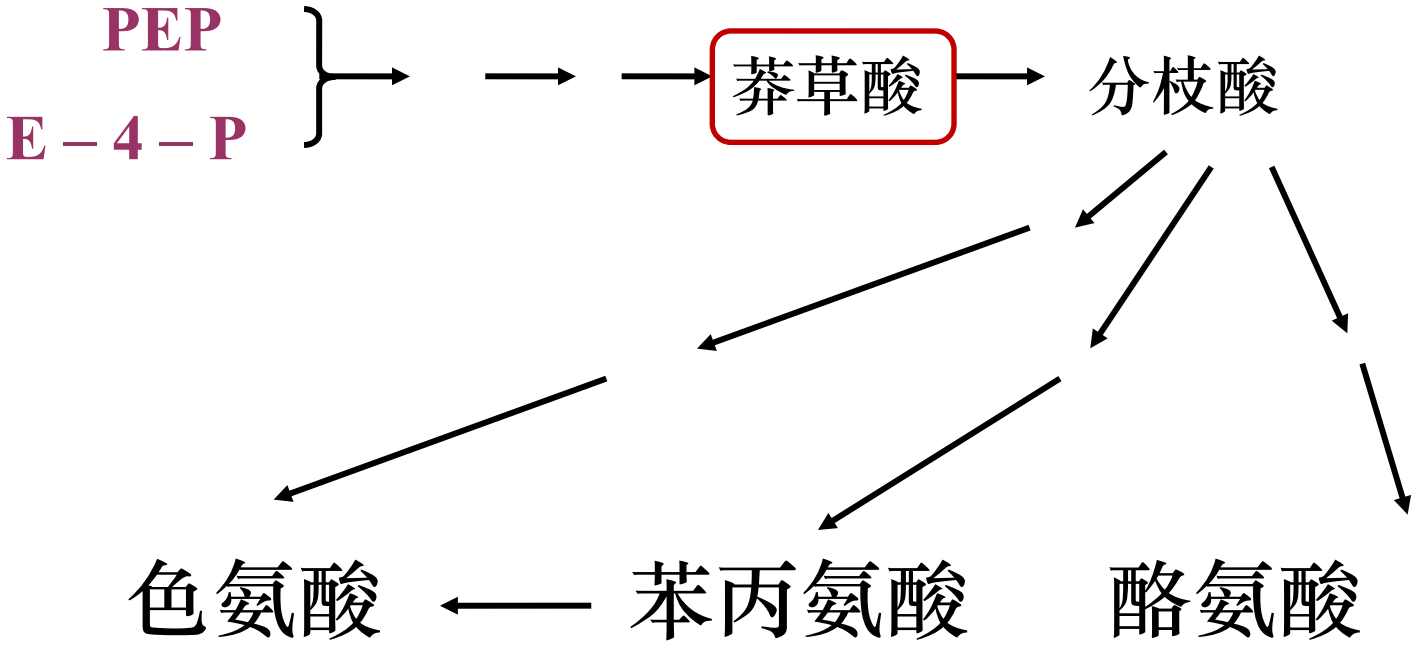
# The *Lac* Operon:

## When Neither Lactose Nor Glucose Is Present

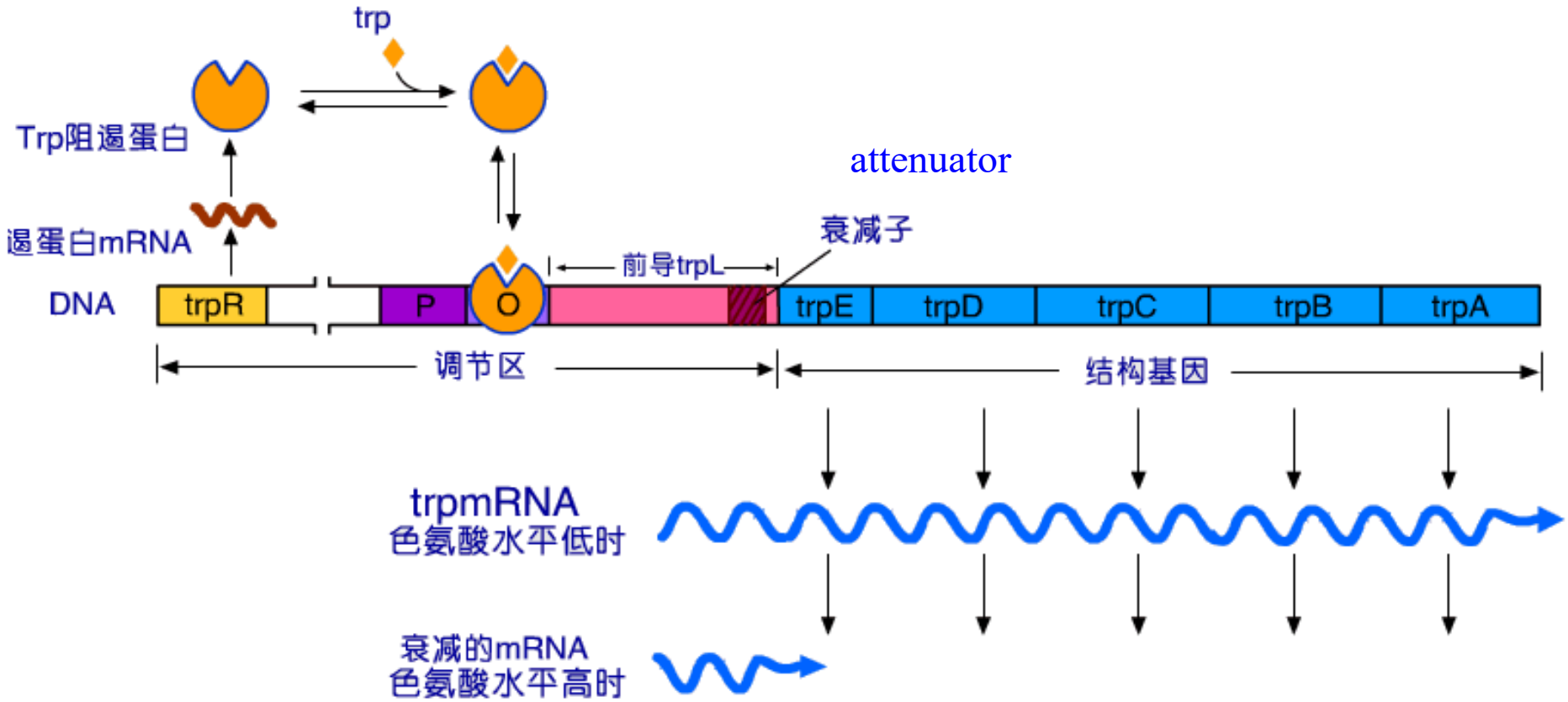


**Phe, F, 苯丙氨酸 ; Tyr, Y, 酪氨酸 ; Trp, W, 色氨酸**



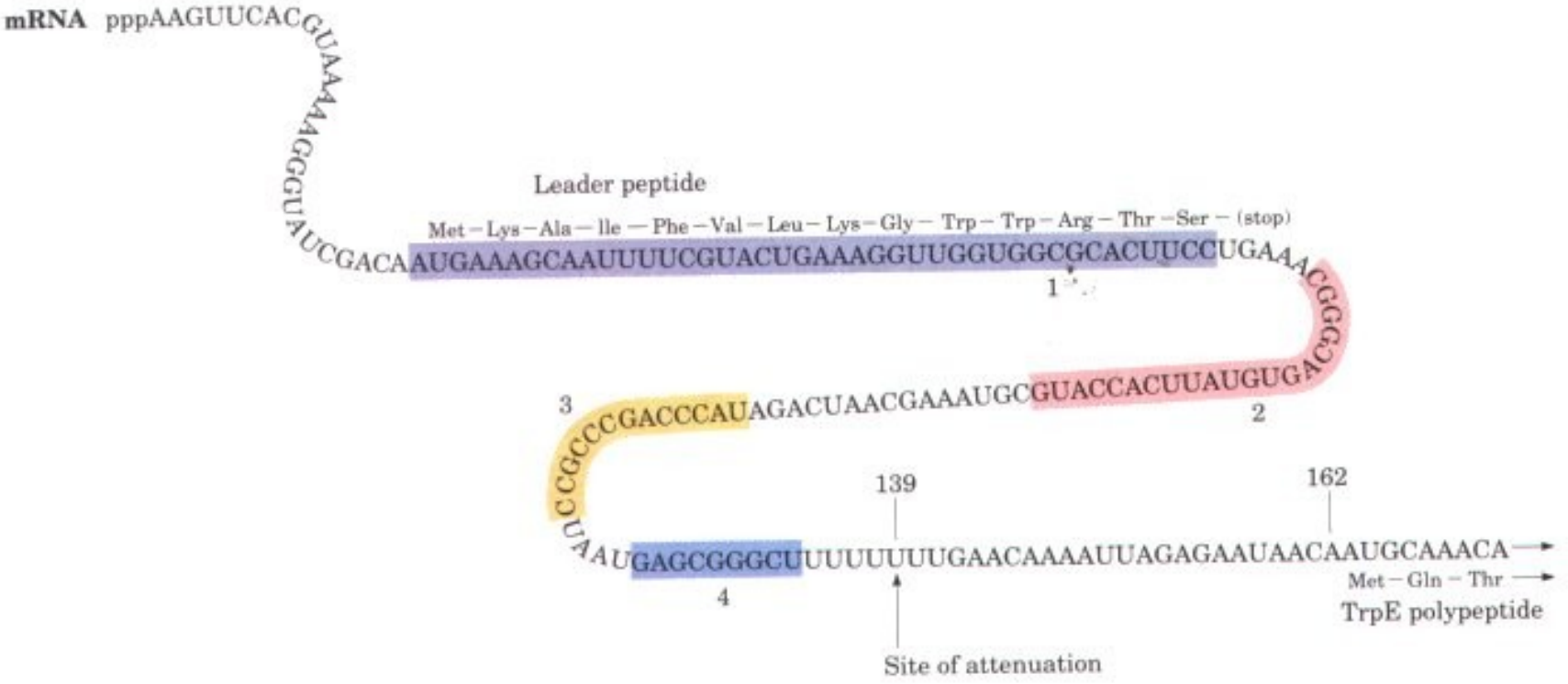

PEP
E – 4 – P
莽草酸
分枝酸
色氨酸
苯丙氨酸
酪氨酸

# 3. The tryptophan operon



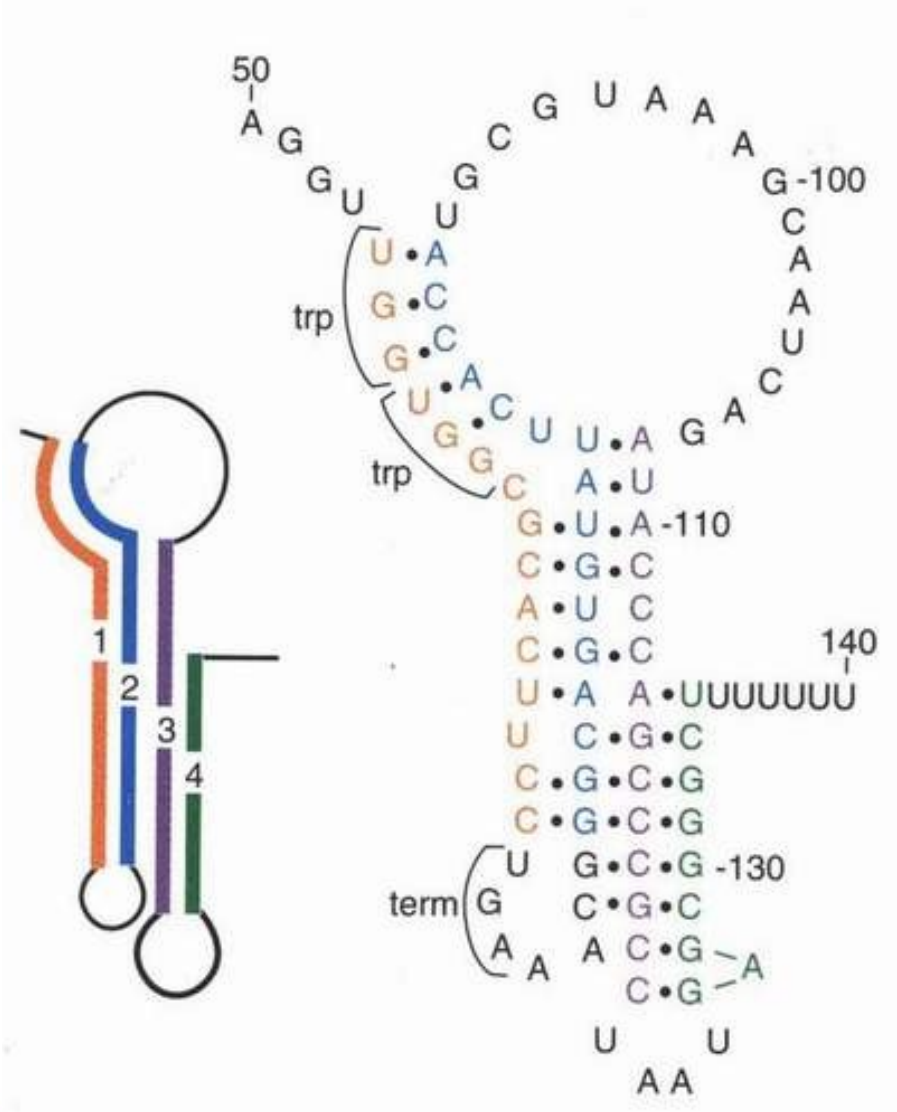

色氨酸操纵子的结构与阻遏蛋白的负性调控

# Leader peptide of trp operon




Attenuator ( 衰减子） (internal terminator), like a trancriptional terminator contains a palindromic GC-rich （回文结构） and a poly (U).

# Leader peptide of trp operon



# 转录、翻译偶联

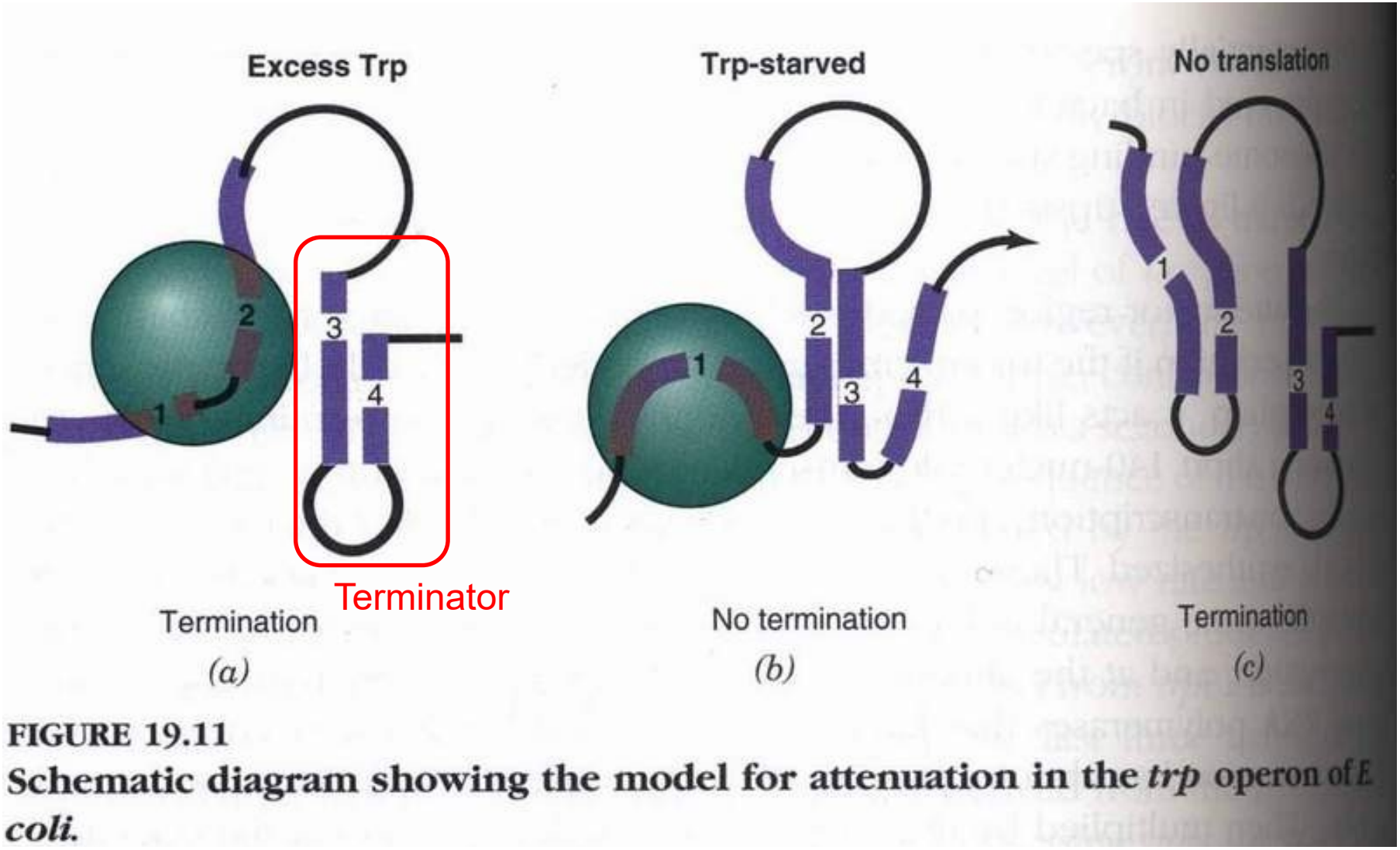
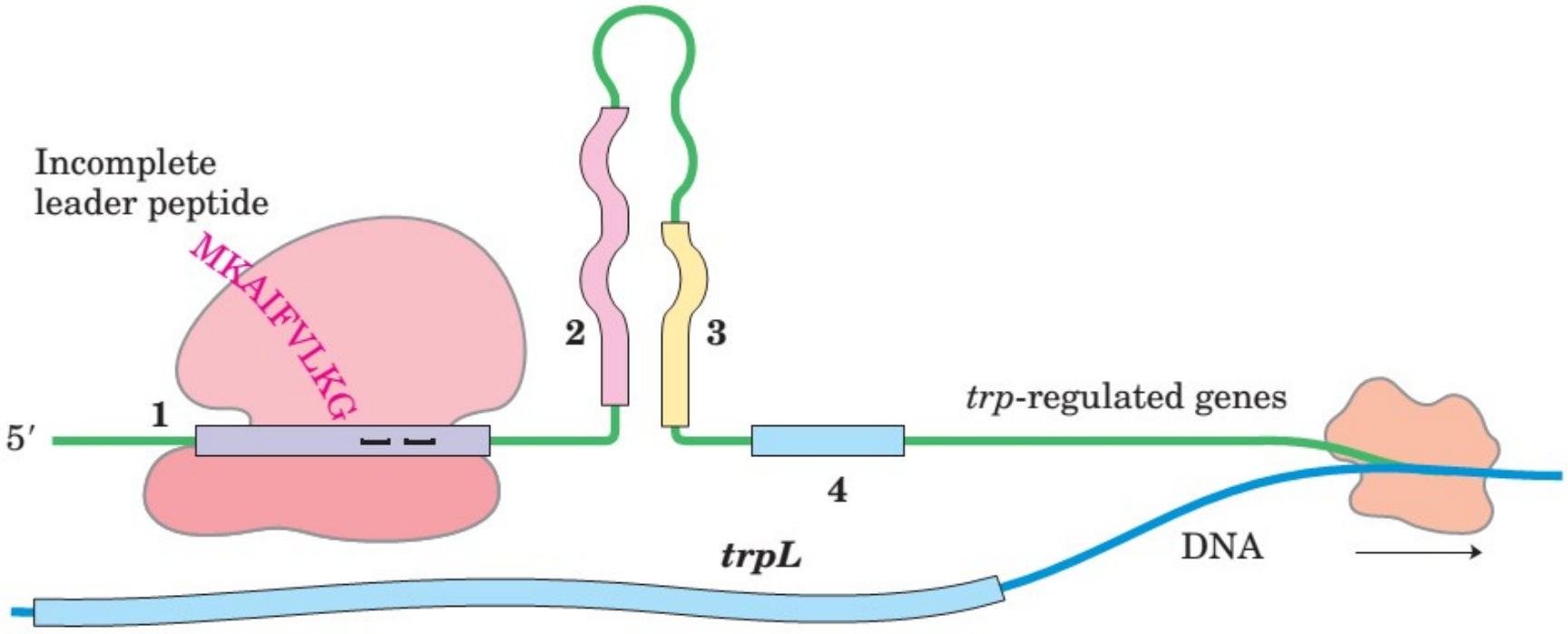
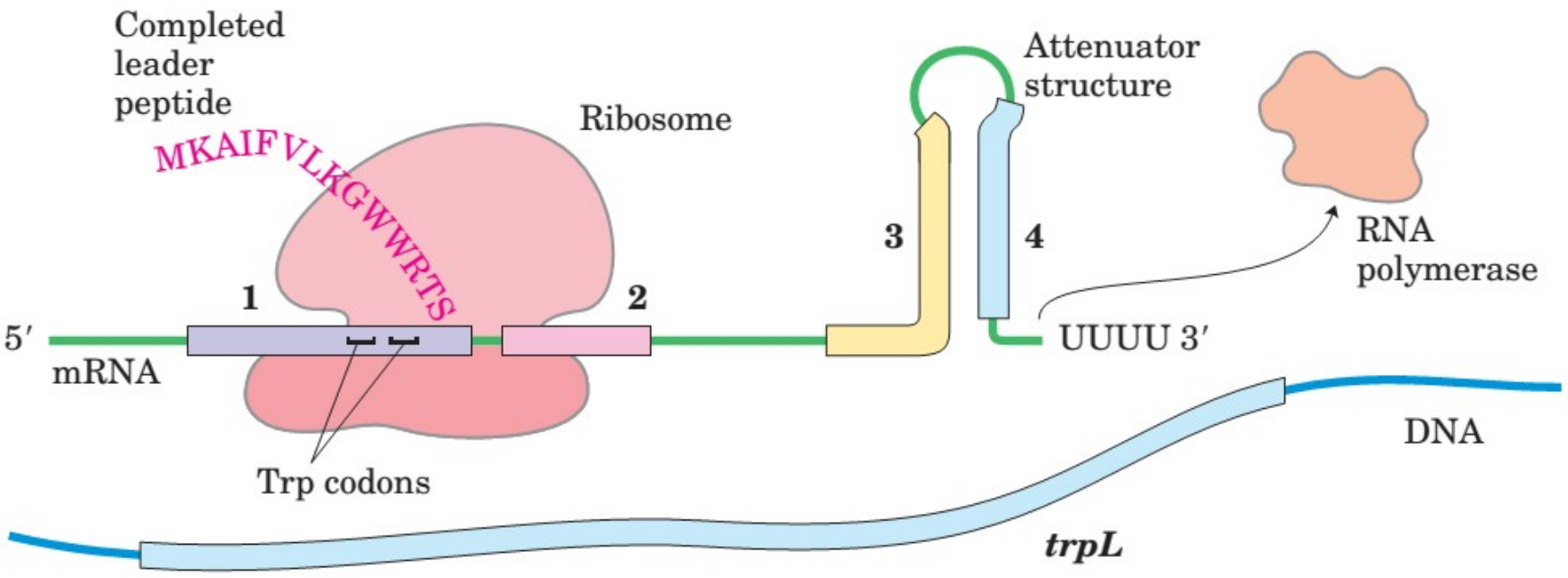



**FIGURE 19.11**
**Schematic diagram showing the model for attenuation in the *trp* operon of *E coli*.**

When Trp levels are low, the ribosome pauses at the trp codons in sequence 1. Formation of the paired structure between sequence 2 and 3 prevents attenuation, because sequence 3 is no longer available to form the attenuator structure with sequence 4.



When Trp levels are high, the ribosome quickly translates sequence 1 and blocks sequence 2 before sequence 3 is transcribed. Continued transcription leads to the terminator-like structure formed by sequence 3 and 4



# G5 Transcription in Eukaryotes: an Overview

# G6 Transcription of Protein-Coding Genes in Eukaryotes





# 1. Three kinds of RNA polymerases (I, II, and III) have been revealed for making RNAs in the nucleus of eukaryotic cells

Each is responsible for the transcription of a certain groups of genes: Ribosomal RNA （rRNA）, Messenger RNA （mRNA） or Transfer RNA （tRNA） genes.

The enzymes are often identified by examining their sensitivity towards α-amanitin （鹅膏蕈碱） (from a toxic mushroom).







*Amanita Phalloides*



*Amanita bisporigera*

真核生物 RNA 聚合酶Ⅰ，II，III 对 α- 鹅膏蕈素的敏感程度不同。

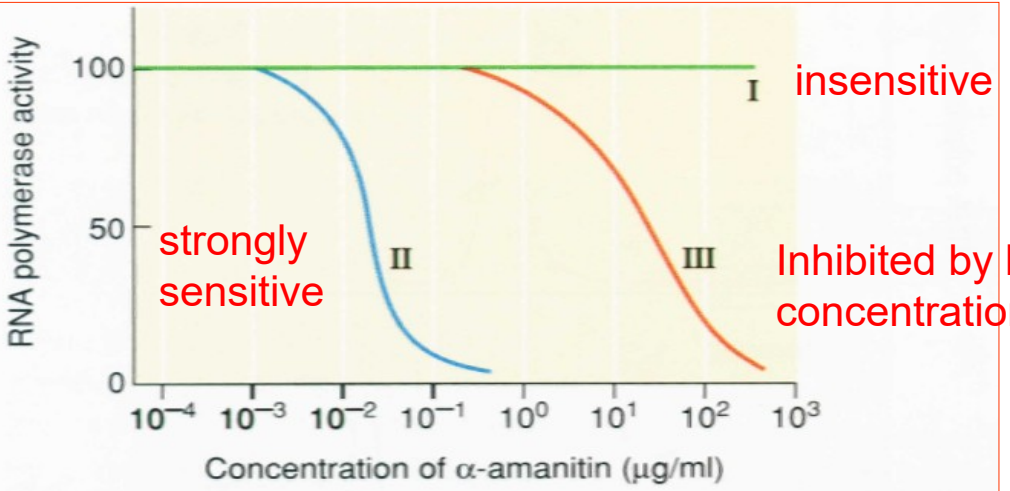



**Figure 10-26** ***Effects of α-Amanitin on Eukaryotic RNA Polymerases I, II, and III*** *RNA polymerase II is very sensitive to inhibition by α-amanitin, RNA polymerase III is moderately sensitive, and RNA polymerase I is not inhibited at all.*

α- 鹅膏蕈素对真核生物 RNA 聚合酶Ⅰ，II，III 的影响

## Eukaryotic RNA Polymerases

| Type | Localization | Cellular transcripts |
|---|---|---|
| I | Nucleolus | 18S, 5.8S, and 28S ribosomal RNAs (rRNA) |
| II | Nucleoplasm | mRNA precursors and small nuclear RNAs (snRNA) |
| III | Nucleoplasm | tRNAs and 5S rRNAs |





**Each of the three RNA polymerases contains 12 or more subunits, some of which are similar to those of *E. coli* RNA polymerase. However, four to seven subunits in each enzyme are unique to that enzyme.**



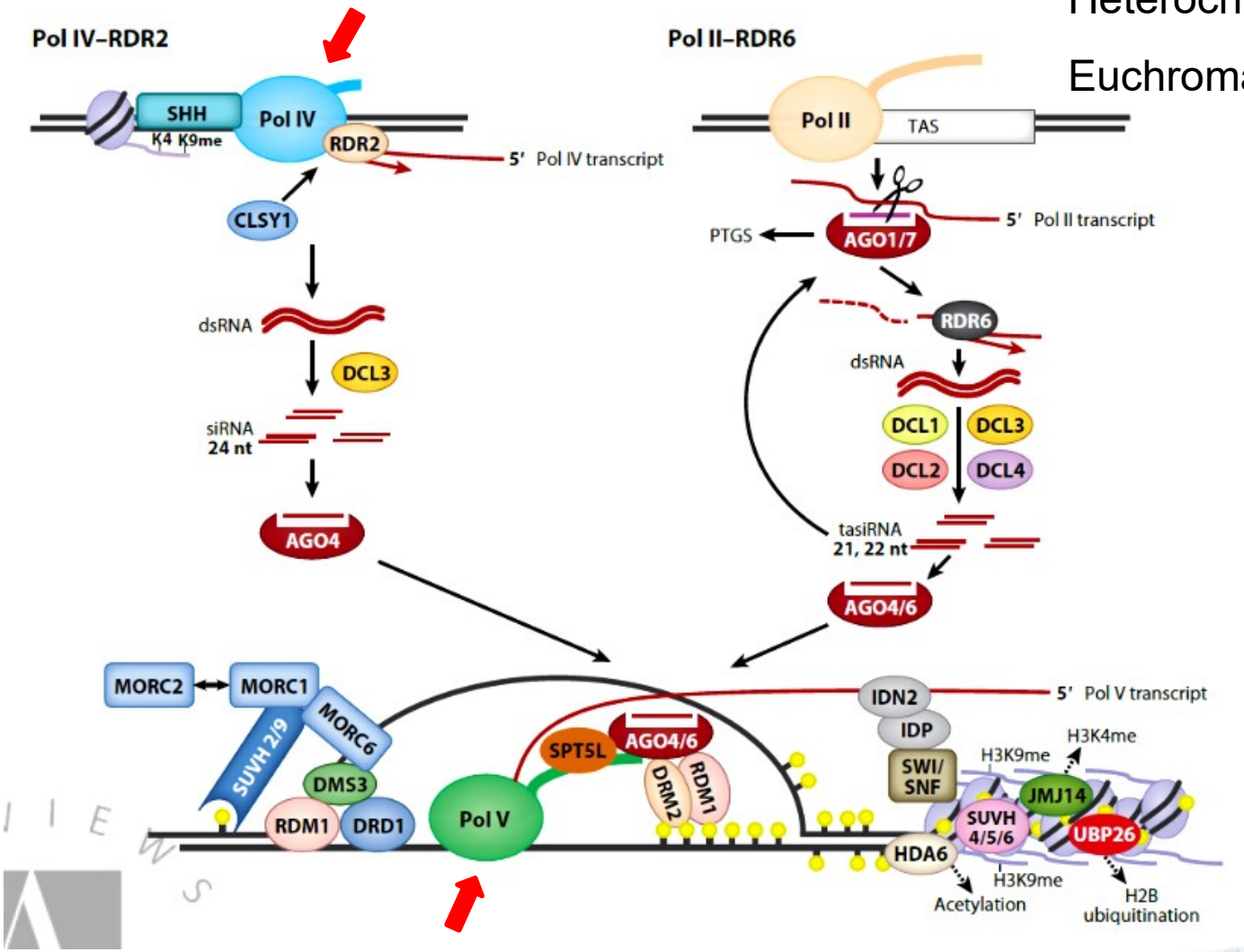


# Pol IV and pol V in plants

DNA methylation

Heterochromatin

Euchromatin




Marjori A. et.al 2015, Annu. Rev. Plant Biol.
ΔOI: 10.1146/αννυρεϖ–αρπλαντ–043014–114633

Biochemistry

## Transposon 转座子





麦克林托克
Barbara Mclintock

https://www.163.com/dy/article/HBOAVDSM0512TP34.html

Biochemistry

# 2. RNA polymerase II (Pol II) binds to promoters of thousands of protein-coding genes

**Many Pol II promoters contain a TATAAA sequence (called a TATA box) at -25 (-30) position and an initiator sequence (Inr) at +1 position.**

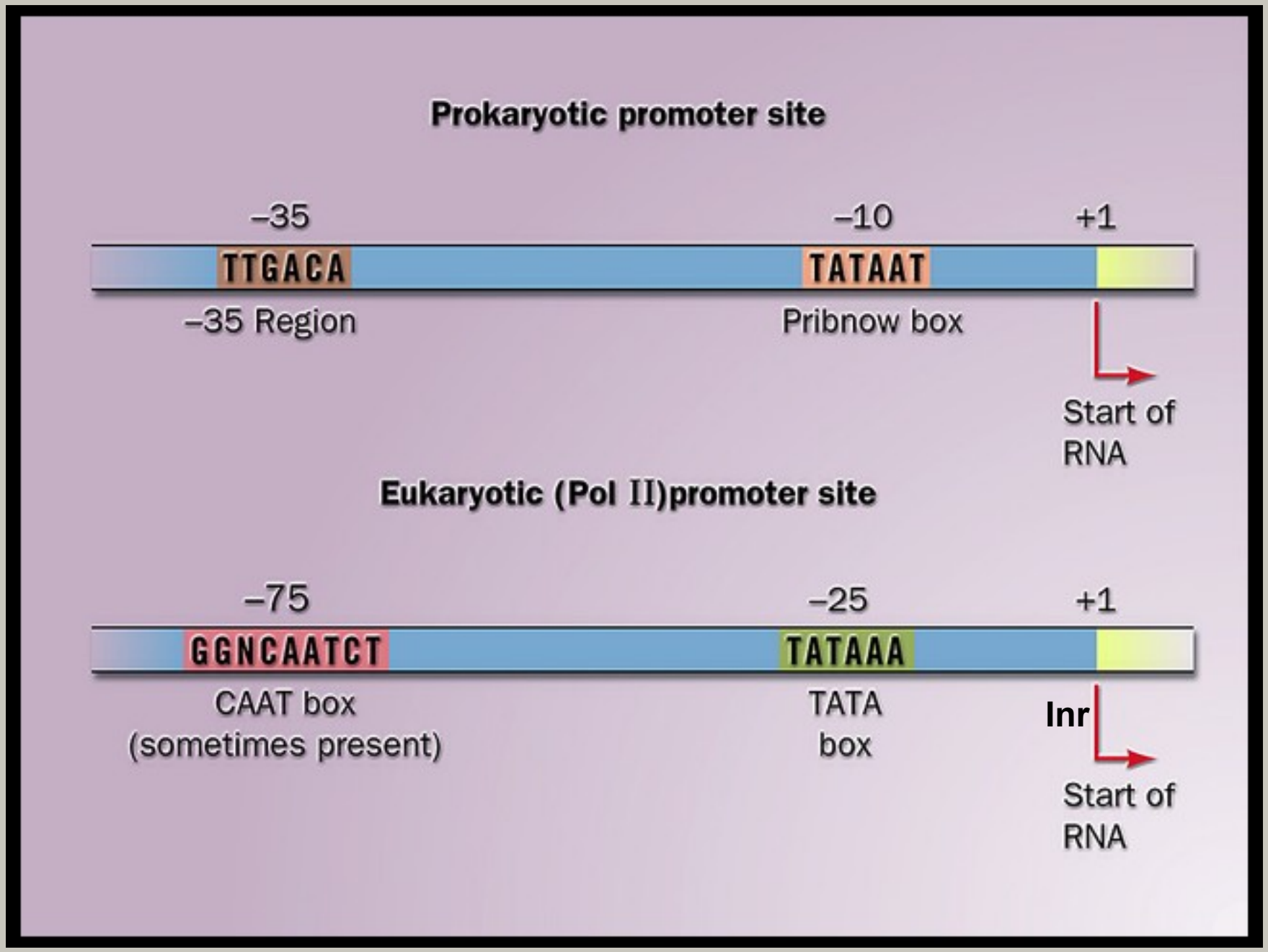
**The preinitiation complex (including Pol II) is believed to assemble at the TATA box, with DNA unwound at the**

**Inr sequence.**

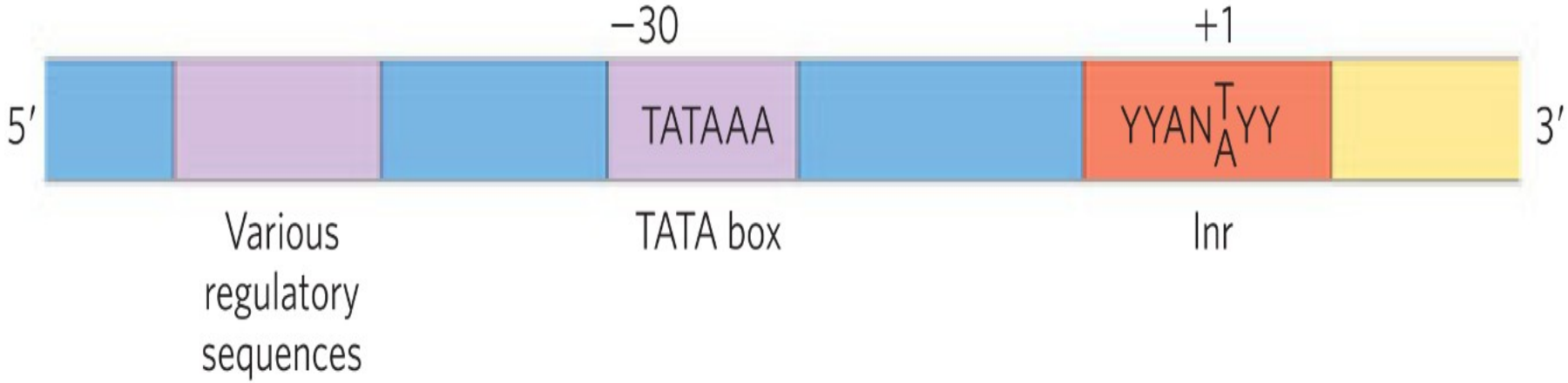
**However, many Pol II promoters lack a TATA or Inr or both sequences!**



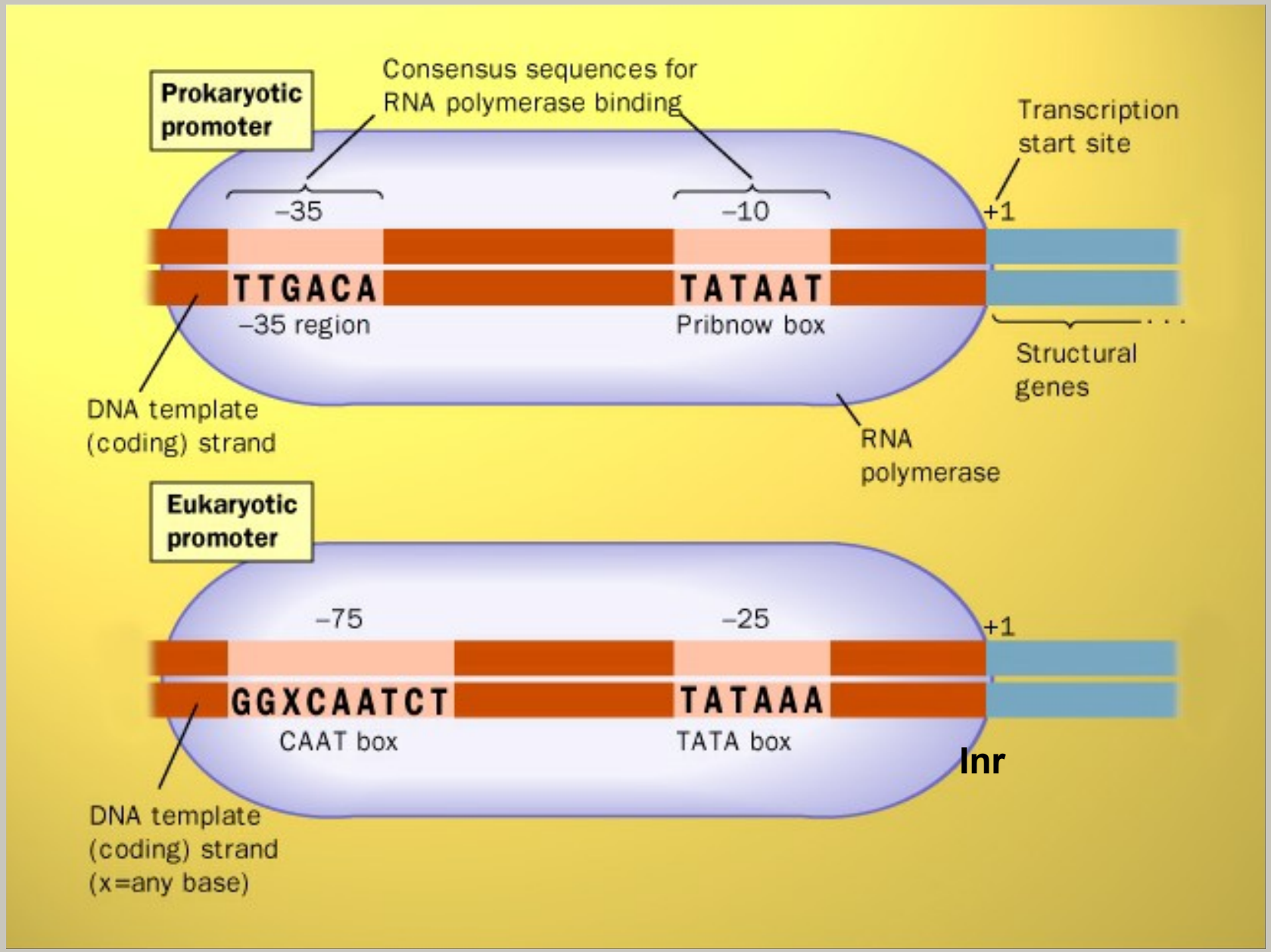



Prokaryotic promoter site
–35
–10
+1
TTGACA
TATAAT
–35 Region
Pribnow box
Start of RNA
Eukaryotic (Pol II)promoter site
–75
–25
+1
GGNCAATCT
TATAAA
CAAT box (sometimes present)
TATA box
Inr
Start of RNA

# General features of promoters for protein-coding genes in higher eukaryotes




| 简并碱基符号对应表 | | |
|---|---|---|
| 简并碱基 | | 正常碱基 |
| R | | A/G |
| Y | | C/T |
| N | | A/G/C/T |


Prokaryotic promoter
Consensus sequences for RNA polymerase binding
Transcription start site
–35
–10
+1
TTGACA
TATAAT
–35 region
Pribnow box
Structural genes
DNA template (coding) strand
RNA polymerase
Eukaryotic promoter
–75
–25
+1
GGXCAATCT
TATAAA
CAAT box
TATA box
Inr
DNA template (coding) strand (x=any base)

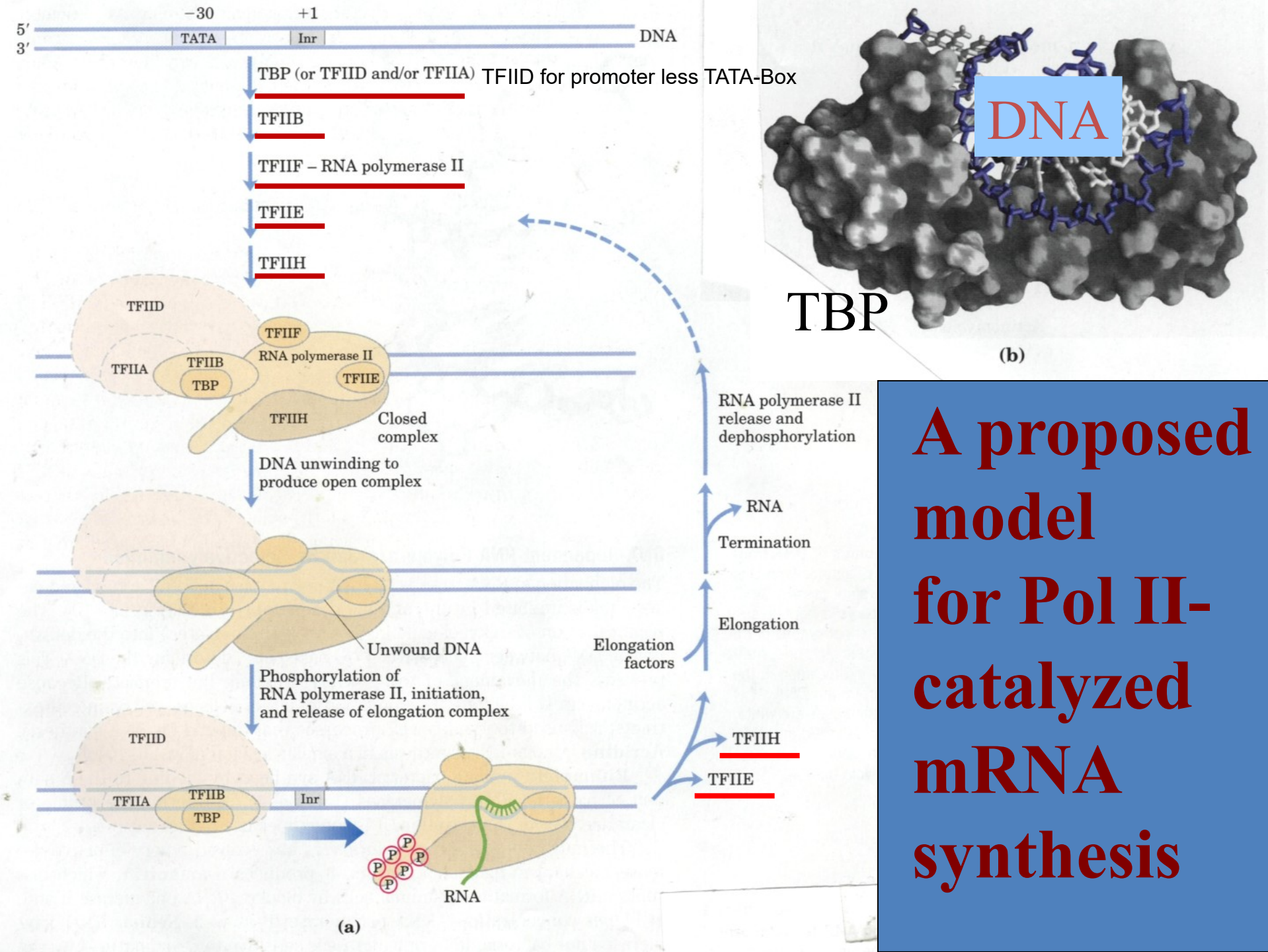
# 3. Pol II is helped by an arrays of protein factors (called general transcription factors(TF)) to form an active transcription complex at a promoter

**Firstly, the TATA-binding protein (TBP) binds to the TATA box, then TFIIB, TFIIF-Pol II, TFIIE, and TFIIH (TFII, Transcription Factor for RNA polymerase II) will be added in order forming the closed complex at the promoter.**

**TFIIH then acts as a helicase to unwind the DNA duplex at the Inr site, forming the open complex.**








# A proposed model for Pol II-catalyzed mRNA synthesis

**A kinase activity of TFIIH will phosphoryate the C-terminal domain of Pol II, which will initiate RNA synthesis and release the elongation complex. TFIIE and TFIIH will be released after the elongation complex moves forward for a short distance.**





- Transcriptional regulation in a eukaryotic cell is mediated by transcription factors (other than general transcription factors ) which recognize and bind to short regulatory DNA sequences (control elements) associated with the gene. These sequences are also called cis-acting elements (cis-elements) ( 顺式作用元件） . And the transcription factors are also known as trans-acting factors ( 反式作用因子） .



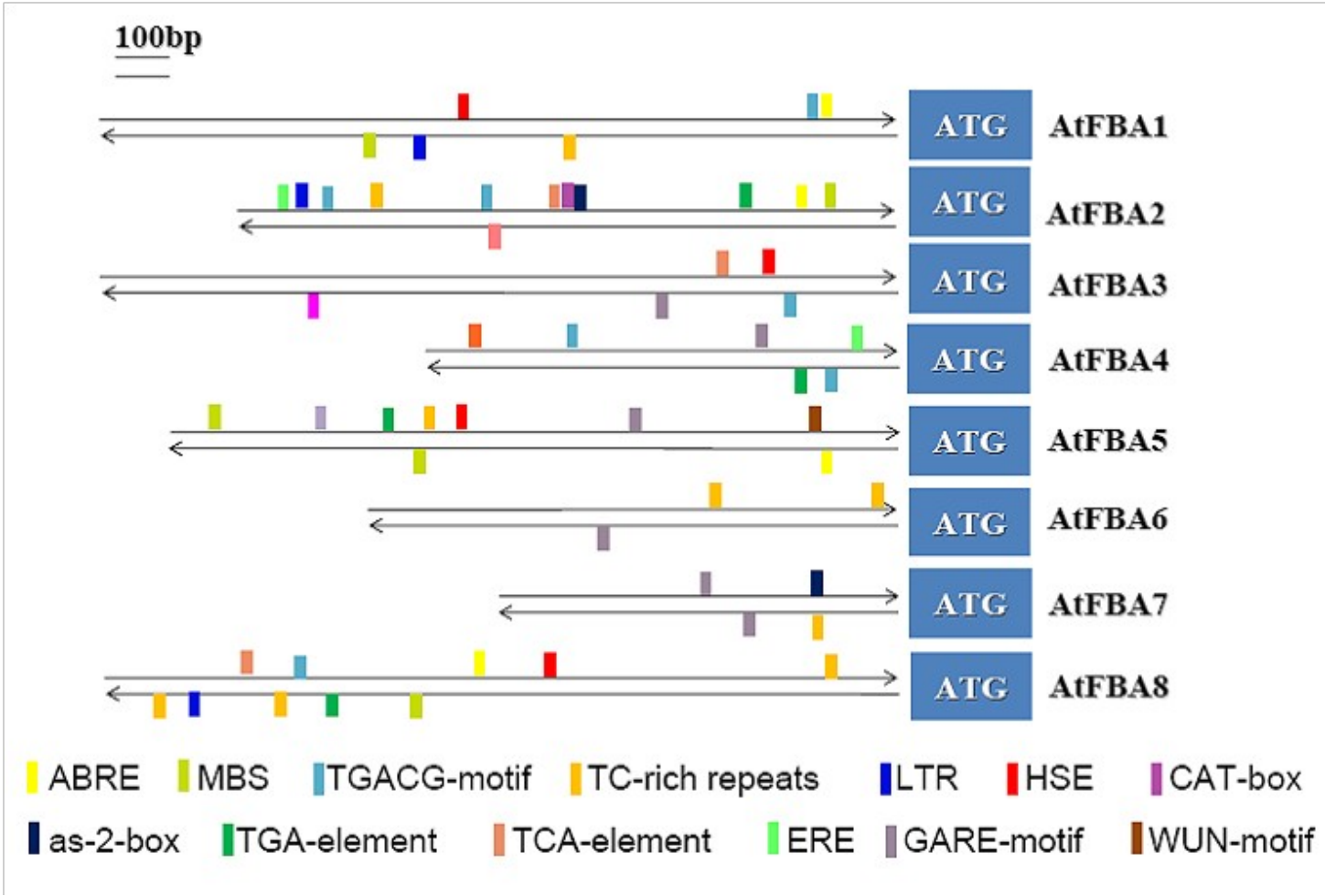


- The transcription factors regulate specific gene transcription by interacting with the proteins of the transcription initiation complex and either **increase or decrease** the rate of transcription of the target gene.





# Many regulatory elements in a promoter




顺式作用元件 (cis-action elements)

# Each cis-acting element has a specific sequence composition

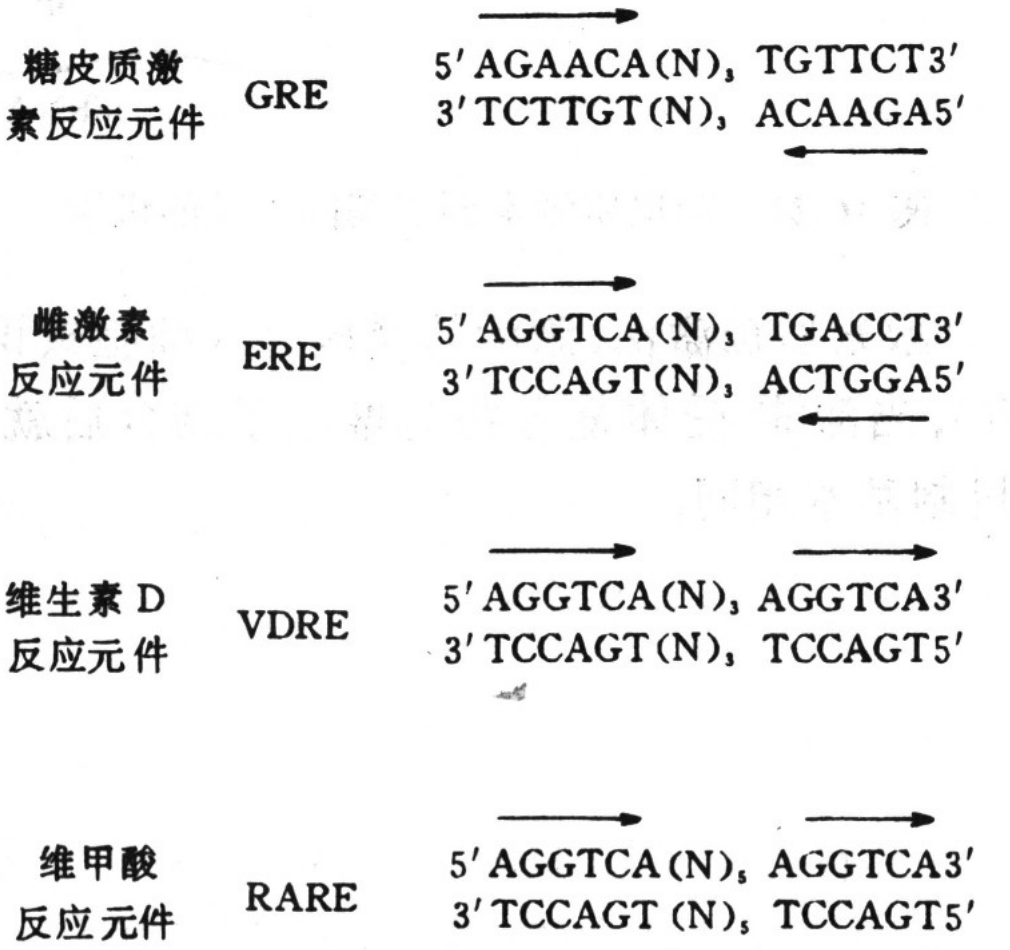
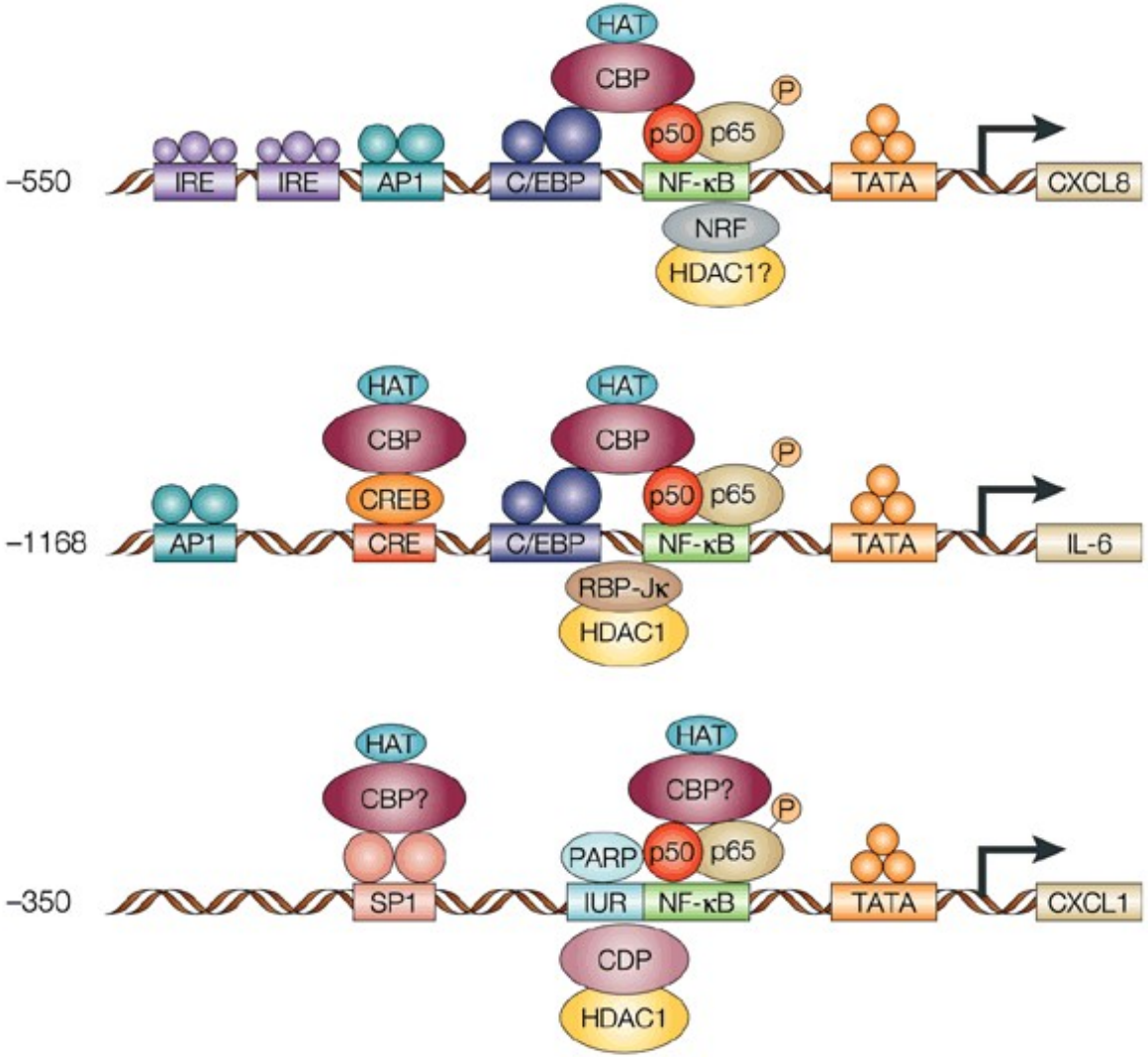


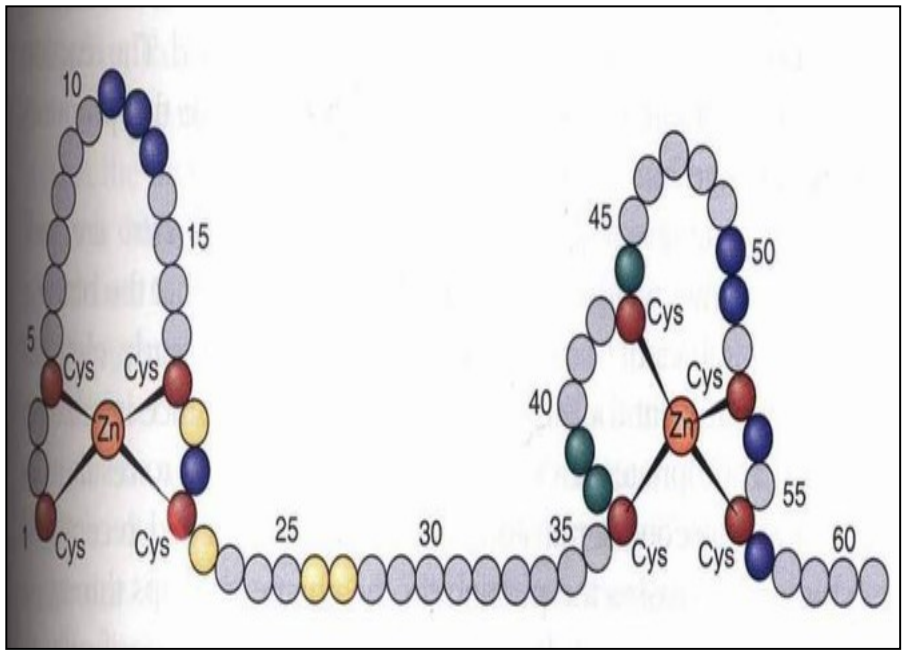

**图 16-20** 几种脂溶性细胞外分子的 DNA 反应元件

# TFs contain a DNA-binding domain (BD) and an activation domain (AD)

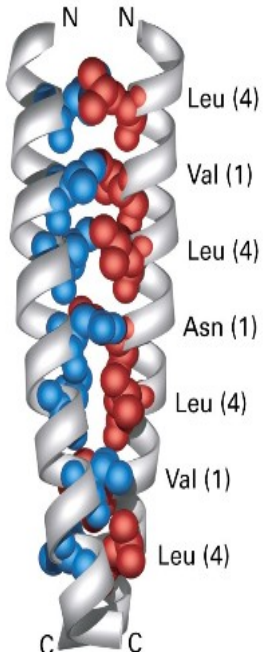


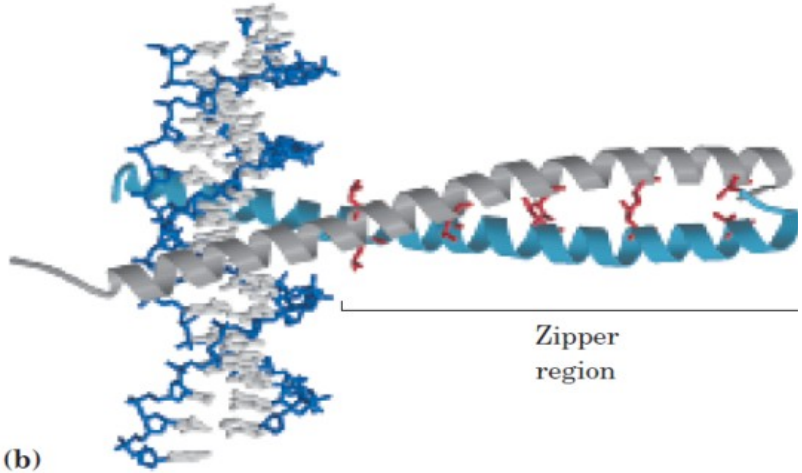
# DNA-binding domains

- The BD of most of the TFs can forms a specific structure.
- zinc finger ( 锌指）



- Basic leucine zipper

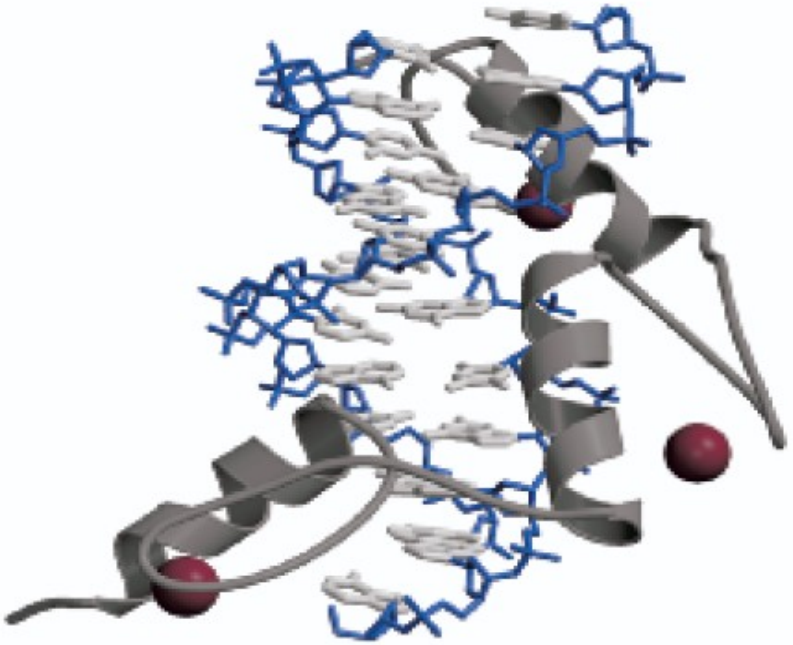



- 一段肽链中每隔 7 个 AA 即有一个 Λευ， α 螺旋
  亲水面：亲水 AA 组成
  疏水面： Λευ 组成
  （亮氨酸拉链）
- 可形成二聚体发挥作用
  （同二聚体 / 异二聚体）
- ΔNA 的结合域：
  拉链区的氨基端约 30AA 的碱性区（富含 Λψσ 和 Αργ ）



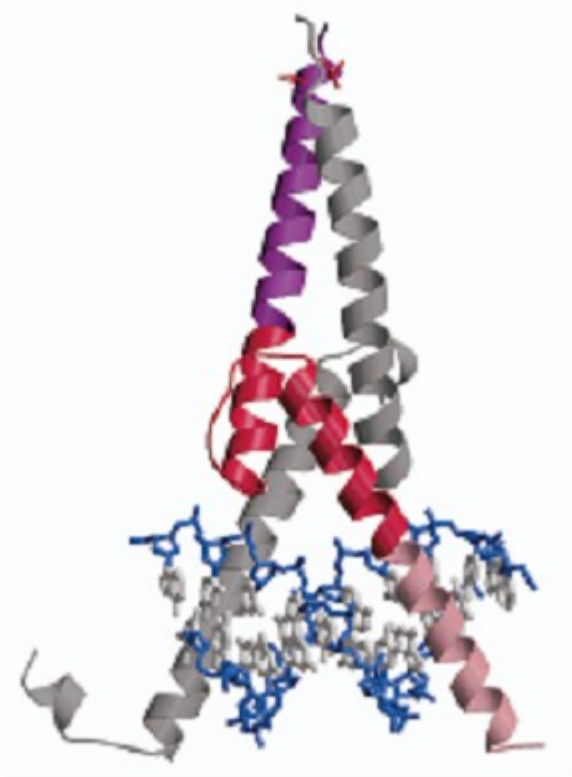
# Helix-turn-helix



# Homeodomain



# Helix~loop~helix

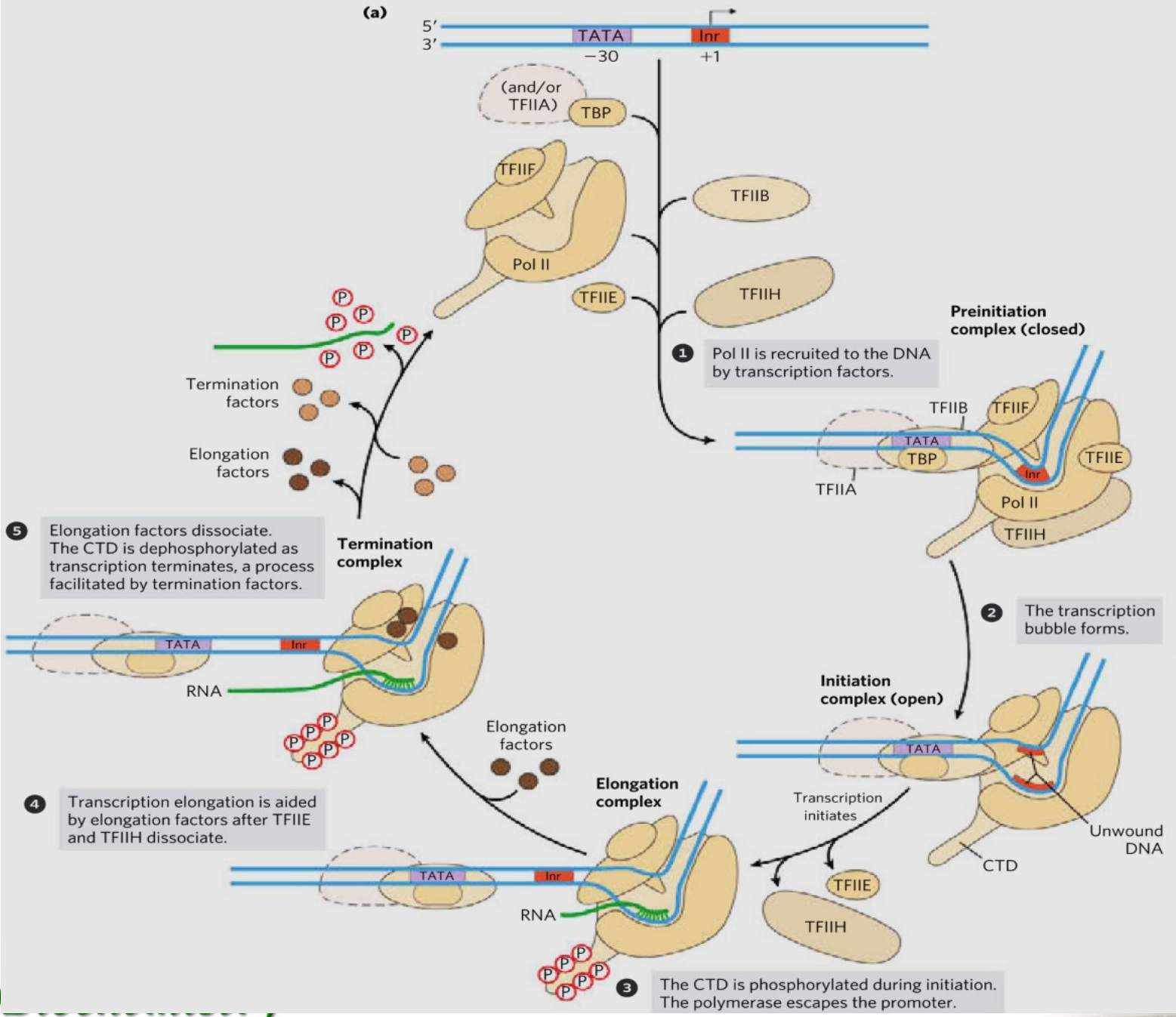


**Transcription in Eukaryotes : imitation, elongation, termination**

**Elongation factors will then join the elongation complex.**
**Unlike RNA polymerase in prokaryotes, RNA polymerase II does not terminate transcription at specific sites but rather transcription stops at varying distances downstream of the gene with help from termination factors.**






(a)
5′
3′
TATA
Inr
−30
+1
(and/or TFIIA)
TBP
TFIIF
TFIIB
Pol II
TFIIE
TFIIH
Preinitiation complex (closed)
1 Pol II is recruited to the DNA by transcription factors.
TFIIB
TFIIF
TATA
TBP
TFIIE
Inr
TFIIA
Pol II
TFIIH
2 The transcription bubble forms.
Initiation complex (open)
TATA
Transcription initiates
Unwound DNA
CTD
TFIIE
TFIIH
3 The CTD is phosphorylated during initiation. The polymerase escapes the promoter.
Elongation complex
TATA
Inr
RNA
Elongation factors
4 Transcription elongation is aided by elongation factors after TFIIE and TFIIH dissociate.
Termination complex
TATA
Inr
RNA
5 Elongation factors dissociate. The CTD is dephosphorylated as transcription terminates, a process facilitated by termination factors.
Termination factors
Elongation factors

- In prokaryotes, RNA transcribed from protein-coding genes (messenger RNA, mRNA), requires little or no modification prior to translation
- However, mRNA in eukaryotes, ribosomal RNA (rRNA) and transfer RNA (tRNA) are synthesized as precursor molelules that do require post-transcriptional processing) (G7, G8, G9)





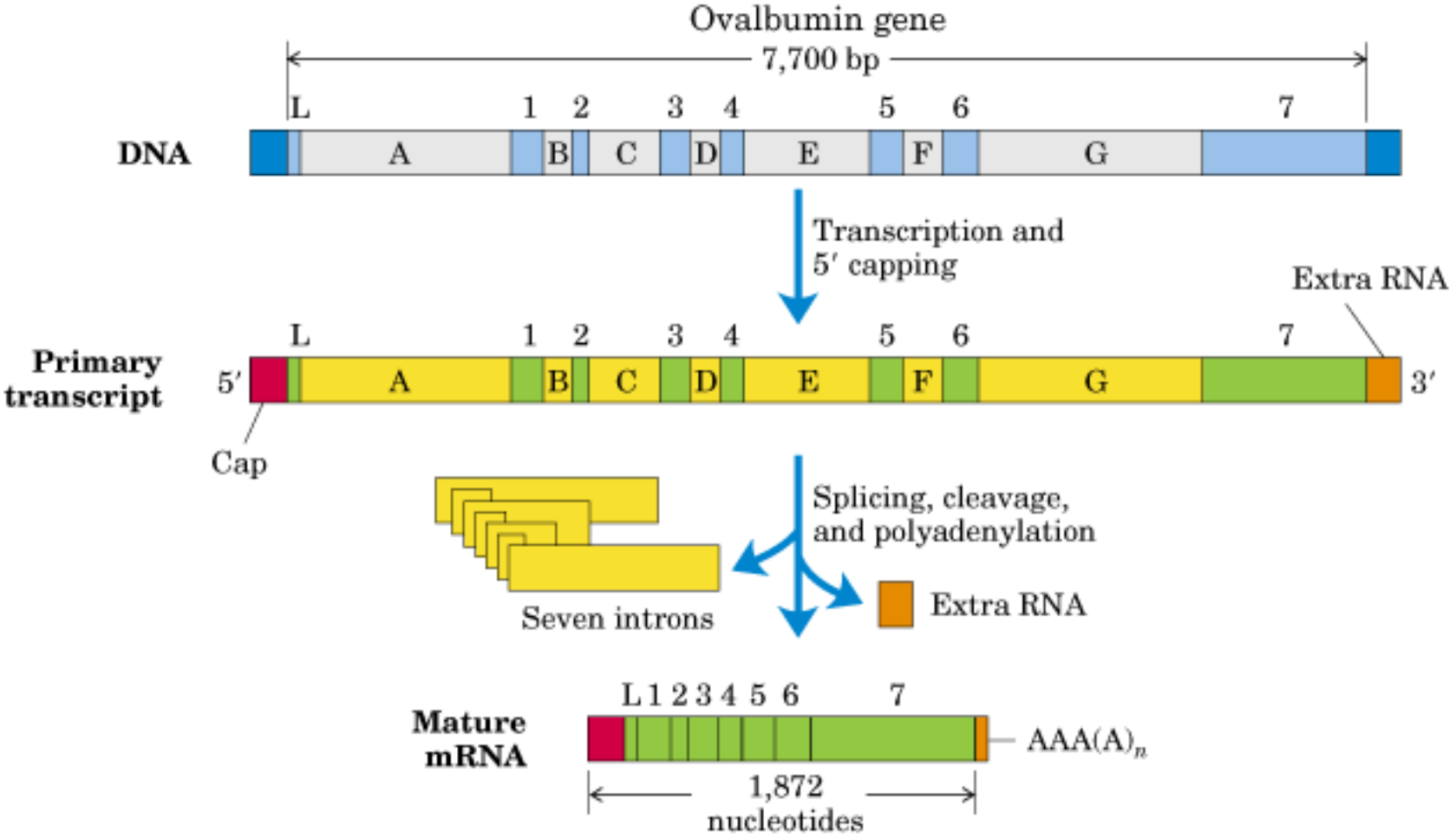
# G7 Processing of eukaryotic pre-mRNA

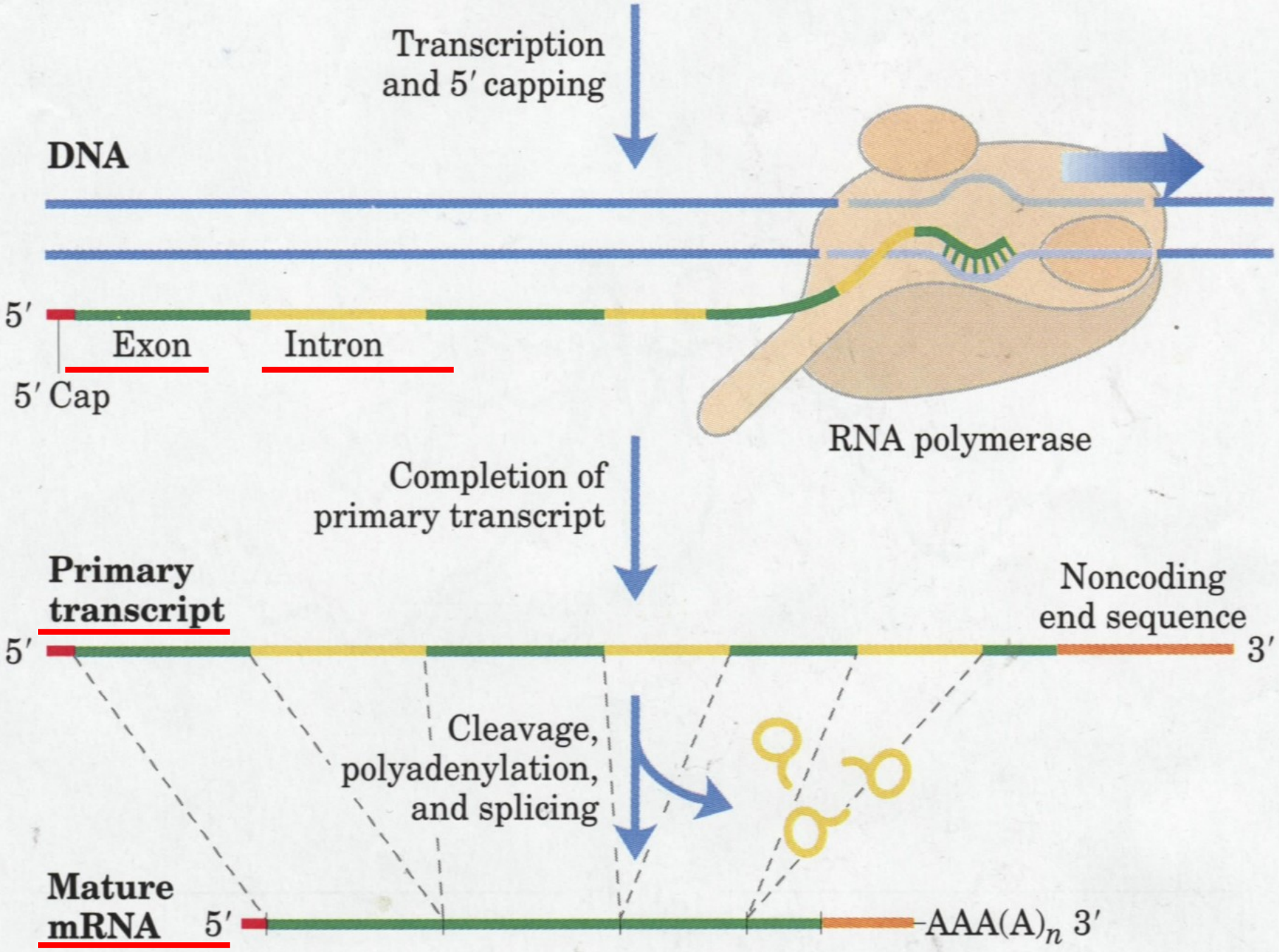
The primary transcripts of eukaryotic mRNAs are often capped at the 5` end, spliced （剪接） in the middle (introns removed and exons linked), polyadenylated at the 3` end.





# Most protein-coding genes in eukaryotes are discontinuous




Transcription
and 5′ capping
DNA
5′
Exon
Intron
5′ Cap
RNA polymerase
Completion of
primary transcript
Primary
transcript
Noncoding
end sequence
5′
3′
Cleavage,
polyadenylation,
and splicing
Mature
mRNA
5′
AAA(A)n 3′

The 5’ cap is added before synthesis of the primary transcript is completed.

A noncoding sequence following the last exon is shown in orange.

Splicing can occur either before or after the cleavage and polyadenylation steps.

All of the processes shown here take place within the nucleus.

真核生物中的所有 RNA 分子在合成后都会在一定程度上进行加工。
有趣的是，催化这些反应的酶有时由 RNA 而不是蛋白质组成。

催化 RNA 或核酶的发现为对 RNA 功能和生命起源的思考带来了一场革命。

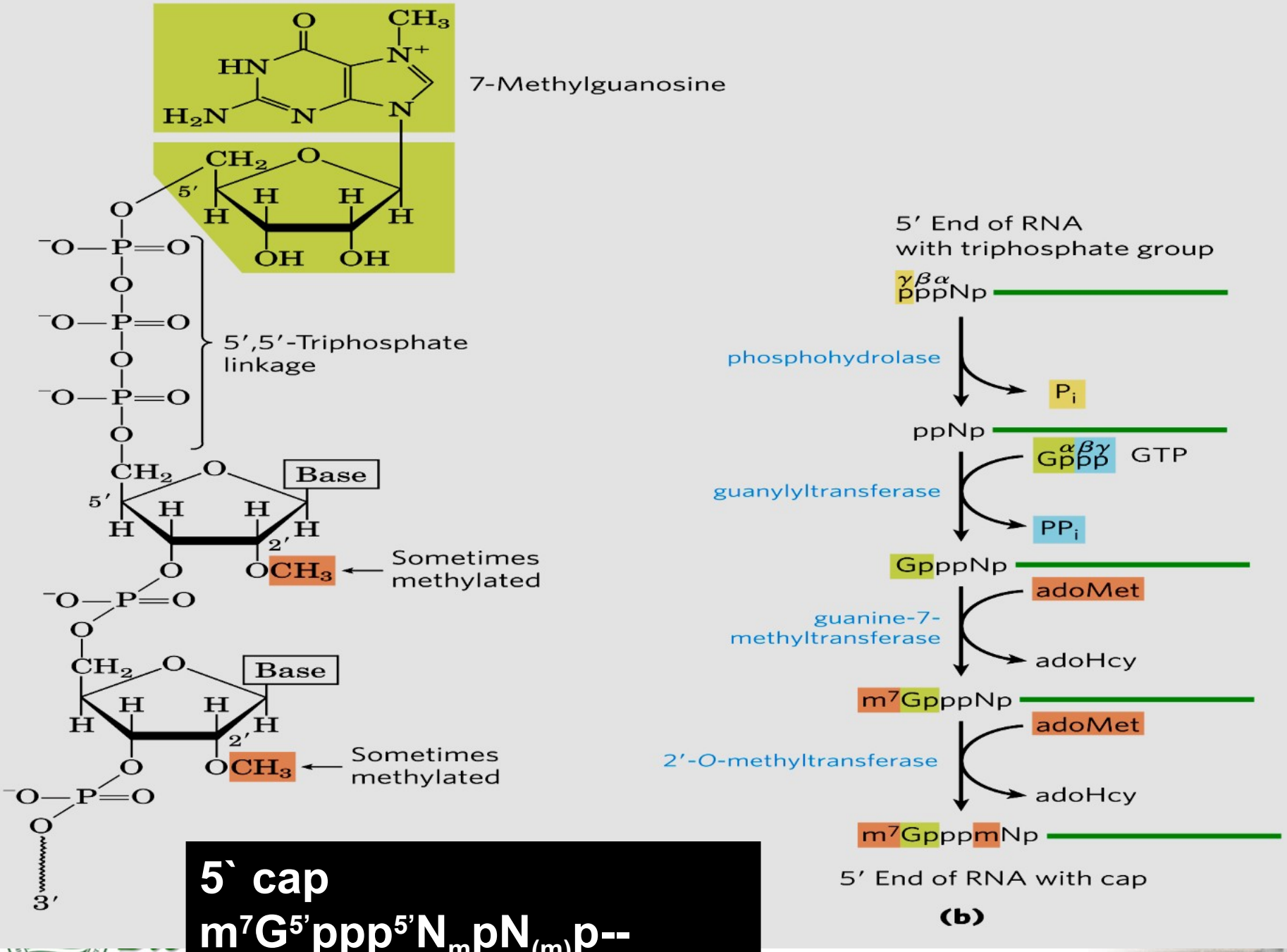




# 1. An eukaryotic mRNA precursor acquire a 5` cap shortly after transcription initiates

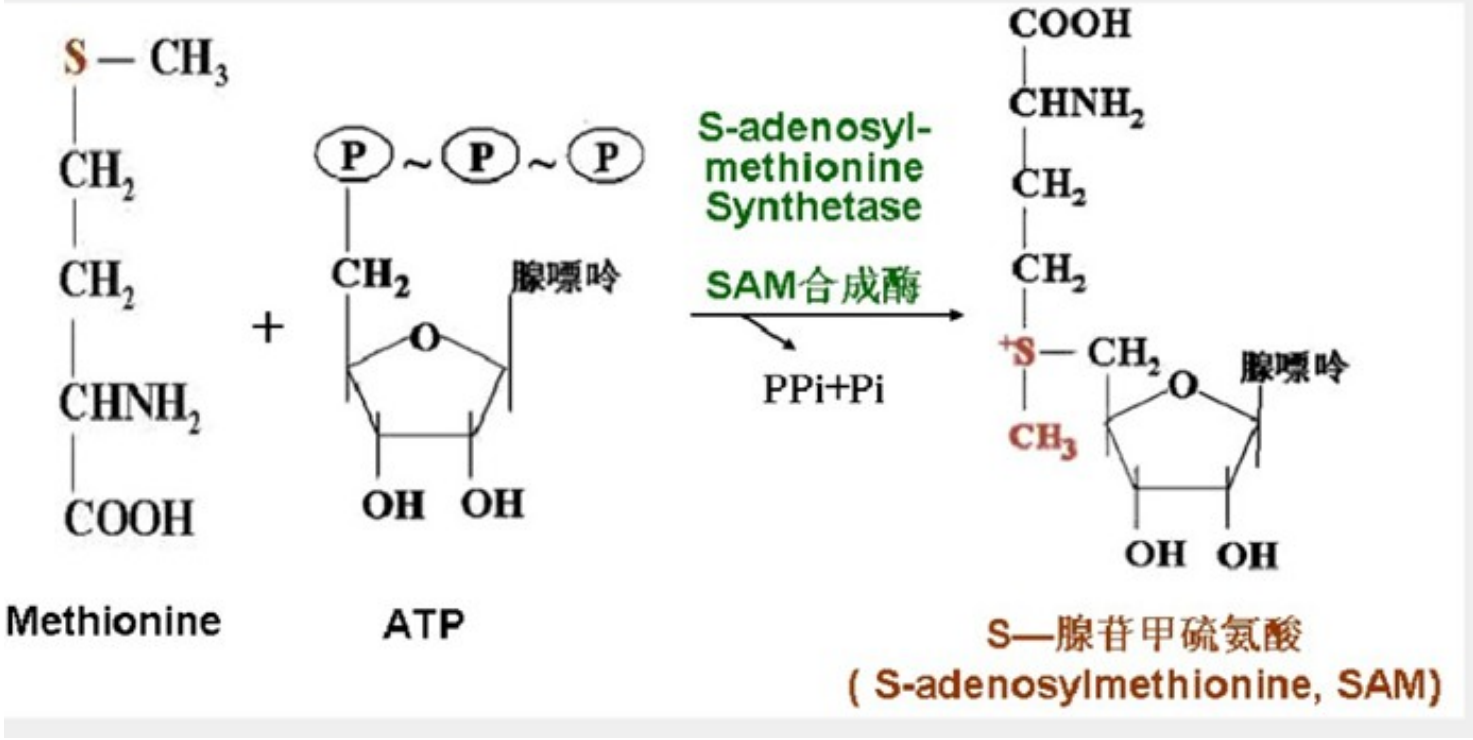




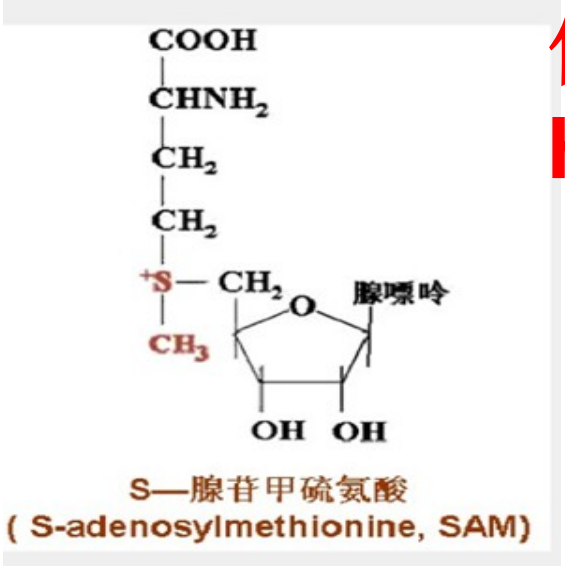



**5` cap**

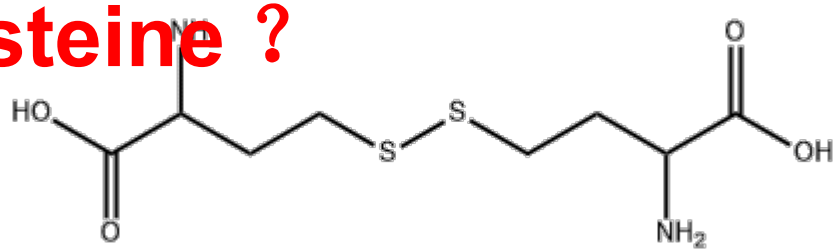
**$m^7G^{5'}ppp^{5'}N_mpN_{(m)}p$--**


S — CH3
CH2
CH2
CHNH2
COOH
Methionine
+
P ~ P ~ P
CH2
腺嘌呤
O
OH OH
ATP
S-adenosyl-
methionine
Synthetase
SAM合成酶
PPi+Pi
COOH
CHNH2
CH2
CH2
+S — CH2
CH3
腺嘌呤
O
OH OH
S—腺苷甲硫氨酸
( S-adenosylmethionine, SAM)

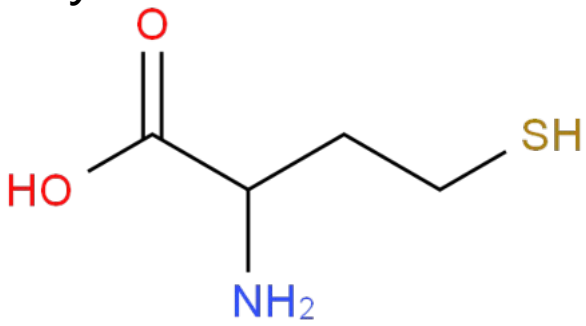
# 何为 Homocysteine ？

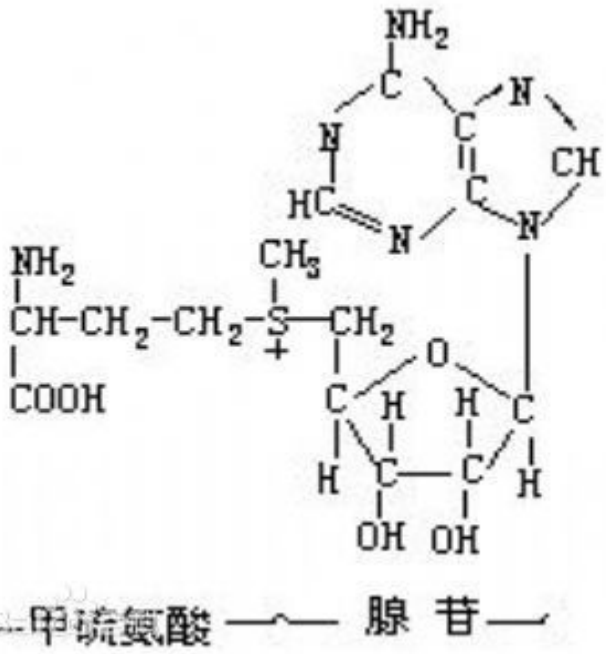


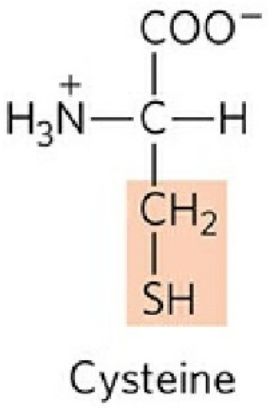

Homocystine

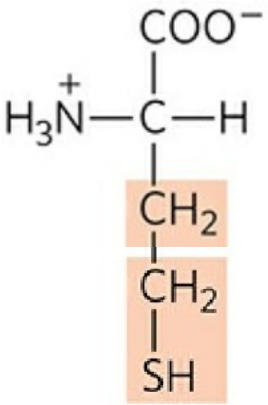


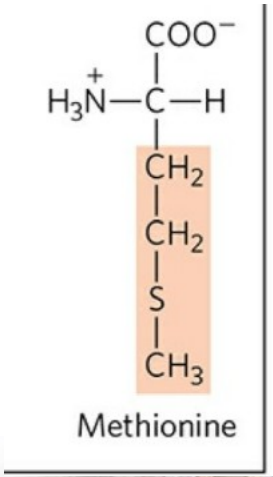

Homocysteine



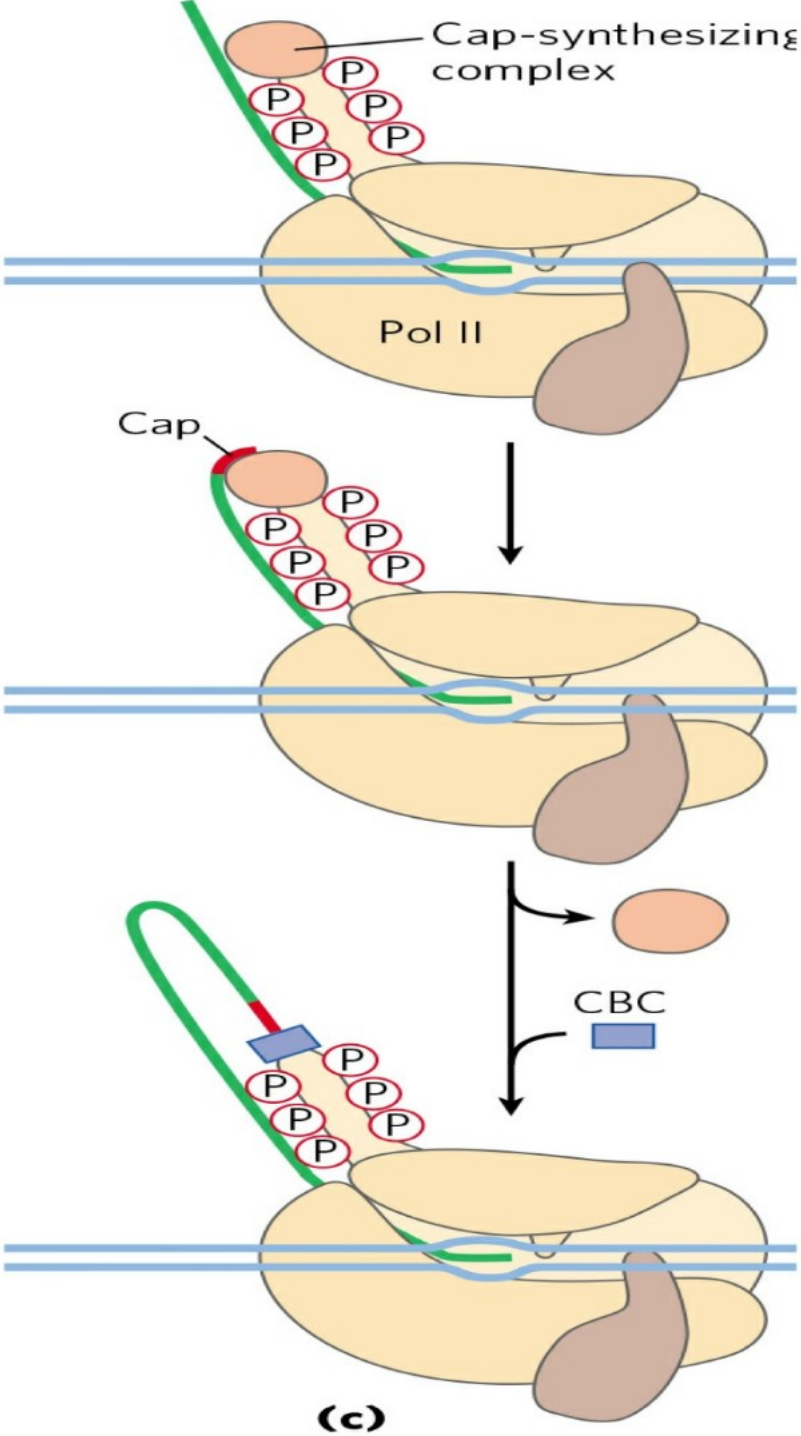



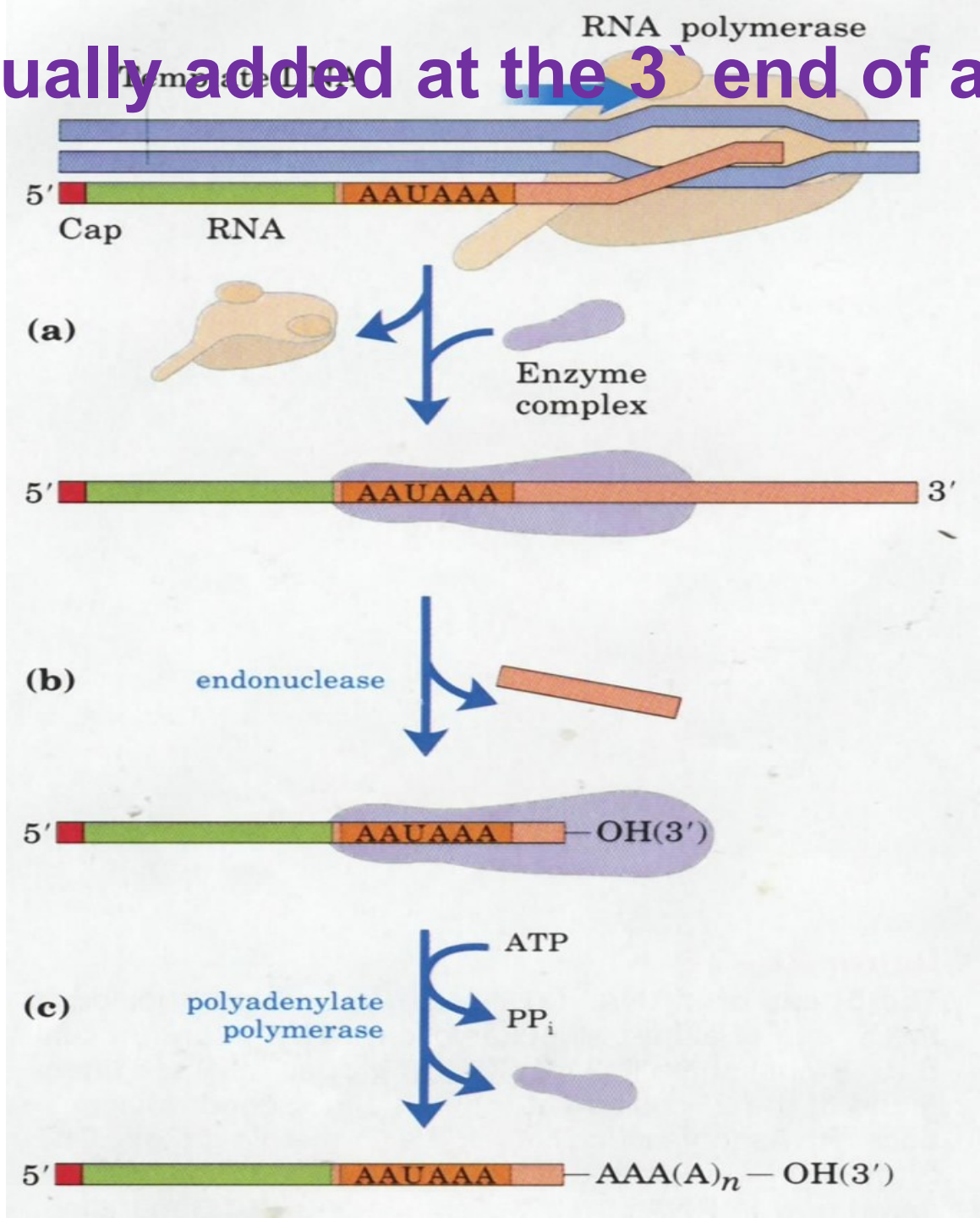


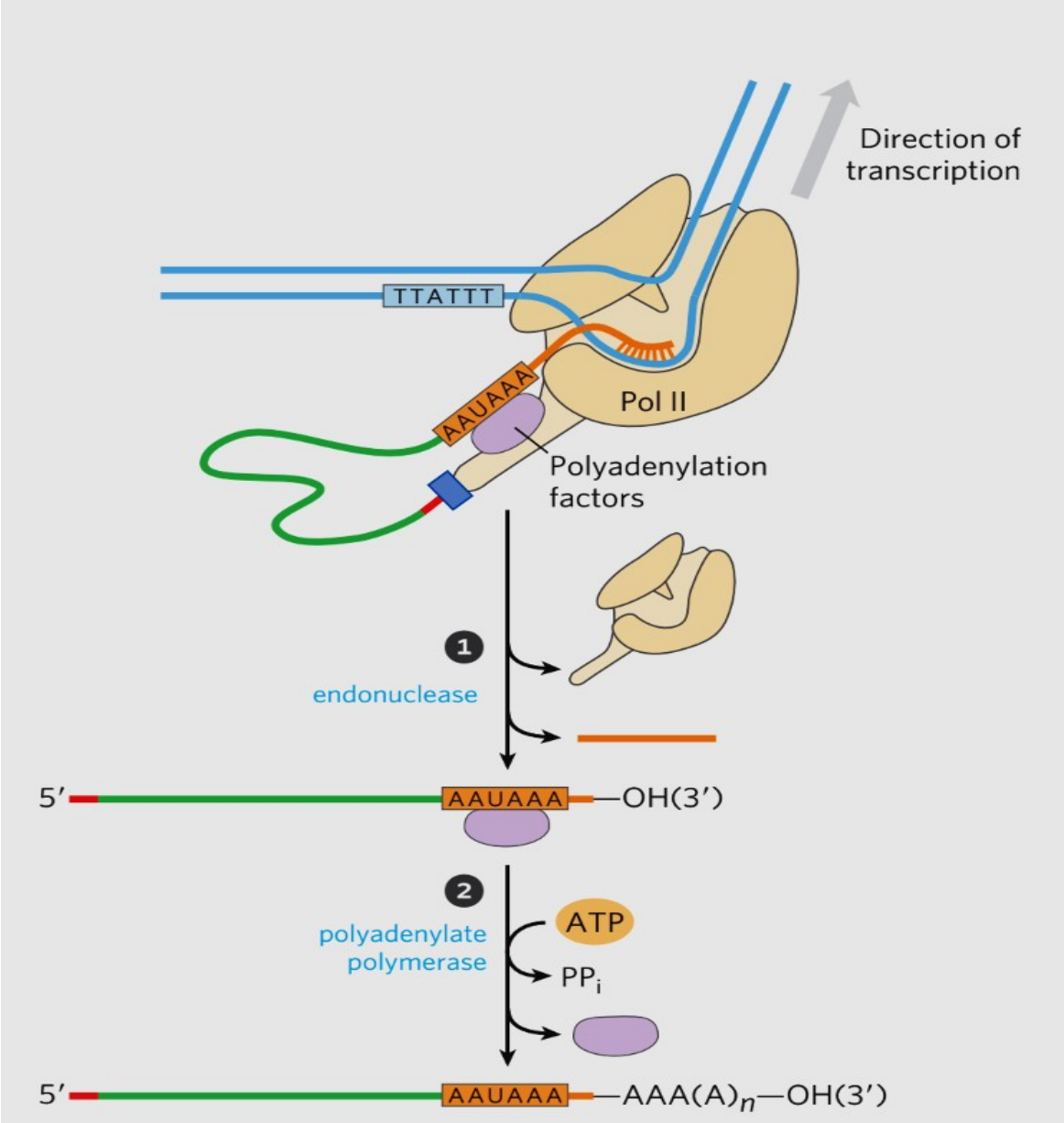

Biochemistry


Cap-synthesizing complex
P
Pol II
Cap
CBC
(c)

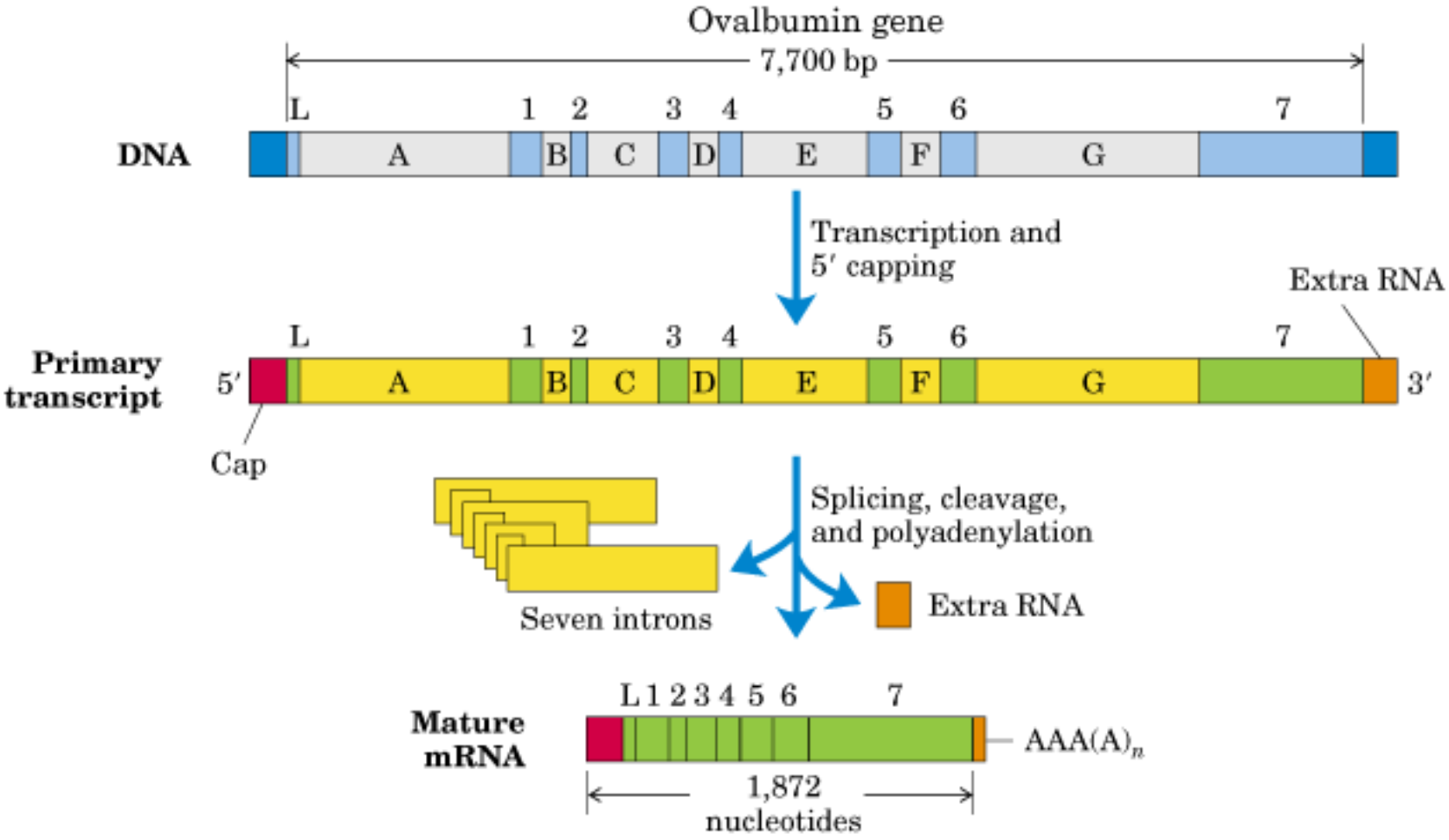
- The terminal 5` phosphate is first removed by a phosphatase.
- A **GMP** component (from a GTP) is joined to the 5` end of the mRNA catalyzed by guanosyl transferase in a novel **5`,5`-triphosphate linkage**.
- The guanine base is then methylated at the N-7.
- The 2`-OH groups of the 1st and 2nd nucleotides adjacent to the **7-methylguanine cap** may also be **methylated** in certain organisms.
- The methyl groups are transferred from S-adenosyl methionine （SAM S- 腺苷甲硫氨酸）.

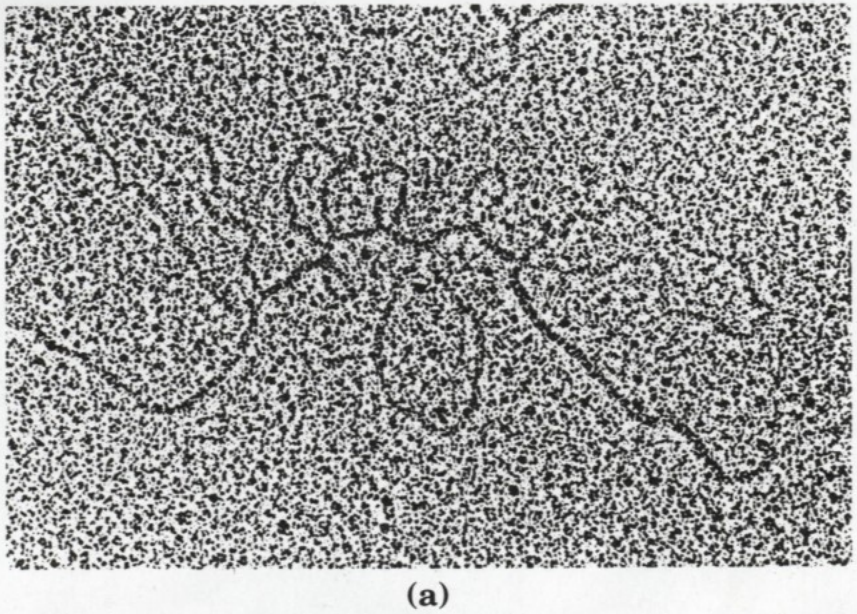
# A poly(A) tail is usually added at the 3` end of a

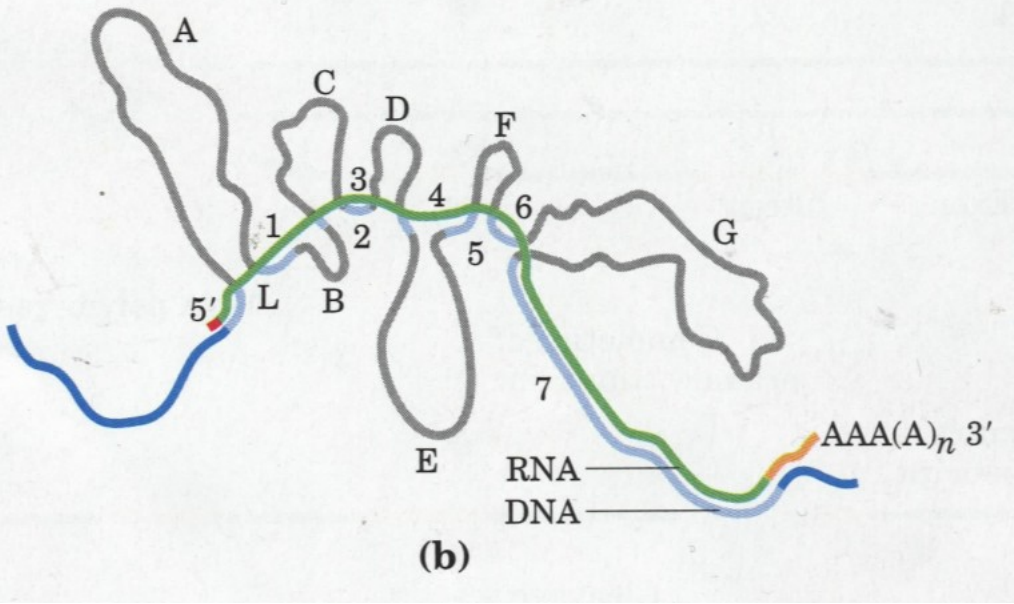


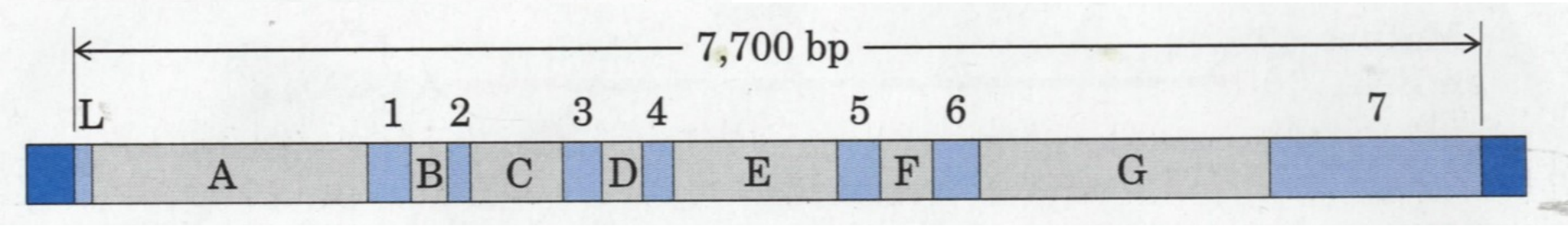

Direction of transcription
TTATTT
AAUAAA
Pol II
Polyadenylation factors
1
endonuclease
5′
AAUAAA
—OH(3′)
2
polyadenylate polymerase
ATP
PPi
5′
AAUAAA
—AAA(A)n—OH(3′)

# 3. The introns transcribed into RNA are removed by splicing






Demonstration of noncoding seque
in the chicken ovalbumin gene by F
DNA hybridization







# The Nobel Prize in Physiology or Medicine 1993

"for their discoveries of split genes"

Press release
Illustrated presentation



**Richard J. Roberts**

Great Britain

New England Biolabs
Beverly, MA, USA

1943 -

Autobiography



**Phillip A. Sharp**

USA

Center for Cancer Research
Massachusetts Institute of Technology
Cambridge, MA, USA

1944 -

Autobiography



# 4. Introns are self-splicing and the ends of the exon sequences ligate together

The intron is removed and the two exons precisely linked to produce a functional mRNA molecule in Eukaryotes.





# Four classes of introns

**Group I introns** nuclear, mitochondrial, and chloroplast genes that code for rRNAs, mRNAs, and tRNAs.

**Group II introns** mitochondrial or chloroplast mRNAs in fungi, algae, and plants.

*self-splicing*

νο ηιγη−ενεργψ (ΑΤΠ) φορ σπλιχινγ

**Group III introns**

**Group IV introns**

Group III: the **spliceosomal introns**
**σπλιχεοσομε** (προτειν χομπλεξ) , χονταινινγ:

**σμαλλ νυχλεαρ ΡΝΑσ (σνΡΝΑσ)** : Φιωε σνPNAσ (Y1, Y2, Y4, Y5, ανδ Y6)

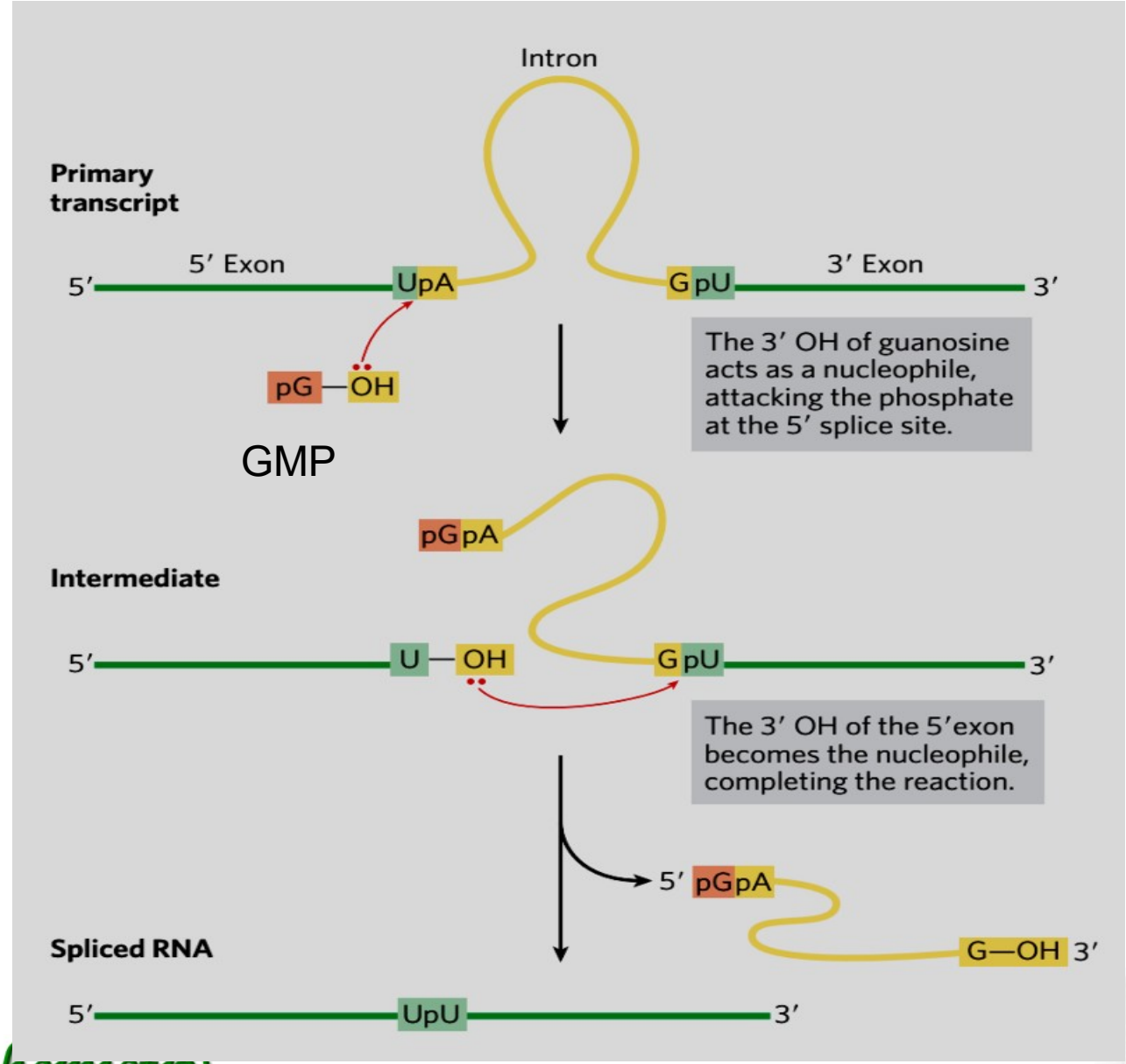
**σμαλλ νυχλεαρ ριβονυχλεοπροτεινσ (σνΡΝΠσ)**

Group IV: found in certain tRNAs, requires ATP and an endonuclease.

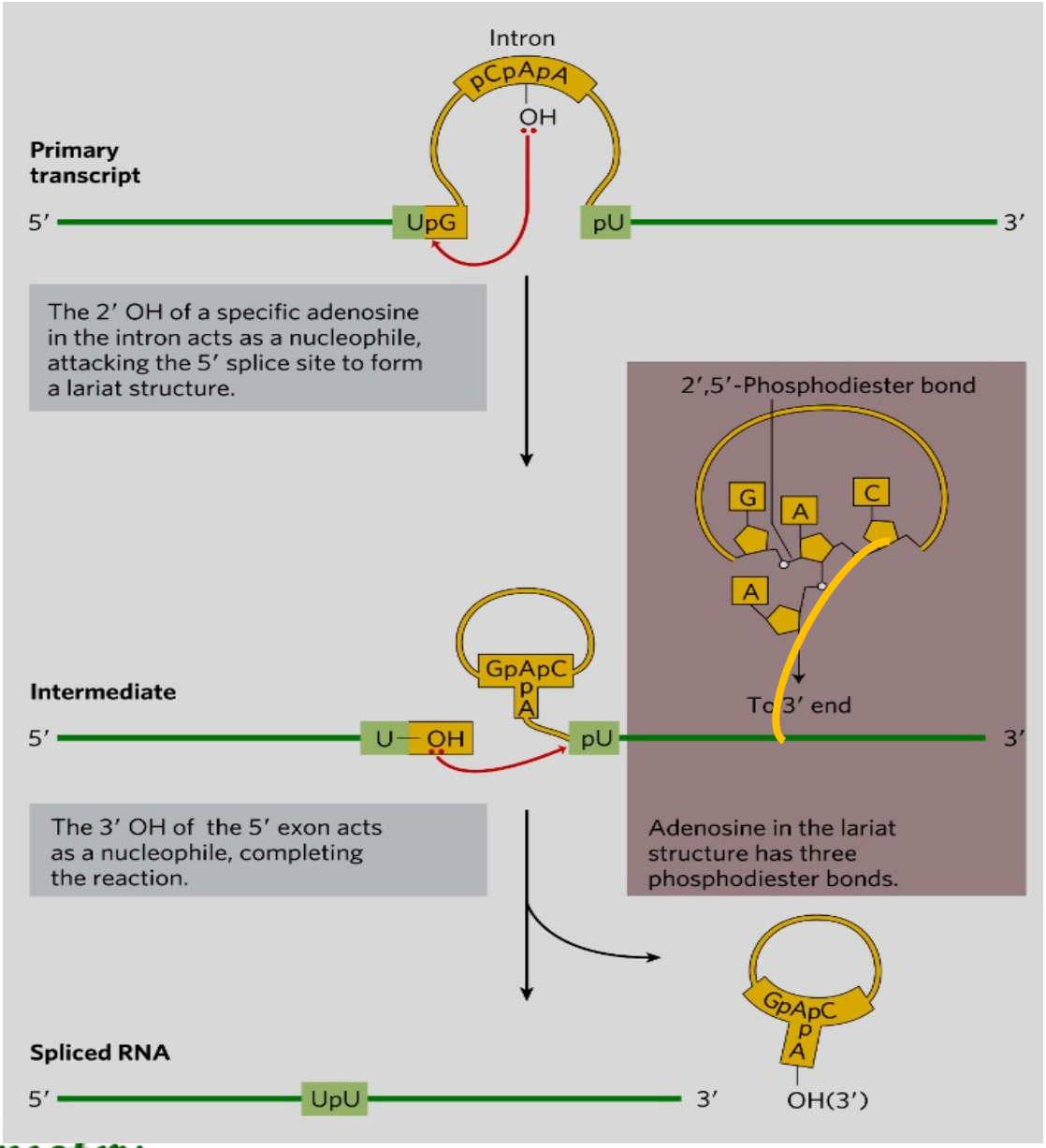




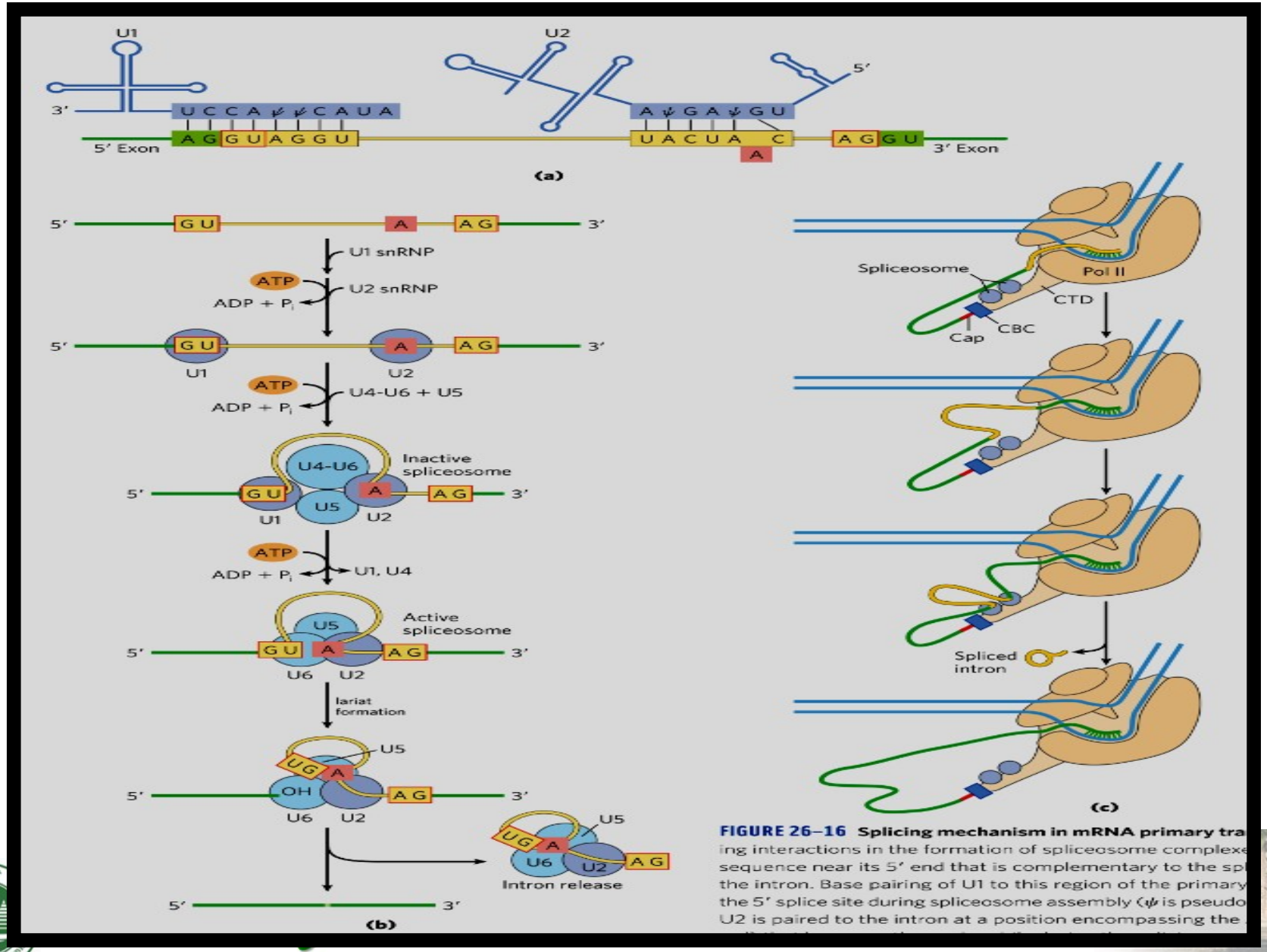
# Splicing mechanism of group I introns



# Splicing mechanism of group II introns



# Splicing mechanism with spliceosome complexes



R- purine 嘌呤
Y- pyrimidine 嘧啶

Group III

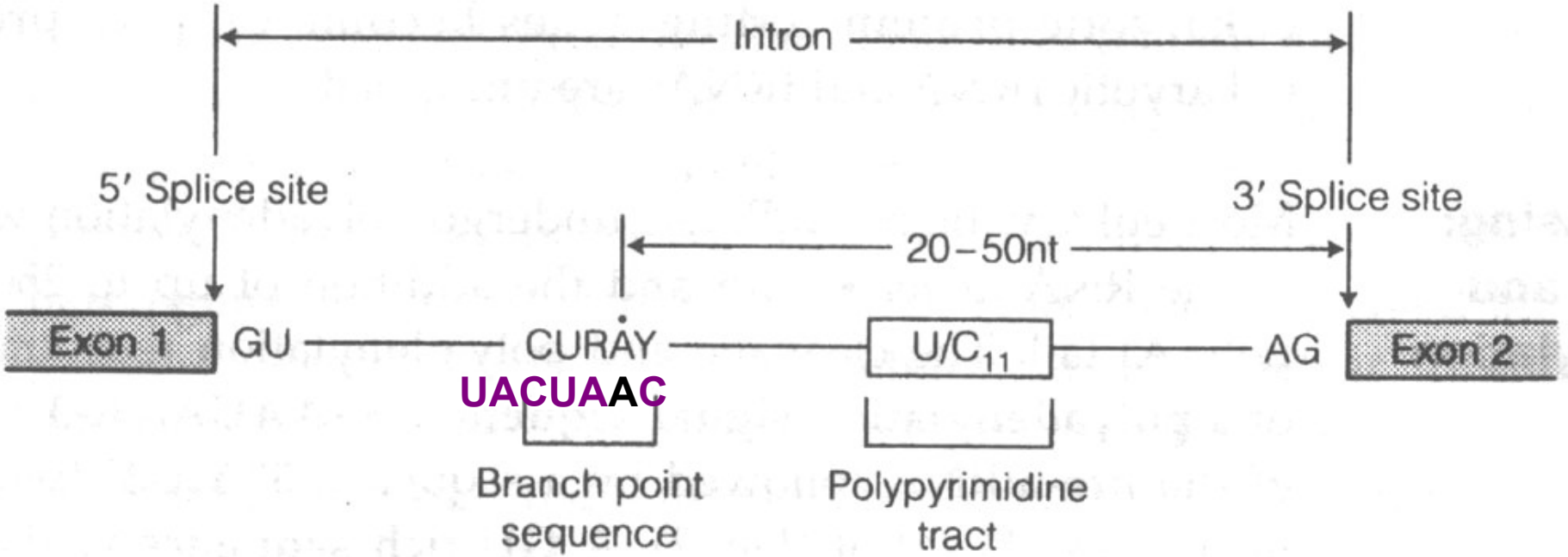



*Fig. 3. Conserved sequences for RNA splicing. The residue marked as Ȧ in the branchpoint sequence is the site of formation of the 2′5′ branch.*

Group III

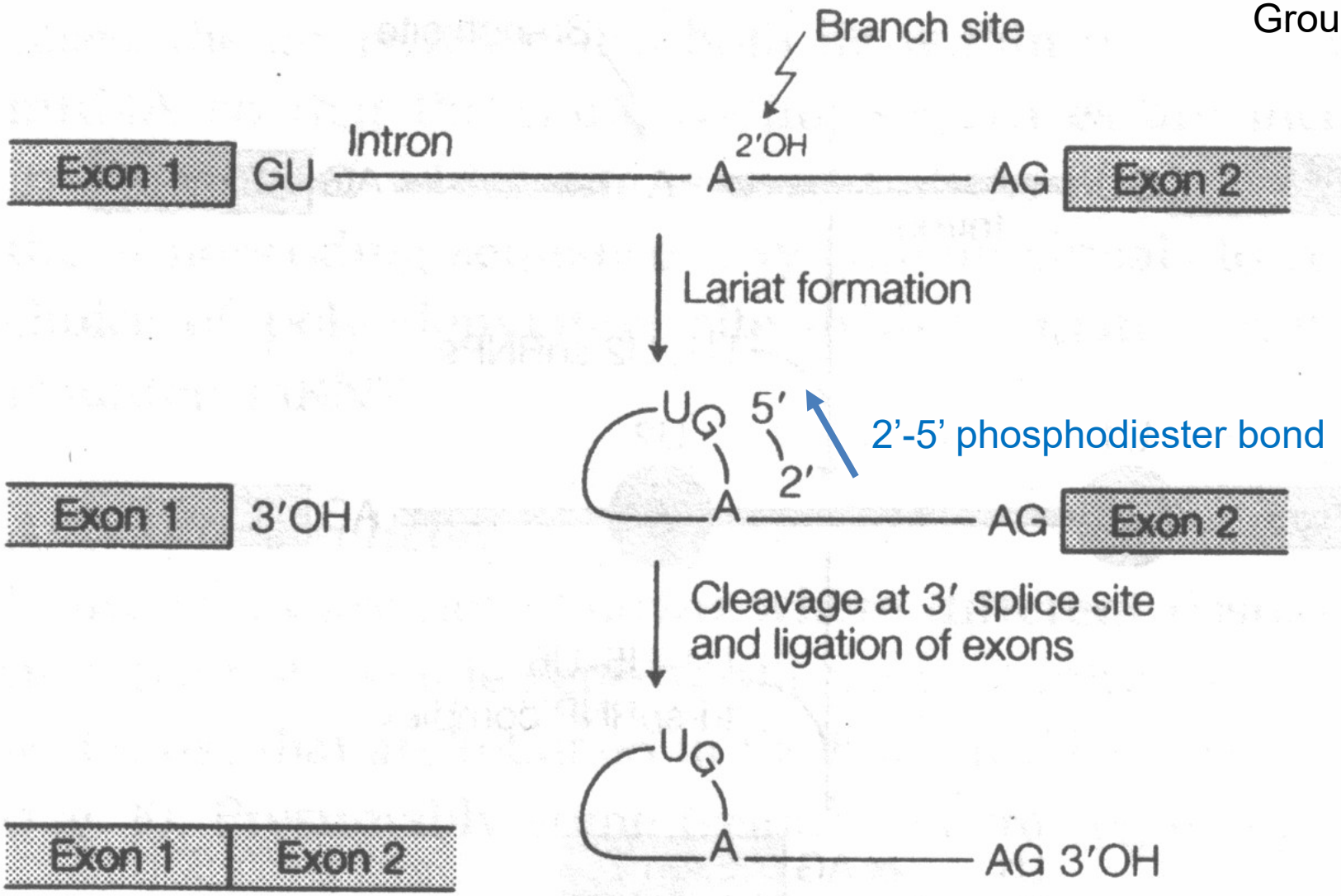



*Fig. 4. The two steps of RNA splicing.*

RNA 剪接分两个步骤进行：
内含子中特定腺苷的 2'OH 充当亲核试剂，攻击 5' 剪接位点以形成套板结构。
5' 外显子的 3'OH 充当亲核试剂，完成反应

在两个剪接反应中的每一个中，一个磷酸酯键被交换为另一个磷酸酯键。





**RNA splicing requires the involvement of several small nuclear RNAs (snRNA) each of which is associated with several proteins to form a small nuclear ribonucleoproteins particle or snRNP(**a large protein complex called a **spliceosome** )





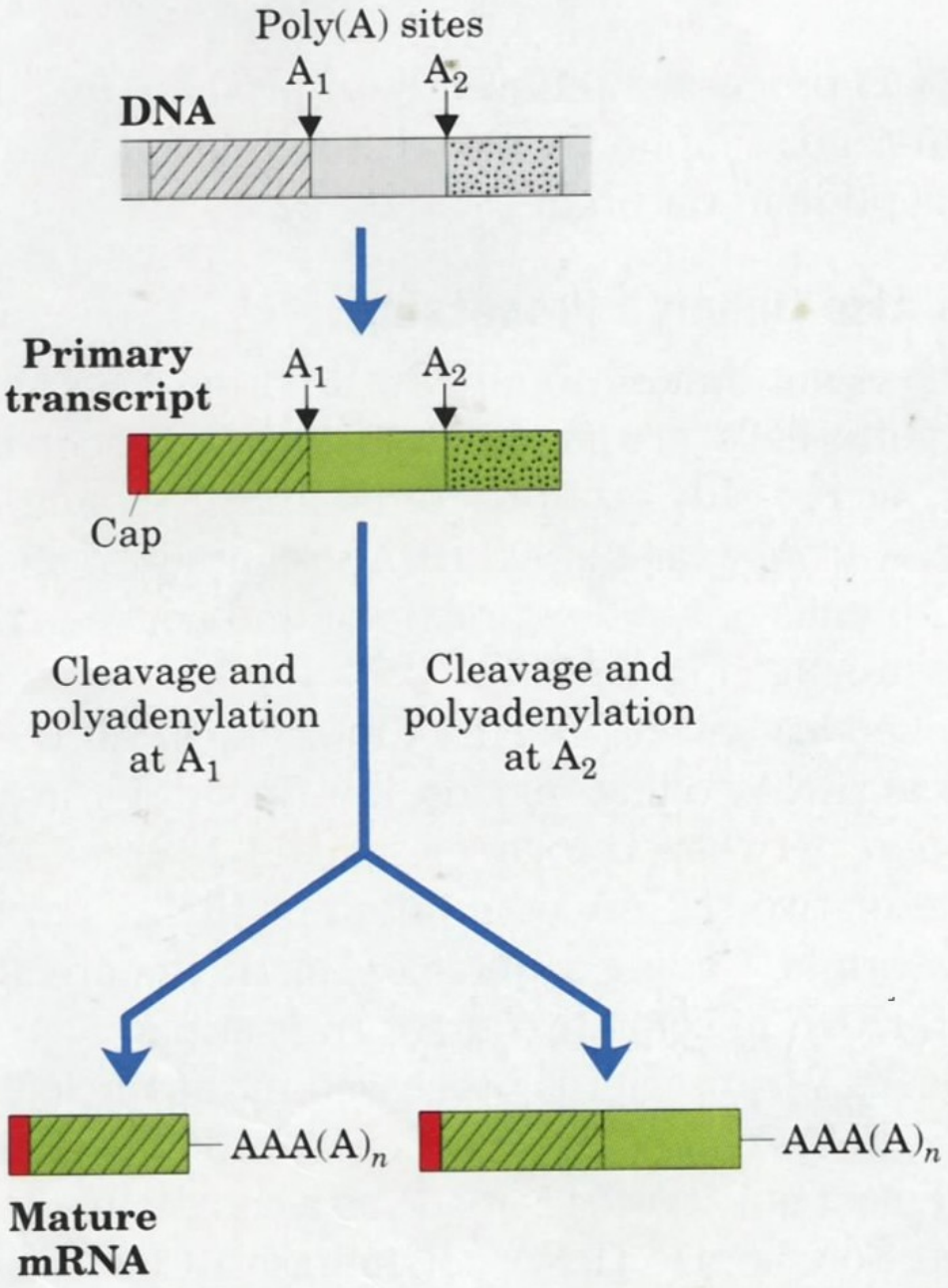
# 5.Mutiple products are derived from one gene by differential RNA processing

The transcription of introns consumes cellular resources and energy without returning any obvious benefit to the organism, but introns may confer an advantage not yet appreciated by scientists.

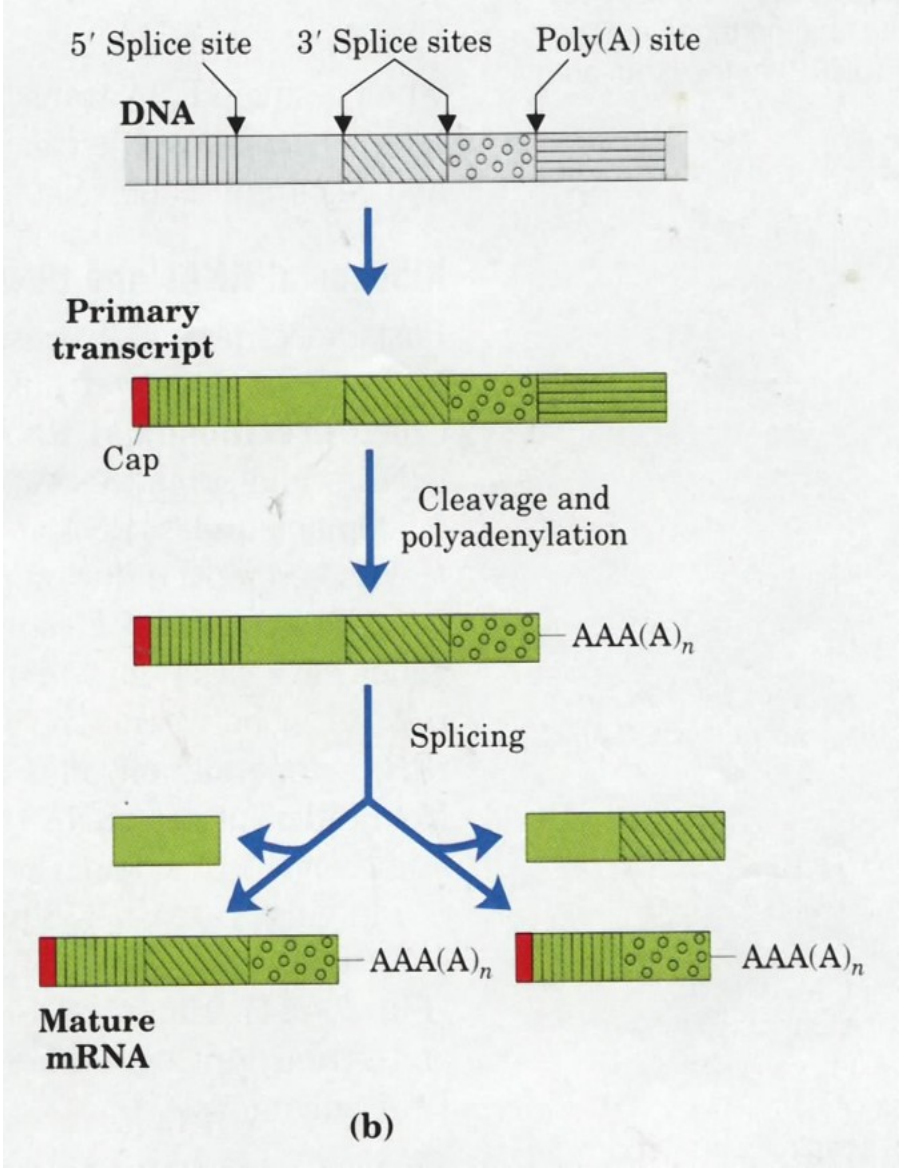
Although most eukaryotic mRNA transcripts produce only one mature mRNA and one corresponding polypeptide, some can be processed in more than one way to produce **different** mRNAs and thus different polypeptides.








(a)




(b)

**Multiple mRNAs can be produced via differential RNA processing.**

**In Fig. a, alternative cleavage and polyadenylation patterns.**

**Two ply(A) sites, A1 and A2, are shown.**

**In Fig. b, alternative splicing patterns.**

**Two different 3' splice sites are shown. In both mechanisms, different mature mRNAs are produced from the same primary transcript.**



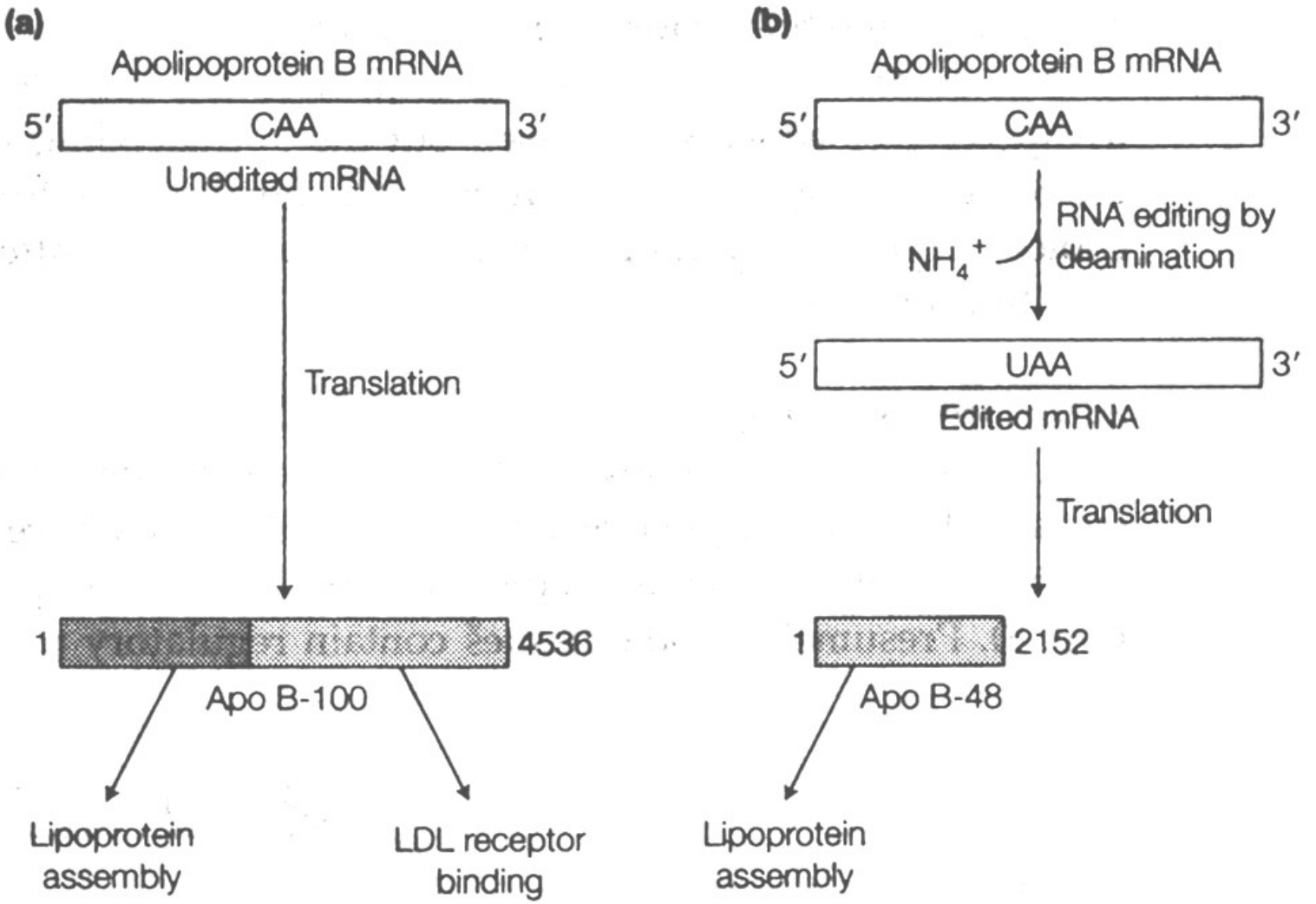


# 6. The sequence of an RNA molecule may be changed after synthesis and processing

Individual nucleotides within the mRNA may be changed to other mucleotides, deleted entirely or additional nucleotides inserted. The effect of RNA editing is to change the coding capacity of the mRNA so that it encodes a different polypeptide than that originally encoded be the gene.



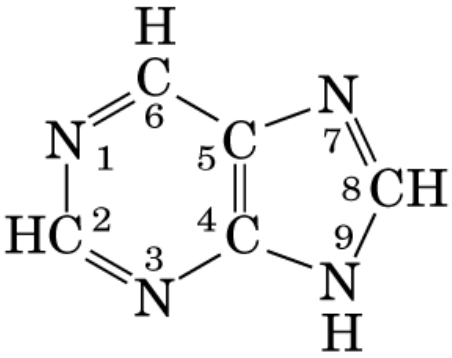




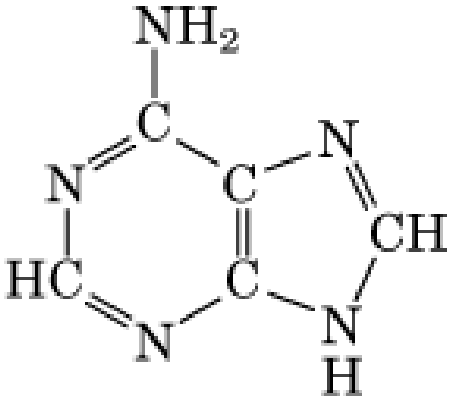

CAA =gln

UAA =stop

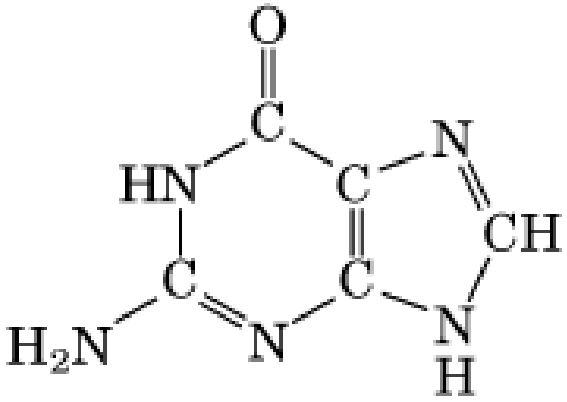
*Fig. 9. RNA editing. (a) Unedited apolipoprotein B mRNA is translated to yield ApoB-100, a 4536-amino acid long polypeptide with structural domains for lipoprotein assembly and receptor binding functions; (b) translation of the edited mRNA yields the shorter ApoB-48 which lacks the receptor binding domain.*



Purine

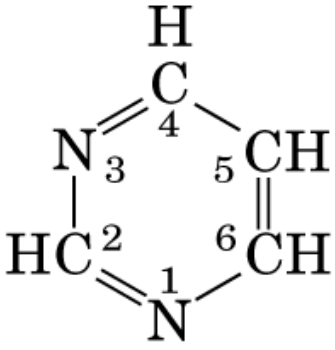


Adenine

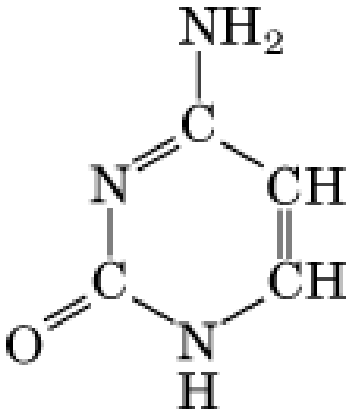


Guanine

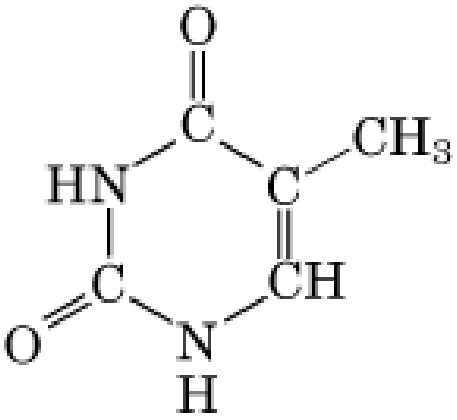
**Purines**



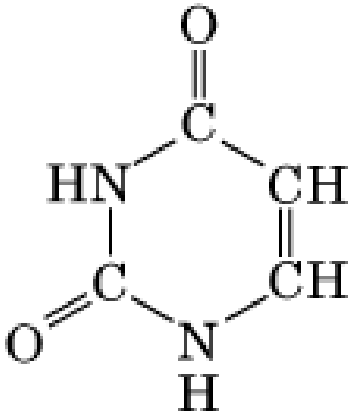
Pyrimidine



Cytosine



Thymine
(DNA)



Uracil
(RNA)

**Pyrimidines**

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Research | 28 February 2018

## The SMAD2/3 interactome reveals that TGFβ controls m$^6$A mRNA methylation in pluripotency

...signalling to diverse effectors including m6A methyltransferase, which has a...

Alessandro Bertero, Stephanie Brown [...] Ludovic Vallier

*Nature* **555**, 256–259

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Amendments and Corrections | 29 November 2017

## Corrigendum: RNA m$^6$A methylation regulates the ultraviolet-induced DNA damage response

Yang Xiang, Benoit Laurent [...] Yang Shi

Biochemistry



methionine_百度搜索 | m6a : nature.com search

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Research | 10 July 2019

## m$^6$A enhances the phase separation potential of mRNA

N6-methyladenosine (m6A) is the most prevalent... ...mRNAs containing at least one m6A. Methylation of mRNA to... form m6A is required for diverse cellular... ...processes3. Although the presence of m6A in an mRNA can affect... ...ways, it is unclear how m6A directs this process and why... ...m6A...

Ryan J. Ries, Sara Zaccara […] Samie R. Jaffrey

*Nature* **571** , 424–428

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Research | 13 March 2019

## Histone H3 trimethylation at lysine 36 guides m$^6$A RNA modification co-transcriptionally

...mRNA, the N6-methyladenosine (m6A) mRNA modification is as... ...However, it is unclear how m6A

海尔生物 搜一下

14:58 2019/10/28

Biochemistry

Keyword Detail | 新标签页 | m6A_百度搜索 | sam 甲基_百度 | ovalbumin_百 | 硫酸_百度百科 | M6A : nature.c | RNA甲基化修饰 | DNA甲基化_百 | DNA甲基化_百 | 姜大卫回应周

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**Histone H3 trimethylation at lysine 36 guides m$^6$A RNA modification co-transcriptionally**

...mRNA, the N6-methyladenosine (m6A) mRNA modification is as... ...However, it is unclear how m6A is specifically and dynamically deposited... ...marker for transcription elongation, guides m6A deposition globally. We show... that m6A modifications are enriched in the... ...the m6A... ...the m6A...

Huilin Huang, Hengyou Weng [...] Jianjun Chen

*Nature* **567**, 414–419

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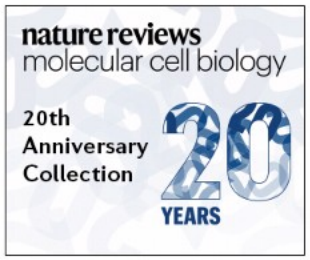
Research | 10 July 2019

**m$^6$A enhances the phase separation potential of mRNA**

N6-methyladenosine (m6A) is the most prevalent... ...mRNAs containing at least one m6A. Methylation of mRNA to... form m6A is required for diverse cellular... ...processes3. Although the presence of m6A in an mRNA can affect... ...ways, it is unclear how m6A directs this process and why... ...m6A...

Ryan J. Ries, Sara Zaccara [...] Samie R. Jaffrey

*Nature* **571**, 424–428




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Biochemistry

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**Research Highlights**
06 Oct 2022
**Nature Cell Biology**
Volume: 24, P: 1448

## m$^6$A in planarian cell fate

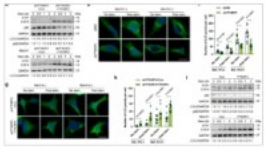
Daryl J. V. David

**Research**
Open Access
04 Oct 2022
**Nature Communications**
Volume: 13, P: 1-23

## Autophagy induction promoted by m$^6$A reader YTHDF3 through translation upregulation of FOXO3 mRNA

The role of eiptranscriptomic modifications in autophagy is unclear. Here, the authors show that the m$^6$A reader YTHDF3 functions as a nutrient responder to recognize upregulated m$^6$A modification, promoting FOXO3 translation to subsequently initiate autophagy.

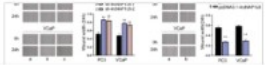
WeiChao Hao, MeiJuan Dian ... Dong Xiao



**Research**
Open Access
04 Oct 2022
**Cancer Gene Therapy**
P: 1-10

## LncNAP1L6 activates MMP pathway by stabilizing the m6A-modified NAP1L2 to promote malignant progression in prostate cancer

Yuxiao Zheng, Feng Qi ... Xiao Li



**Research**
07 Sept 2022
**Nature Genetics**
Volume: 54, P: 1427-1437

## RNA m$^6$A regulates transcription via DNA demethylation and chromatin accessibility

METTL3-dependent RNA $N^6$-methyladenosine (m$^6$A) deposition can lead to DNA demethylation. The m$^6$A reader F recruits DNA 5-methylcytosine dioxygenase TET1 to chromatin, which is linked to chromatin accessibility and gene transcription.

Shuang Deng, Jialiang Zhang ... Jian Zheng

**Research**

## *Mettl3*-dependent m$^6$A modification attenuates the brain stress response in *Drosophila*


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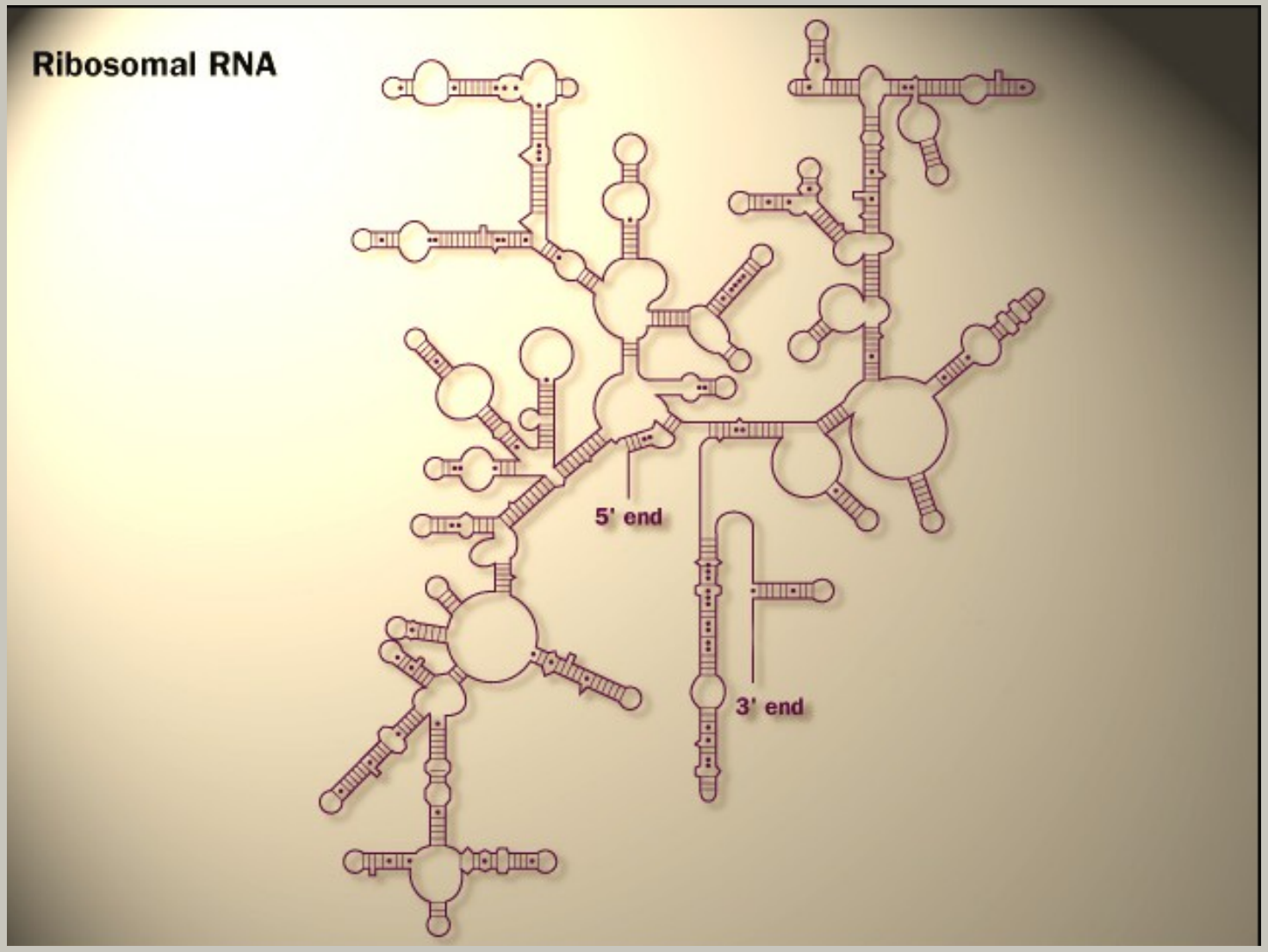
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15:58
2022/10/27

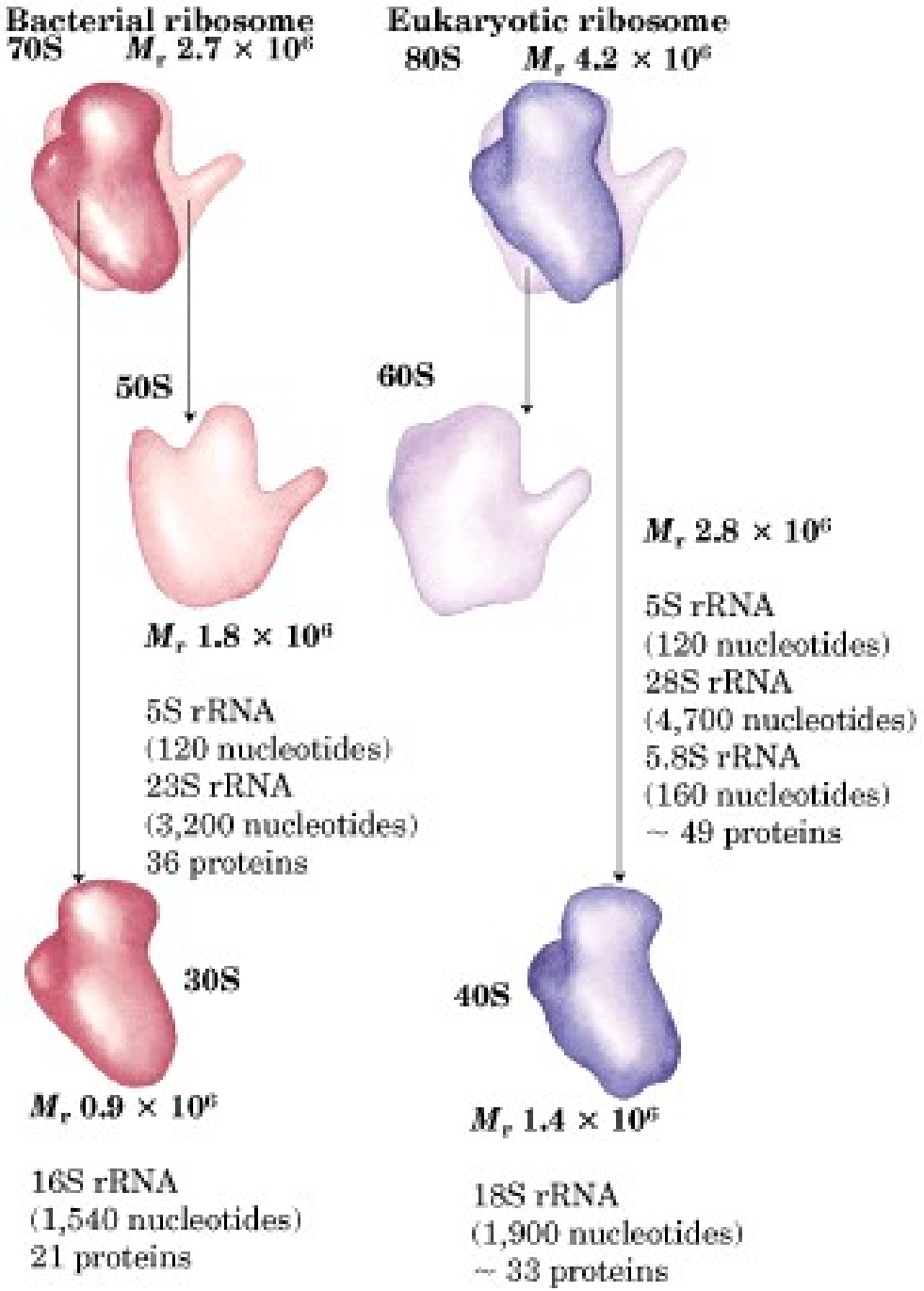

Biochemistry

# G8 Ribosomal RNA



# 1. Ribosomes

沉降系数
S

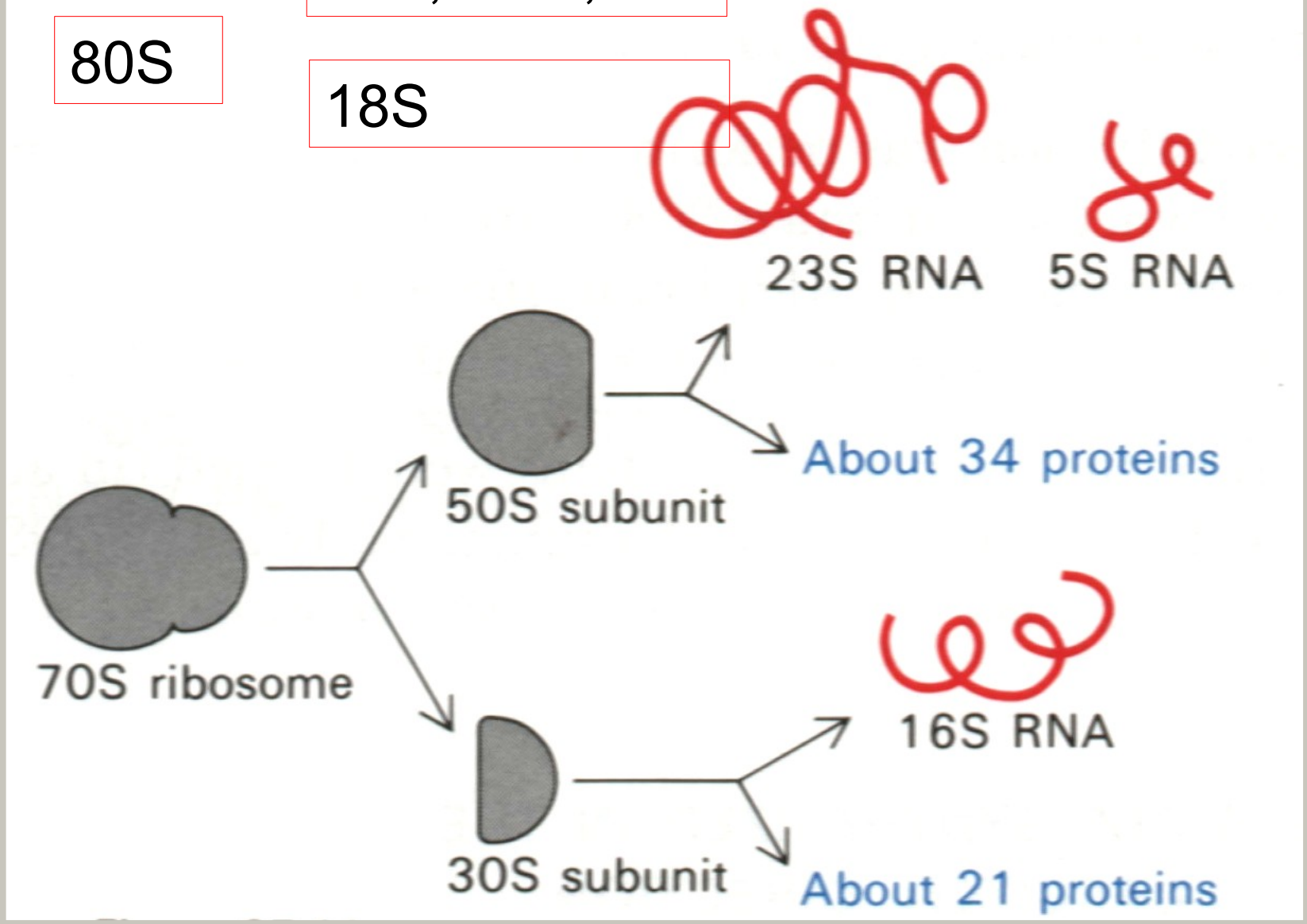



(b)

28S, 5.8S, 5S

80S

18S



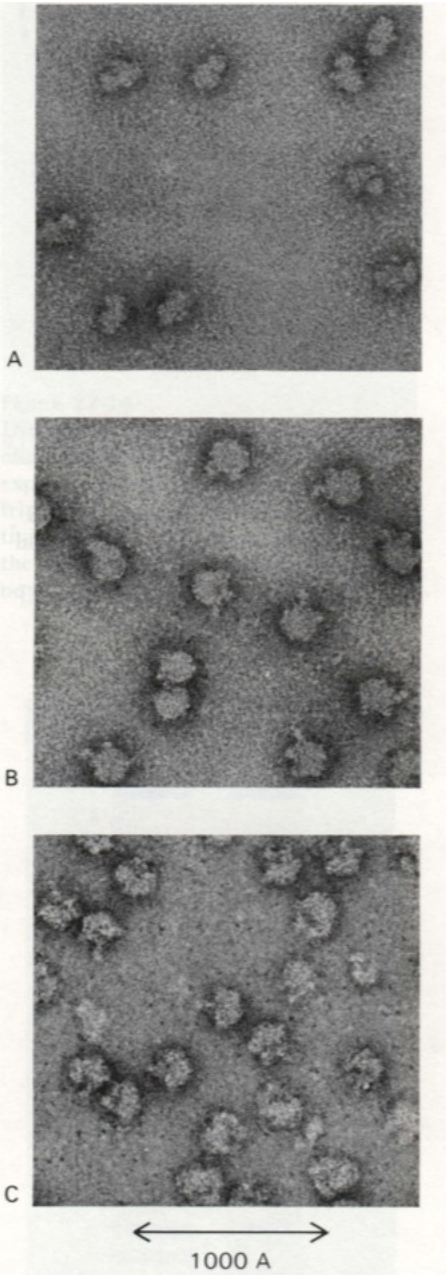
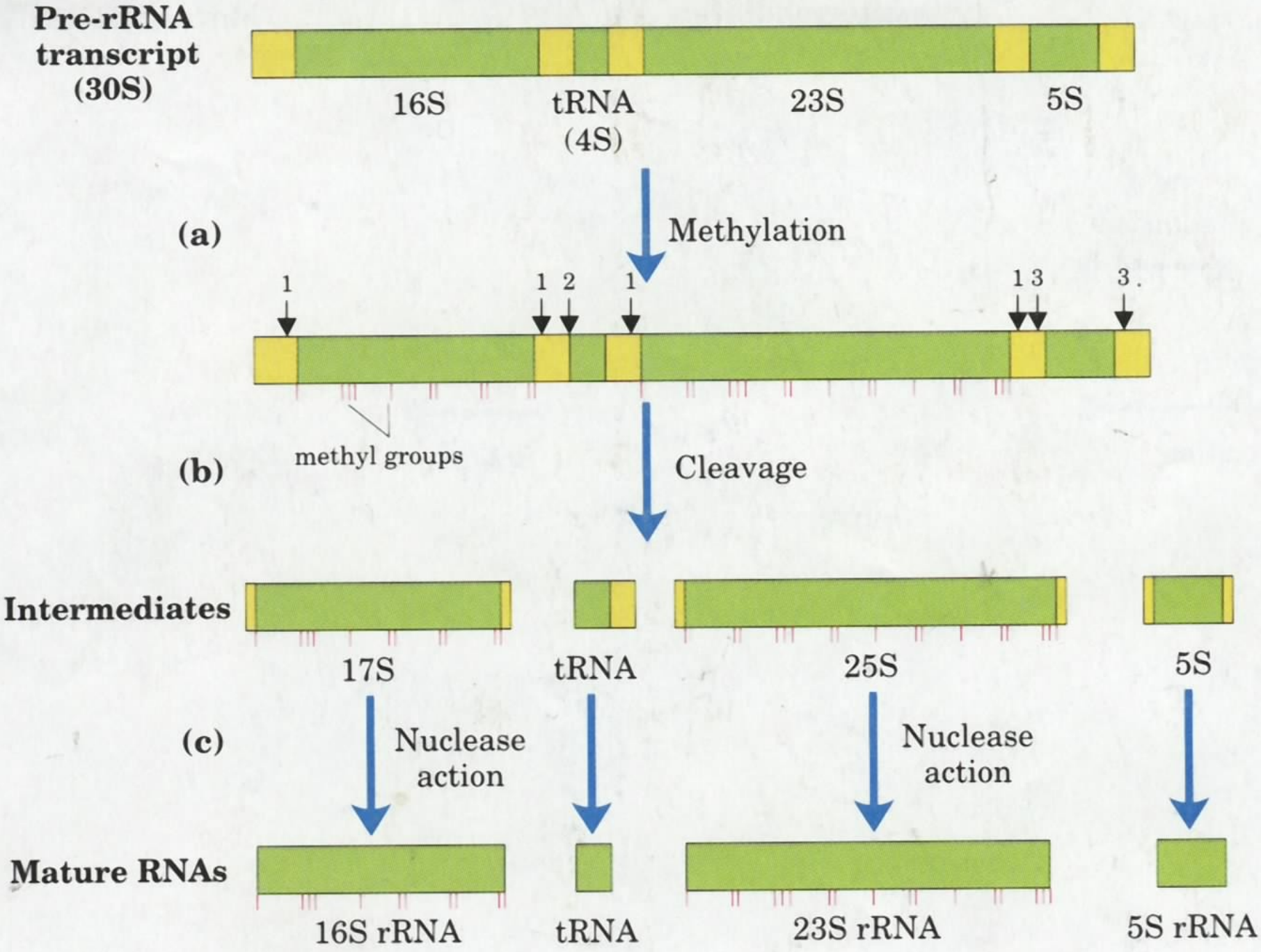
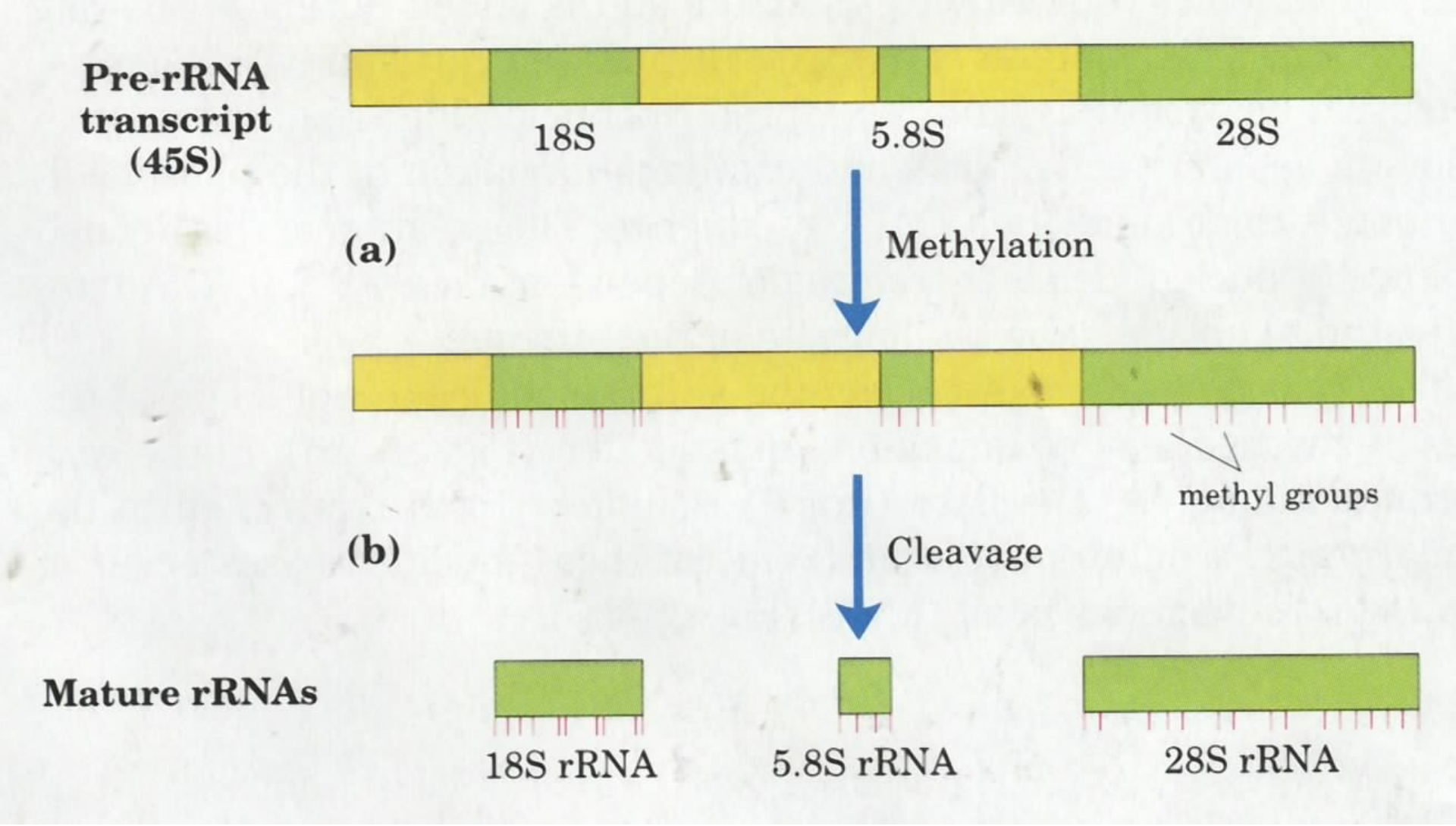



**Figure 27-10**
Electron micrographs of (A) 30S subunits, (B) 50S subunits, and (C) 70S ribosomes. [Courtesy of Dr. James Lake.]

# 2. The different rRNA molecules of both prokaryotes and eukaryotes are generated from single pre-rRNAs

The 16S, 23S and 5S rRNAs in bacteria are all generated from a single 30S pre-rRNA (*about 6.5 kb*).
The 18S, 28S and 5.8S rRNAs in eukaryotes are generated from a single 45S pre-rRNA (~14 kb).
The 5S rRNA in eukaryotic cells is generated separately (*transcribed by RNA polymerase III*).


Pre-rRNA transcript (30S)
16S
tRNA (4S)
23S
5S
(a)
Methylation
1
1 2
1
1 3
3
methyl groups
(b)
Cleavage
Intermediates
17S
tRNA
25S
5S
(c)
Nuclease action
Nuclease action
Mature RNAs
16S rRNA
tRNA
23S rRNA
5S rRNA




**The 18S, 5.8S, and 28S rRNAs in eukaryotic cells are derived from one pre-rRNA molecule (the processing needs small nucleolar RNA-containing proteins).**

# 3. More RNA molecules (ribozymes) were found to be catalytic





# The Nobel Prize in Chemistry 1989

"for their discovery of catalytic properties of RNA"

Press release



Sidney Altman

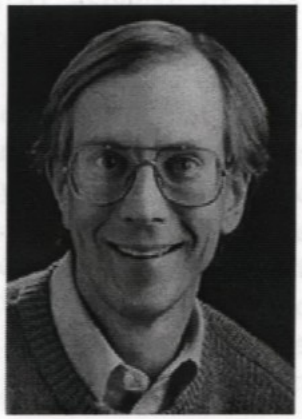
USA and Canada

Yale University
New Haven, CT, USA

1939 -

Autobiography

**The M1 RNA in ribonuclease P is catalytic**



Thomas R. Cech

USA

University of Colorado
Boulder; CO, USA

1947 -

Autobiography
Other Resources

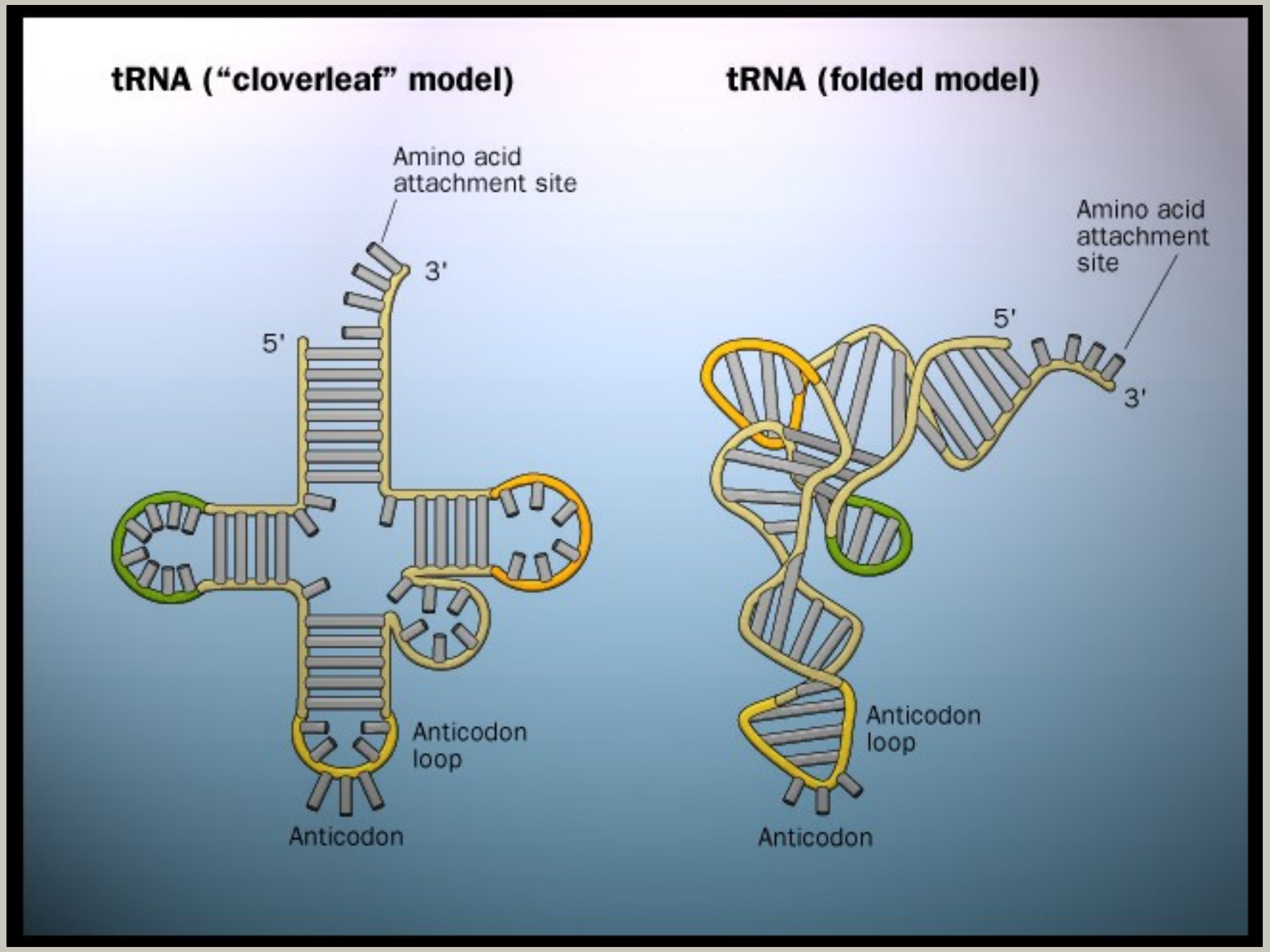
**The intron in the pre-rRNA of *Tetrahemena* is self-spliced**

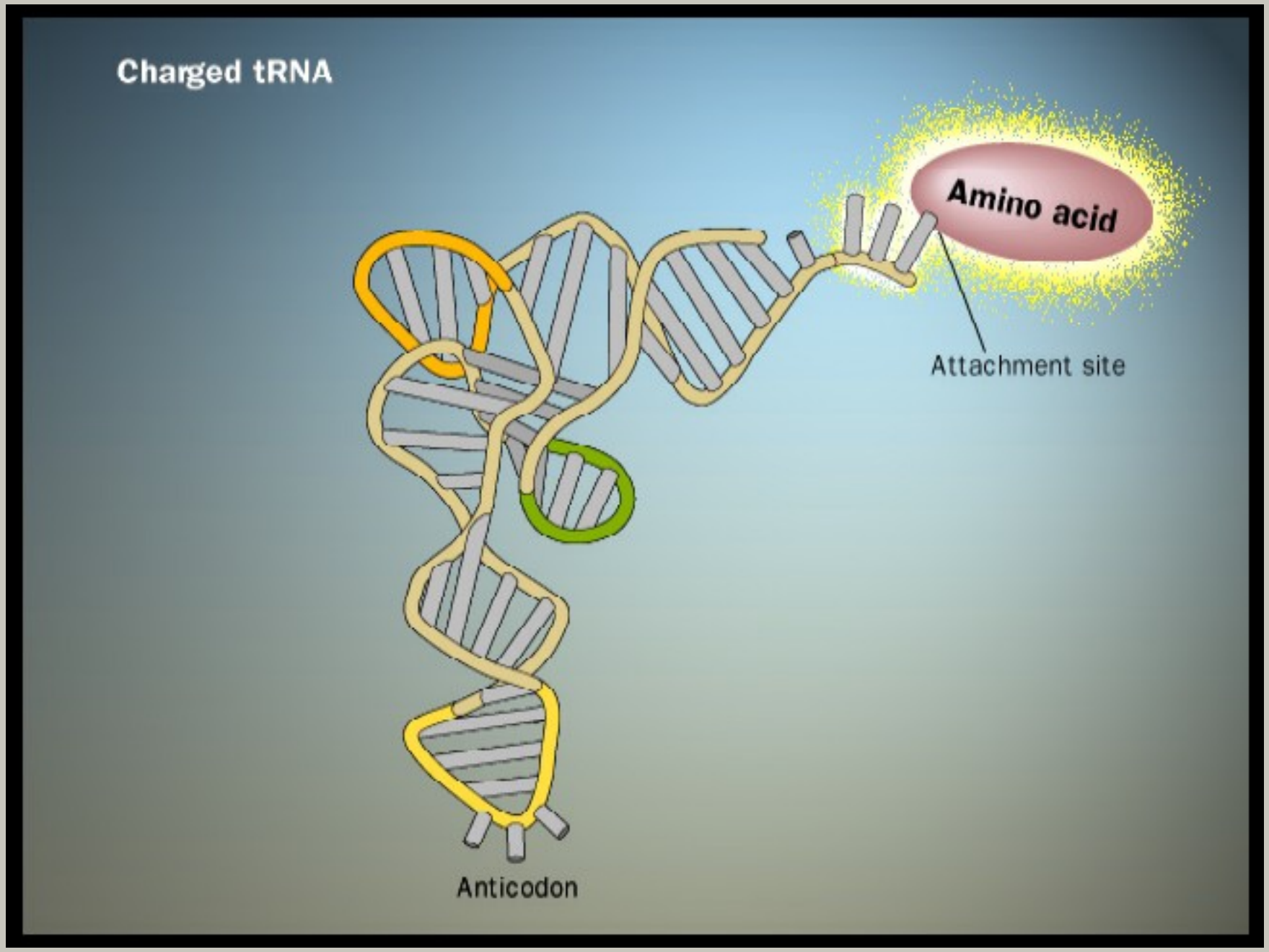


# G9 Transfer RNA

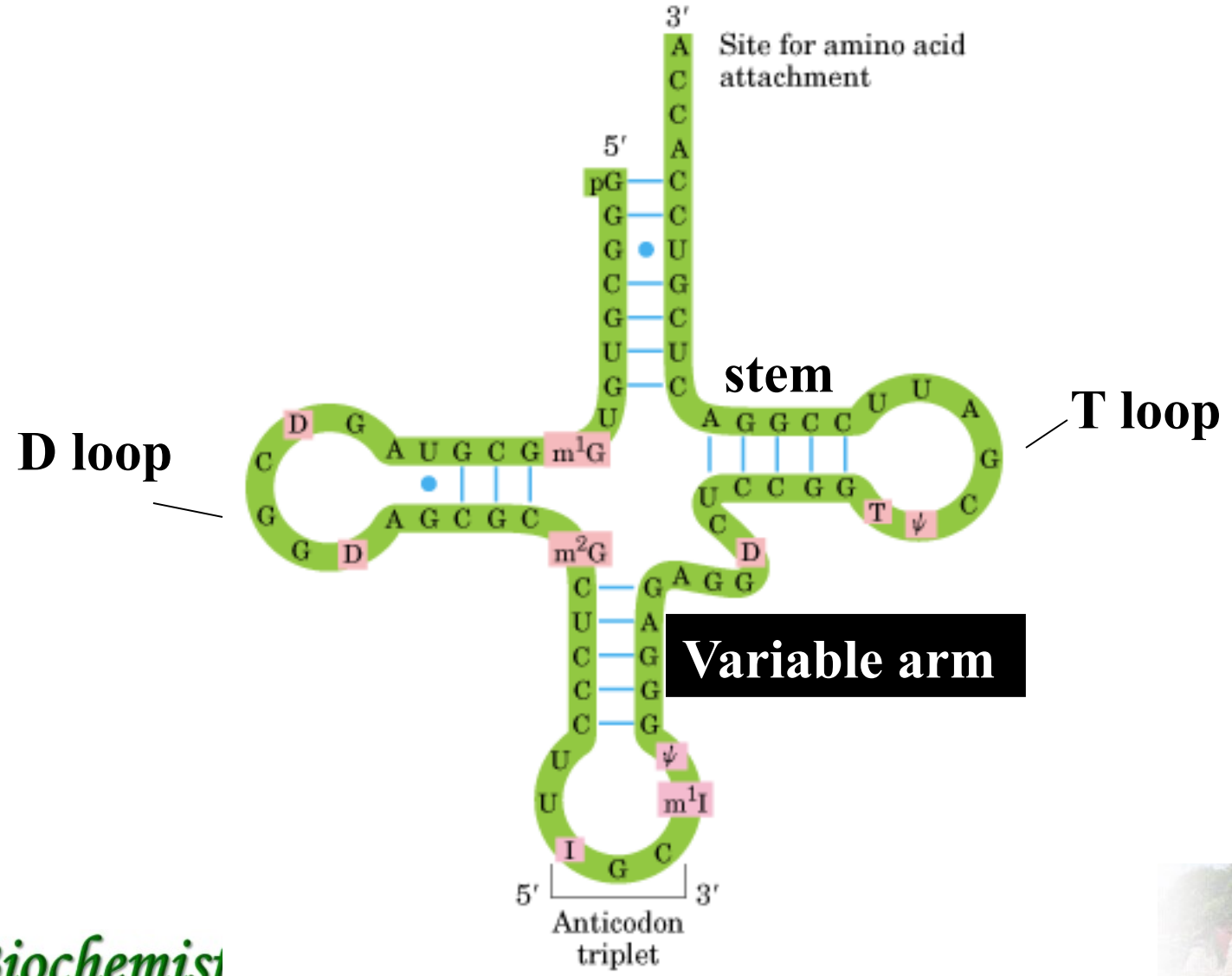


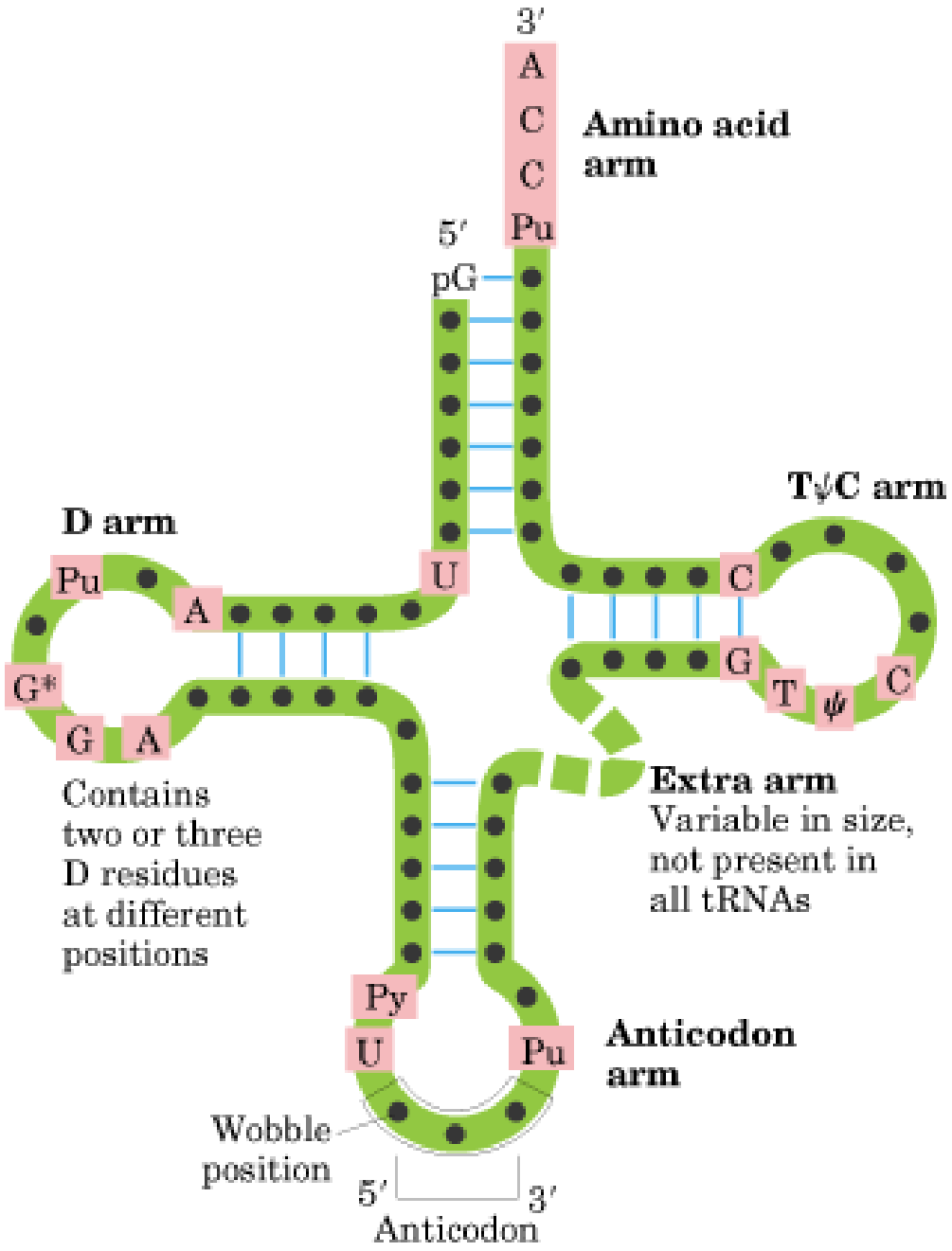



tRNA ("cloverleaf" model)
Amino acid attachment site
3'
5'
Anticodon loop
Anticodon
tRNA (folded model)
Amino acid attachment site
5'
3'
Anticodon loop
Anticodon


Charged tRNA
Amino acid
Attachment site
Anticodon

# Cloverleaf secondary structure of tRNA

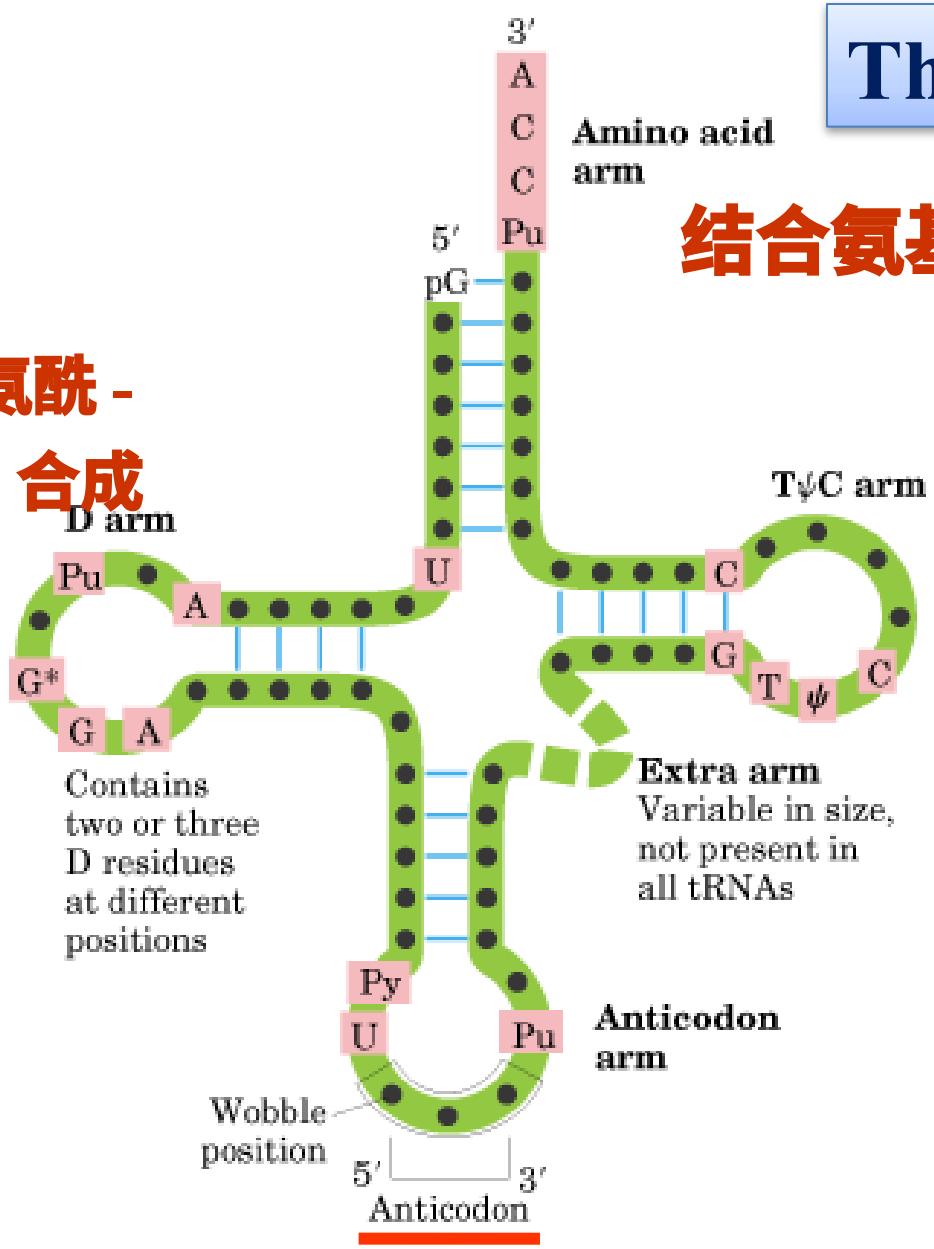



3′
A
C
C
Pu
Amino acid arm
5′
pG
D arm
Pu
A
U
G*
G
A
Contains two or three D residues at different positions
TψC arm
C
G
T
ψ
C
Extra arm
Variable in size, not present in all tRNAs
Py
U
Pu
Anticodon arm
Wobble position
5′
3′
Anticodon

# The function of tRNA

结合氨基酸

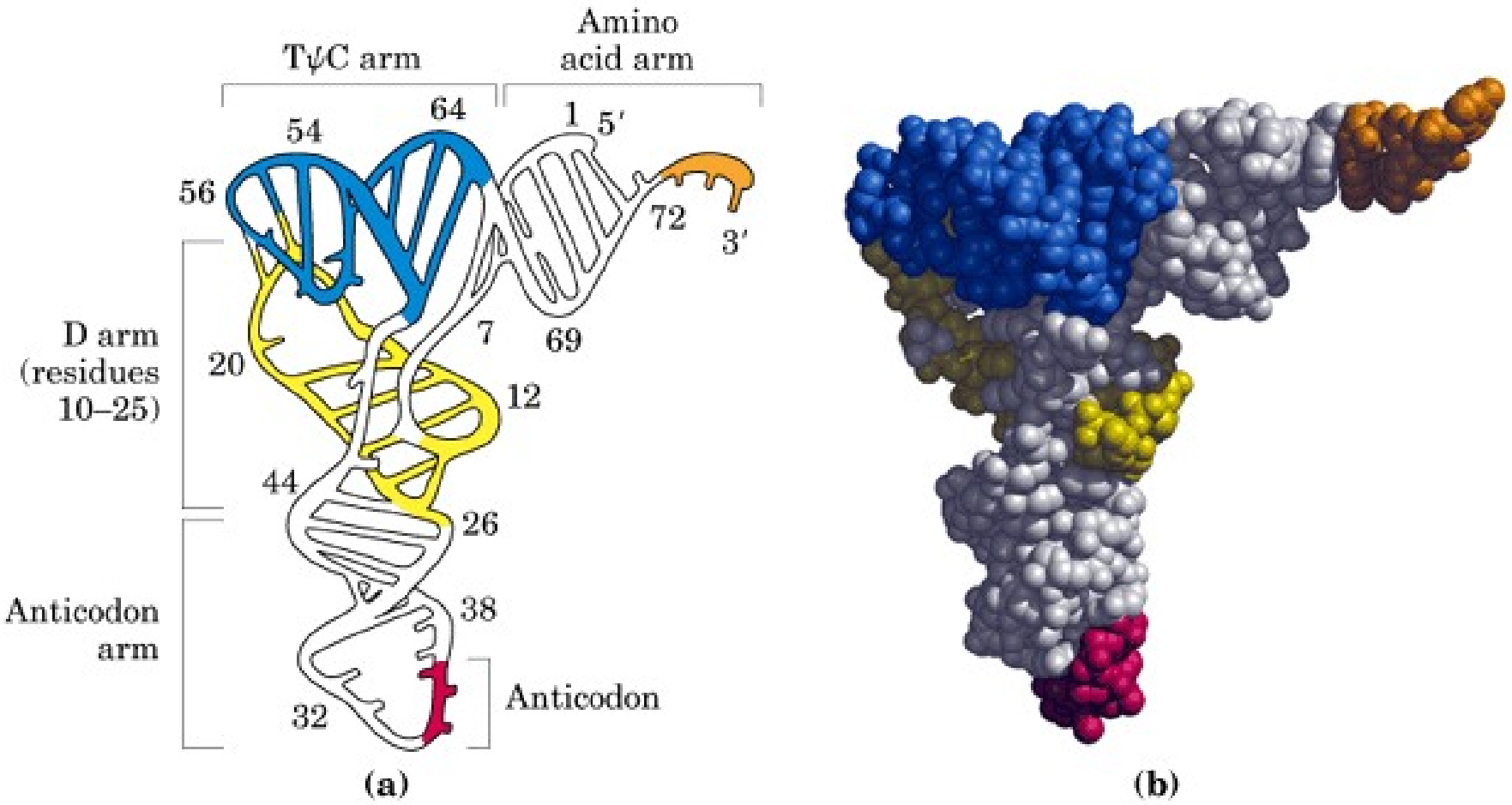
识别氨酰 - tRNA 合成酶



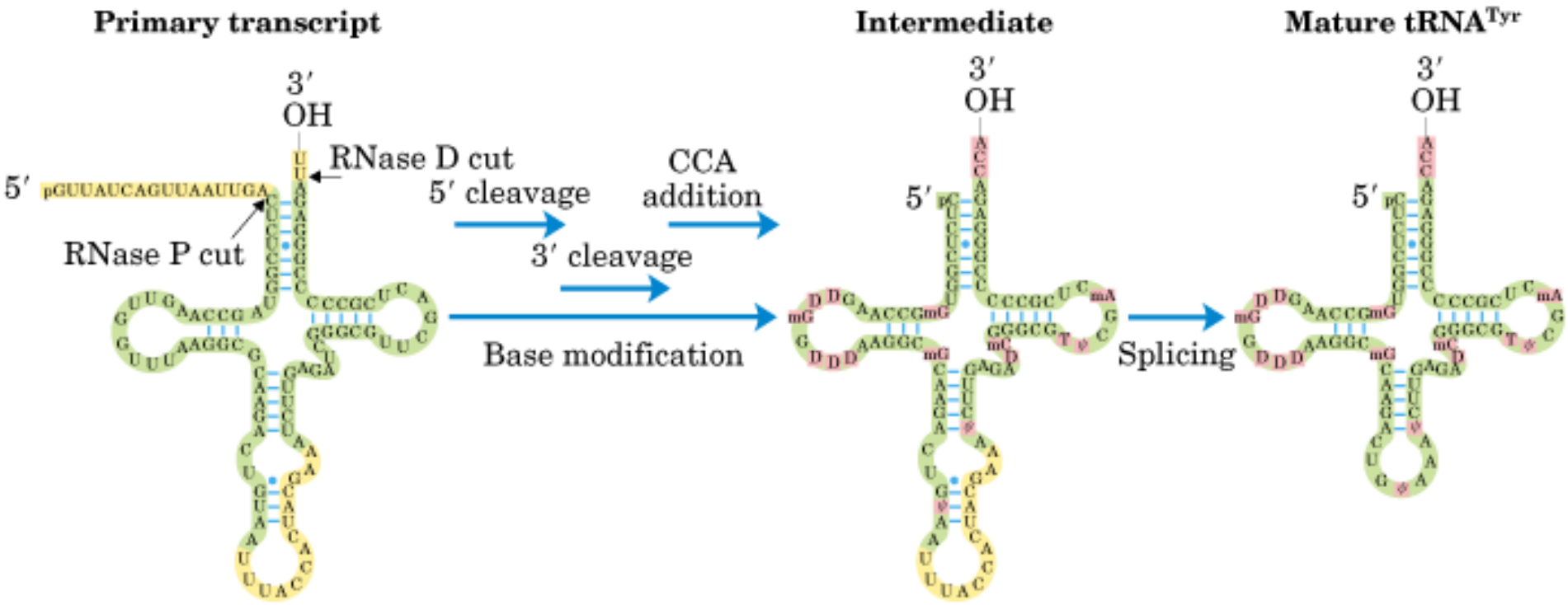

识别核糖体

反密码区正中的 3 个核苷酸残基可以与 mRNA 的三联体密码子形成碱基互补配对，解读遗传密码，称为反密码子（anticodon）。

# Tertiary structure of tRNA



# 1. Primary tRNA transcripts undergo a series of posttranscriptional processing

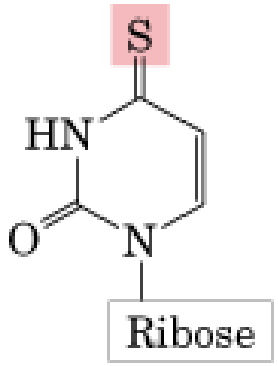


# Primary tRNA posttranscriptional processing

The extra sequences at the 5` and 3` ends are removed by **RNase P** and **RNase D** respectively.
**Intron sequences are spliced**.
The **CCA** sequence is generated at the 3` end.
Some of the bases in tRNA molecules are modified by methylation, deamination, reduction and others.

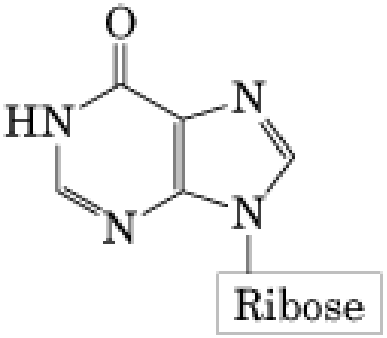






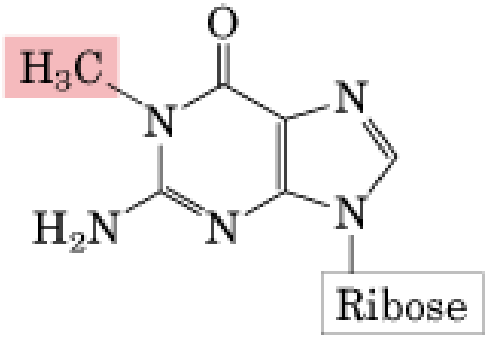

4-Thiouridine ($S^4U$)

4- 硫尿苷

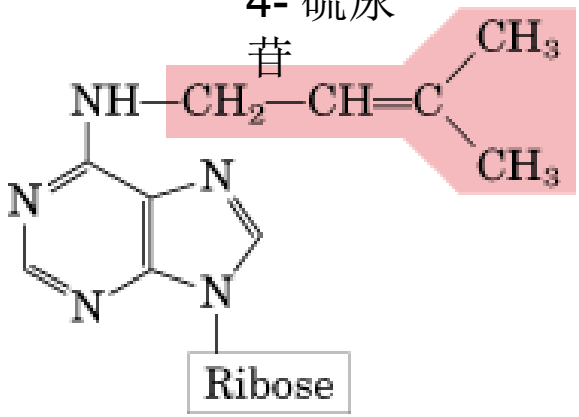



Inosine (I)

次黄（嘌呤核）苷，肌苷

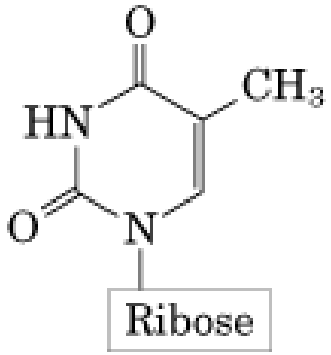



1-Methylguanosine ($m^1G$)

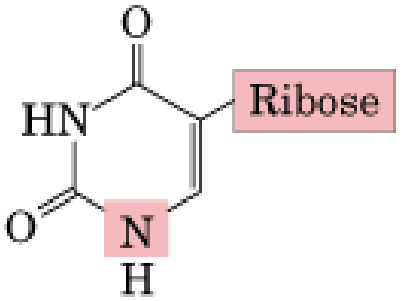



$N^6$-Isopentenyladenosine ($i^6A$)

异戊烯腺苷

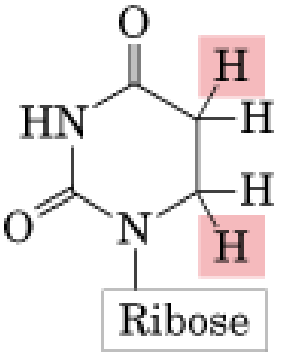



Ribothymidine (T)




Pseudouridine ($\psi$)

. 假尿（嘧啶核）苷



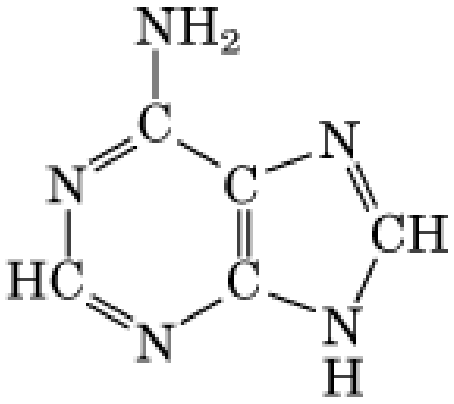

Dihydrouridine (D)

二氢尿嘧啶核苷

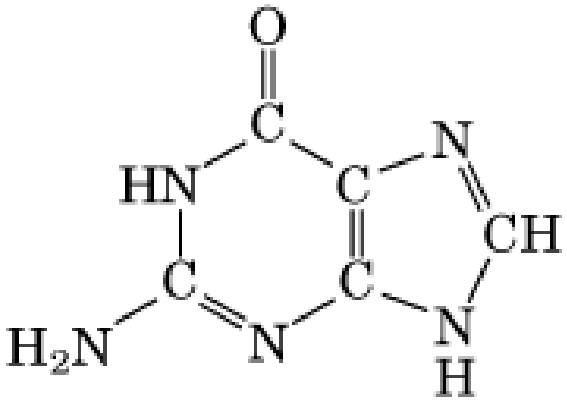
# Some typical modificied bases found In mature tRNA molecules.



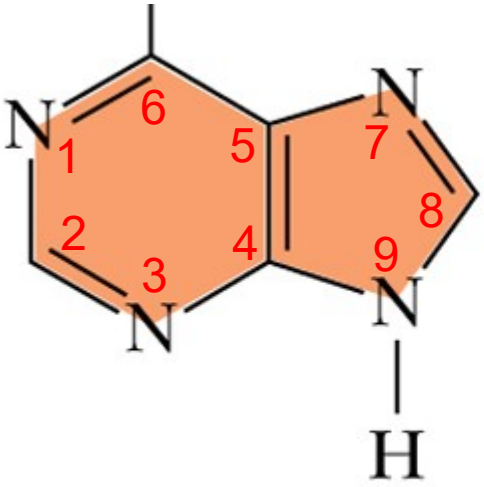




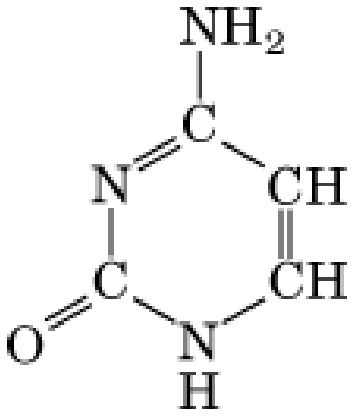
Adenine



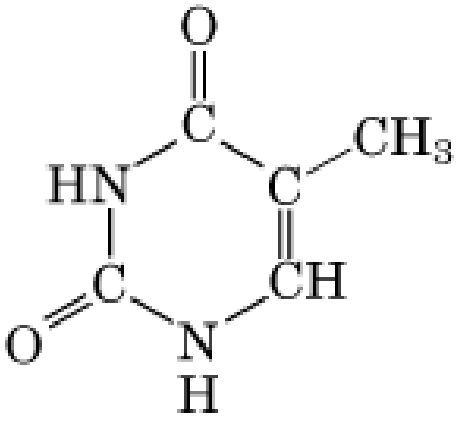
Guanine



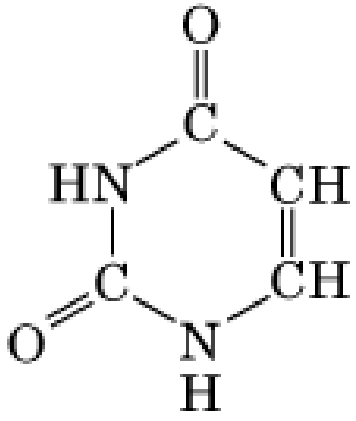

**Purines**



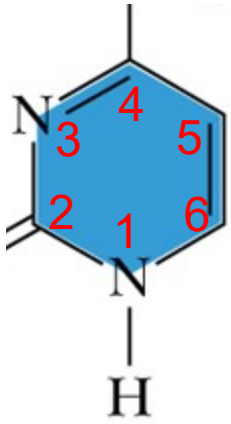
Cytosine



Thymine
(DNA)



Uracil
(RNA)




**Pyrimidines**

# RNA 指导下的 RNA 和 DNA 的合成

# RNA dependent synthesis of DNA and RNA





# Reverse transcriptases catalyze the production of DNA from RNA

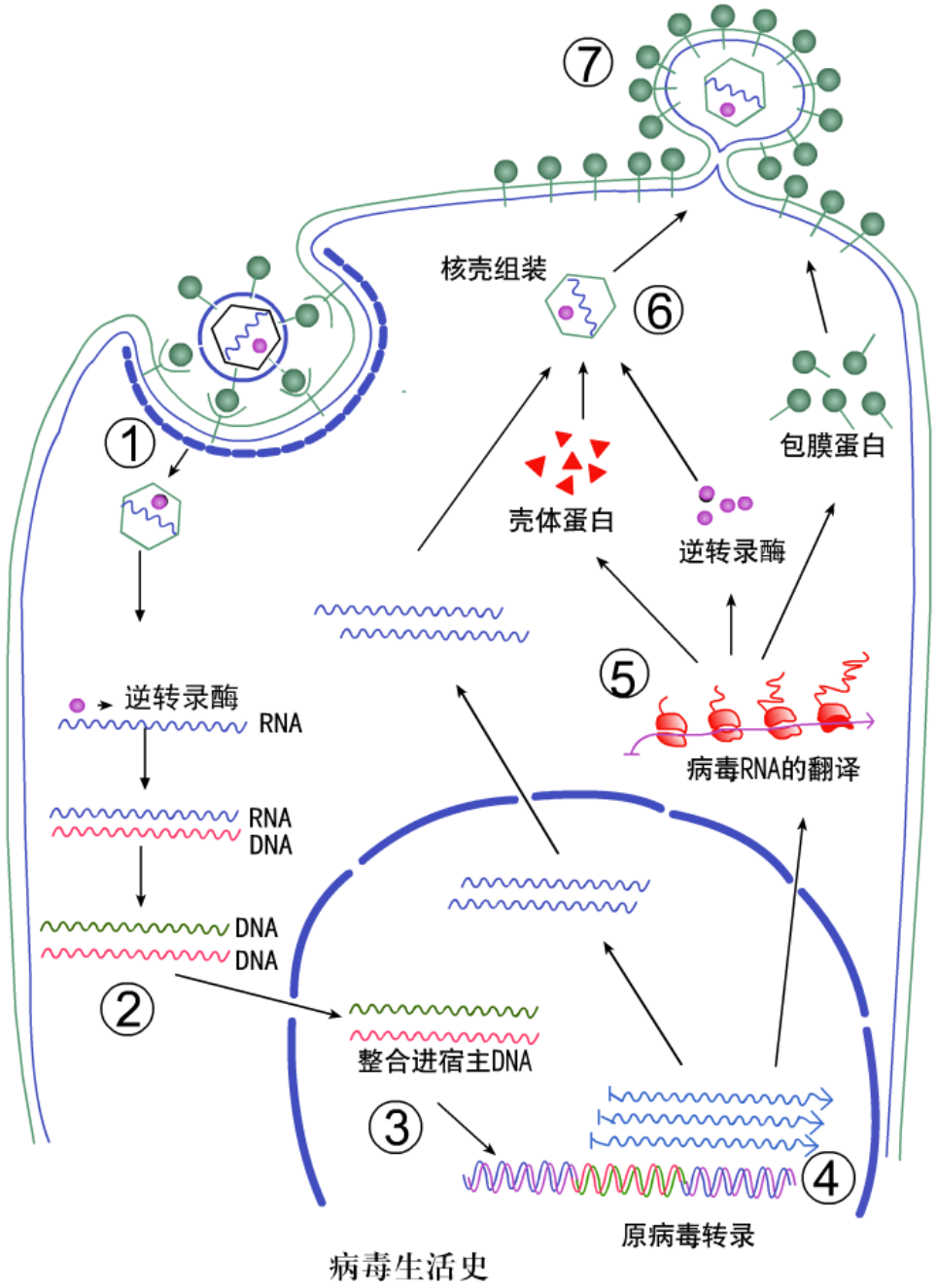
This enzyme catalyzes three reactions:

- RNA-directed DNA synthesis using tRNAs as primers;
- Degradation of the RNA template;
- DNA-directed DNA synthesis;

The enzyme has **no 3`- 5`** proofreading exonuclease activity, thus generating high rate of mutations.

This enzyme is widely used to synthesize **complementary DNAs (cDNAs)** from mRNAs.

# 逆转录病毒


⑦
核壳组装
⑥
包膜蛋白
①
壳体蛋白
逆转录酶
⑤
逆转录酶
RNA
病毒RNA的翻译
RNA
DNA
DNA
DNA
②
整合进宿主DNA
③
④
原病毒转录
病毒生活史

# Some viral RNAs are replicated by RNA-directed RNA polymerase

The RNA genomes of some viruses are replicated using RNA-directed RNA polymerases (or **RNA replicases**).
The RNA replicase consists of subunits encoded both by the viruses and the host genome.
They have features similar to that of DNA-directed RNA polymerases, but are usually specific for the RNA of the specific viruses.

# Summary

Transcription shares the basic chemical mechanisms with replication except: no primers required, no proofreading activity exist.
Transcription begins at specific promoter sequences and ends at specific terminator sequences (being ρ -independent or dependent in bacteria).
One multimeric RNA polymerase catalyzes the synthesis of all the RNA molecules in bacteria and different RNA polymerases are used to synthesize the different types of RNAs in eukaryotes.





Primary RNA transcripts are further processed: capped, tailed, spliced and sometimes edited for the mRNAs; ends removed and modified, bases modified for the tRNAs; cleaved and sometimes spliced for the rRNAs.

DNA can be synthesized by using RNA as templates in a reaction catalyzed by reverse transcriptase.

RNA is very likely the first type of biomacromolecules produced during biochemical evolution.

RNA replicase catalyzes the synthesis of RNA from RNA templates.

# Summary of RNA transcription

**G1 RNA structure: tRNA cloverleaf**
**G2 RNA Transcription in Prokaryotes**
- **1. similar and different aspects between DNA replication and RNA transcription**
- **2. RNA polymerase**
- **3. transcriptional bubble**
- **4. promoter**
- **5. RNA transcription: initiation, elongation, termination**

**G3 the Lac operon**
**G5,G6 Transcription in Eukaryotes**
- **1. Three RNA polymerases**
- **2. Pol II binding promoter, transcription factor**

**G7 Processing of eukaryotic pre-mRNA**
- **1. mRNA processing**
- **2. rRNA processing**
- **3. tRNA processing**





# END

Thank you





# 2006年诺贝尔医学奖



美国 安德鲁·菲尔



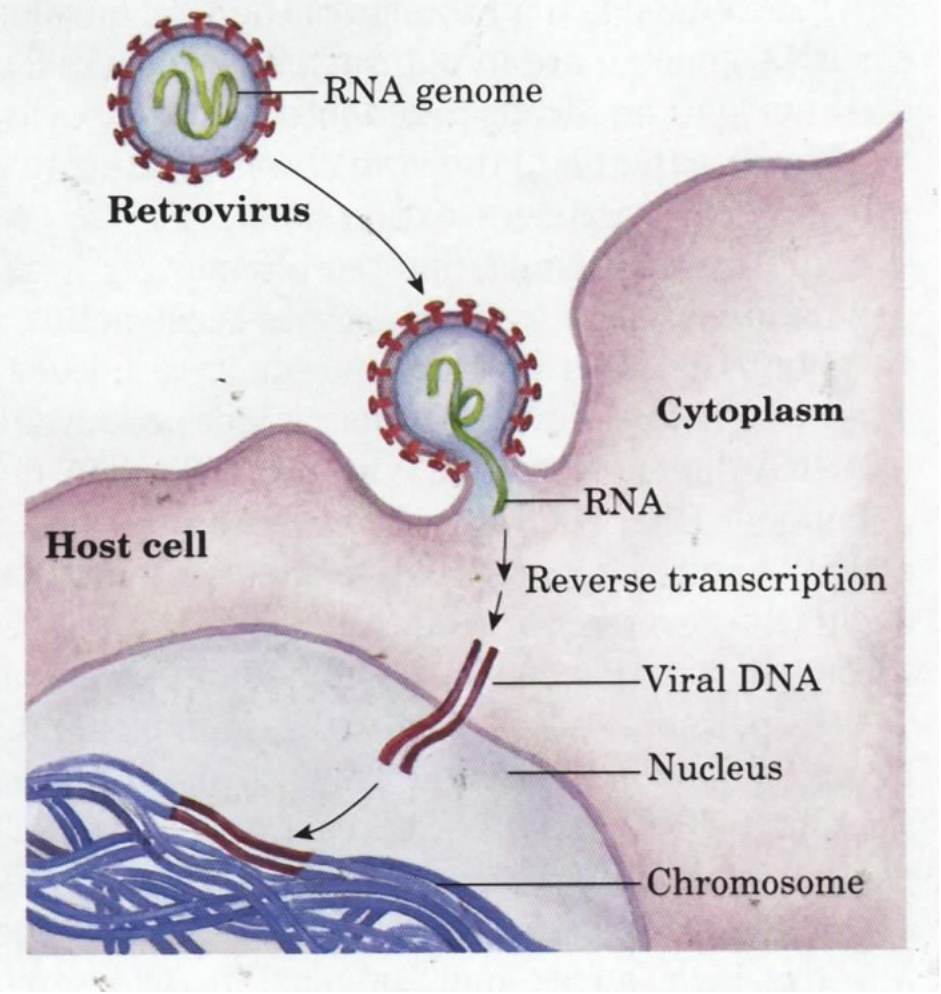
美国 克雷格·米洛

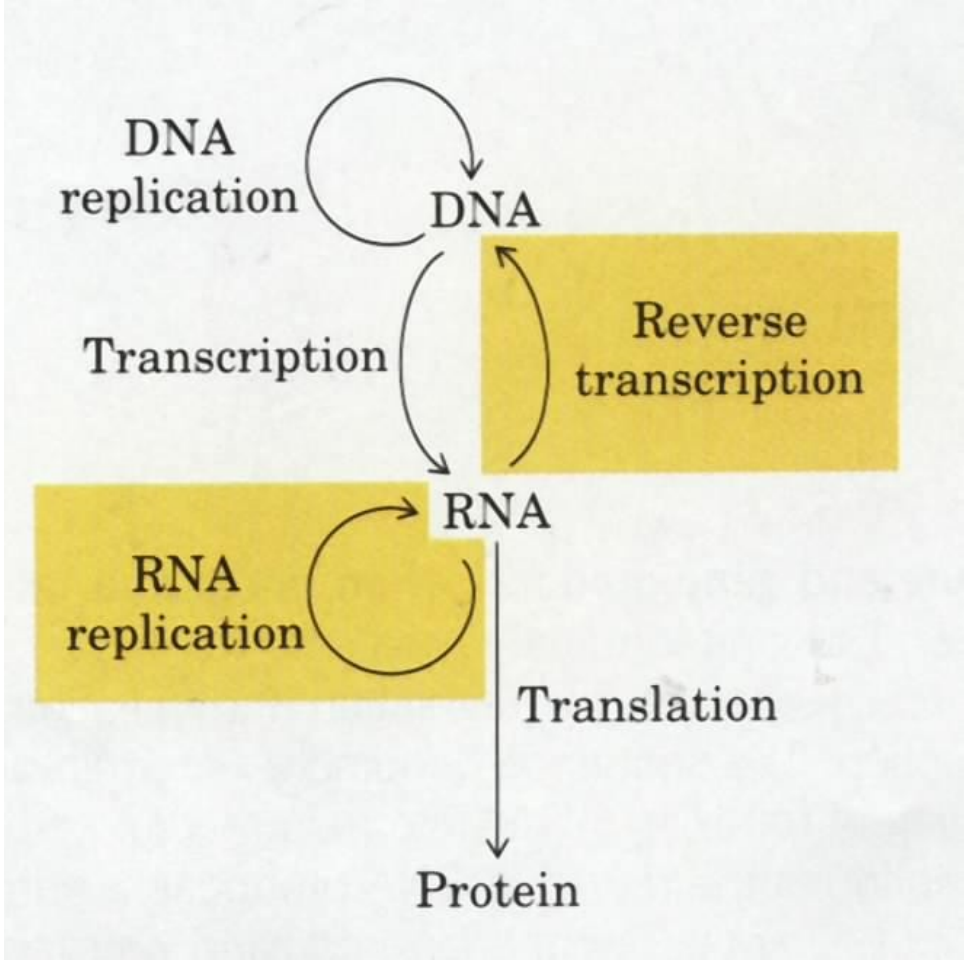
**2006年诺贝尔医学奖得主：**

- ［美］安德鲁·菲尔
- ［美］克雷格·米洛

**获奖原因：**

- 发现了控制基因信息流动的基本机制，因RNA干扰的发现而入选。RNA干扰是一个生物过程，在这个过程中双链RNA以一种非常明确的方式抑制了基因表达。自1998年发现以来，RNA干扰已经作为一种强大的“基因沉默”技术而出现。这项技术被用于全球的实验室来确定各种病症中哪种基因起到了重要作用。[详细]




# Reverse transcriptases catalyzes the sythesis of DNA from RNA template.

# The Nobel Prize in Physiology or Medicine 1975

"for their discoveries concerning the interaction between tumour viruses and the genetic material of the cell"

Press release



David Baltimore

USA

Massachusetts Institute of Technology
Cambridge, MA, USA

1938 -

Autobiography

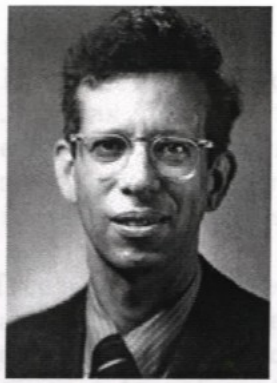
Renato Dulbecco

USA

Imperial Cancer Research Fund Laboratory
London, Great Britain

1914 -





Howard Martin Temin

USA

University of Wisconsin
Madison, WI, USA

1934 - 1994



# 关于沉降系数 S

$$F_c(\text{离心力}) = m\omega^2 x$$

$$F_b(\text{浮力}) = \mathrm{V}\rho\omega^2 x = m\bar{v}\rho\omega^2 x$$

$$F_f(\text{摩擦力}) = fv = f\frac{dx}{dt}$$

$$F_c - F_b = F_f$$





$$F_c - F_b = F_f$$

$$\frac{dx/dt}{\omega^2 x} = \frac{m(1-\bar{v}\rho)}{f}$$

m 是分子颗粒的质量（g）；$\omega$ 是转头的角速度；x 是旋转中心至界面的径向距离（cm）；$\omega^2 x$ 是离心加速度（也称离心场强度或离心场，是单位质量的力）；V 是分子颗粒体积；$\rho$ 是溶剂密度（g/cm3）；$\bar{v}$ 是蛋白质的偏微比容；$V\rho$ 或 $m\bar{v}\rho$ 是被分子颗粒排开的溶剂质量；f 是摩擦系数；v 是沉降速率，即 $dx/dt$ 。



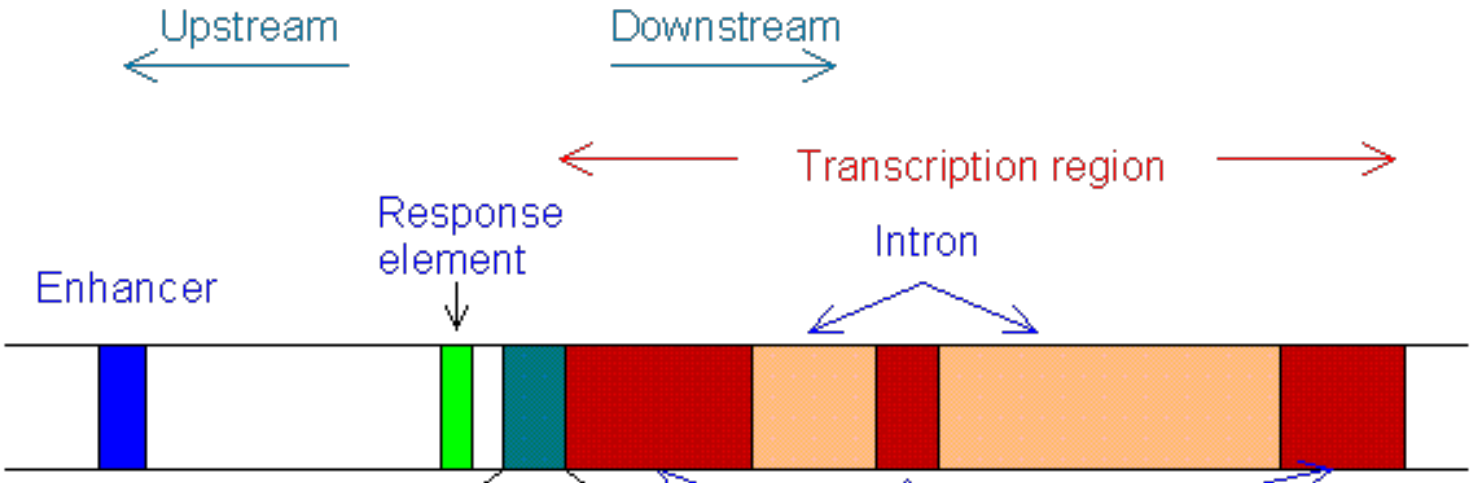


可见单位离心场的沉降速度是个定值，称为沉降系数（sedimentationcoefficient）。用 s（小写）表示。

蛋白质、核酸核糖体和病毒等的沉降系数介于$1\times10^{-13}$到$200\times10^{-13}$秒的范围。为方便起见，$10^{-13}$作为一个单位，称为（Svedberg unit）或沉降系数单位，用 S（大写）表示。

# Enhancers (upstream activator sequences (UASs) in yeast)

- Although many positive control elements lie close to the gene they regulate, others can be located long distances away (10-50kb) either upstream or downstream of the gene.

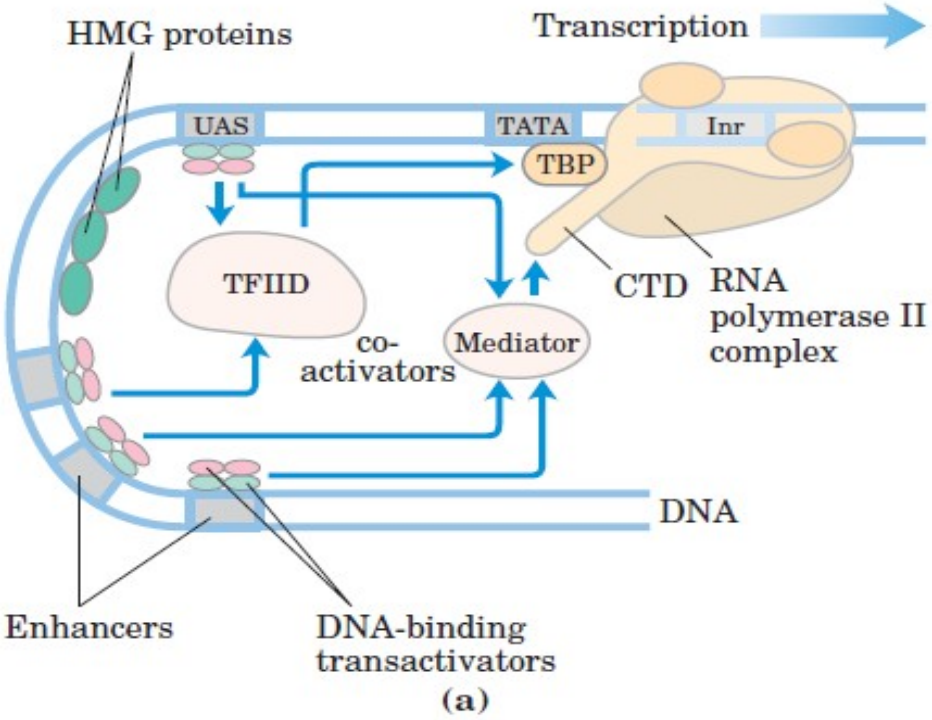


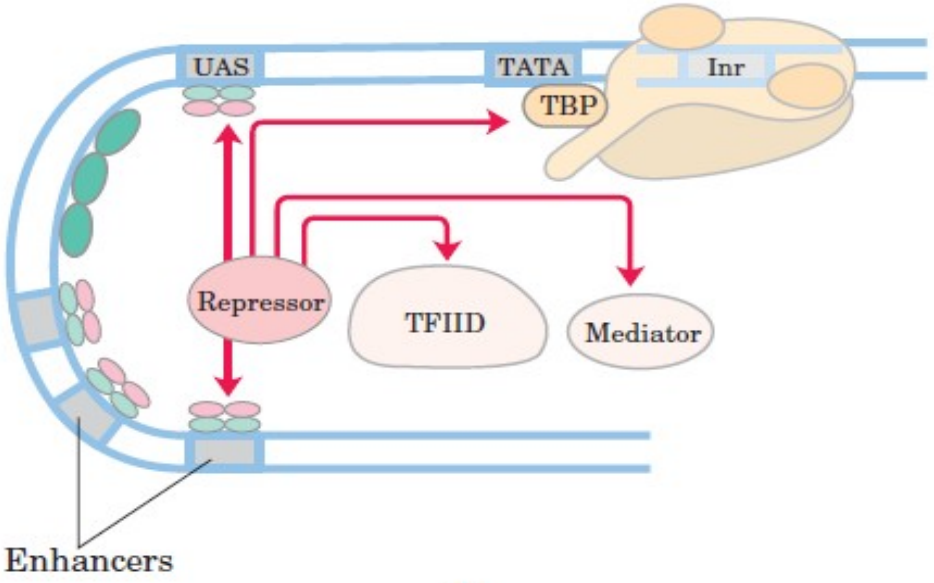
# Repressors

- Repressors near the gene called negative regulators.
- Repressors long distance away from the gene called silencers ( 沉默子） .



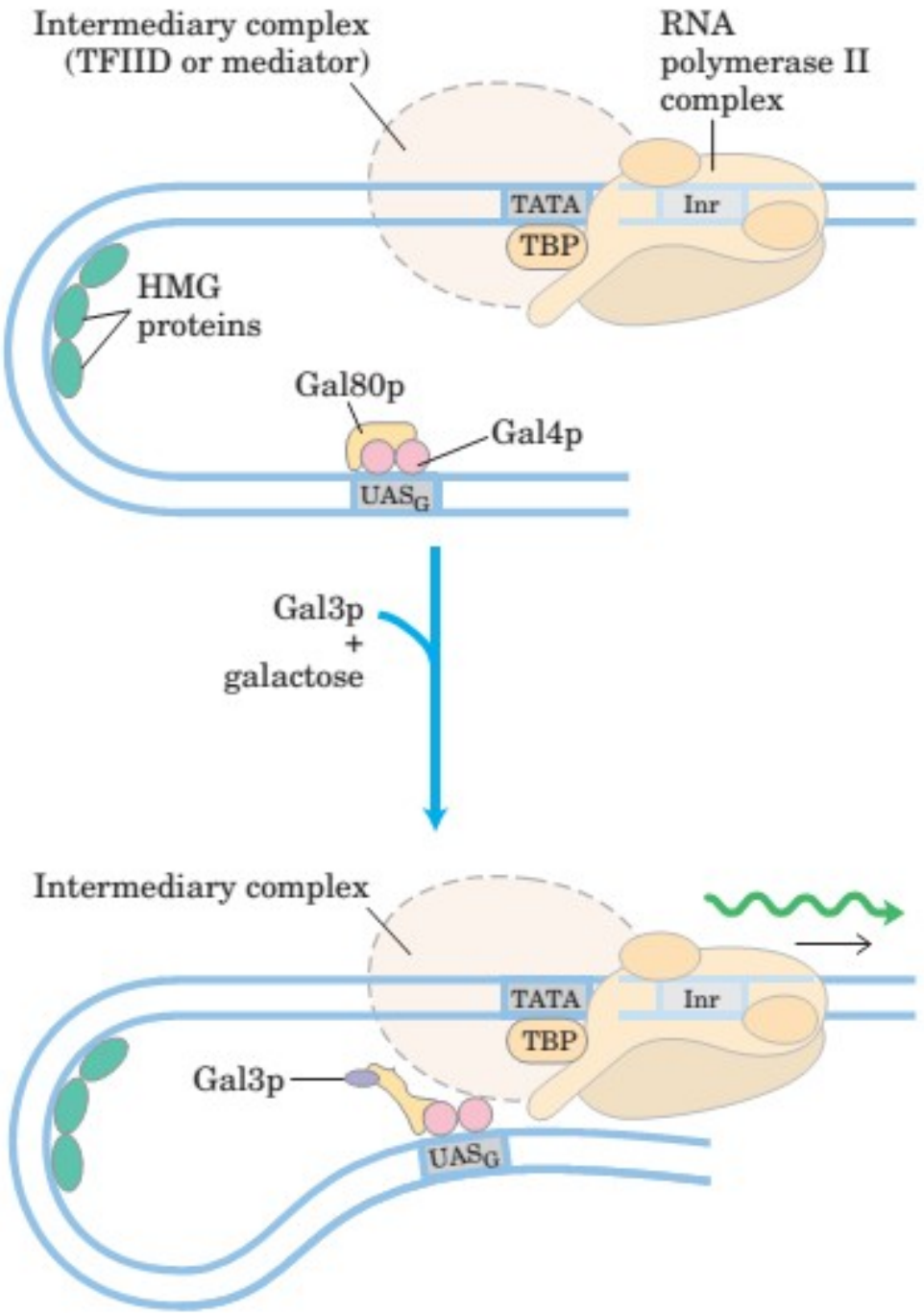




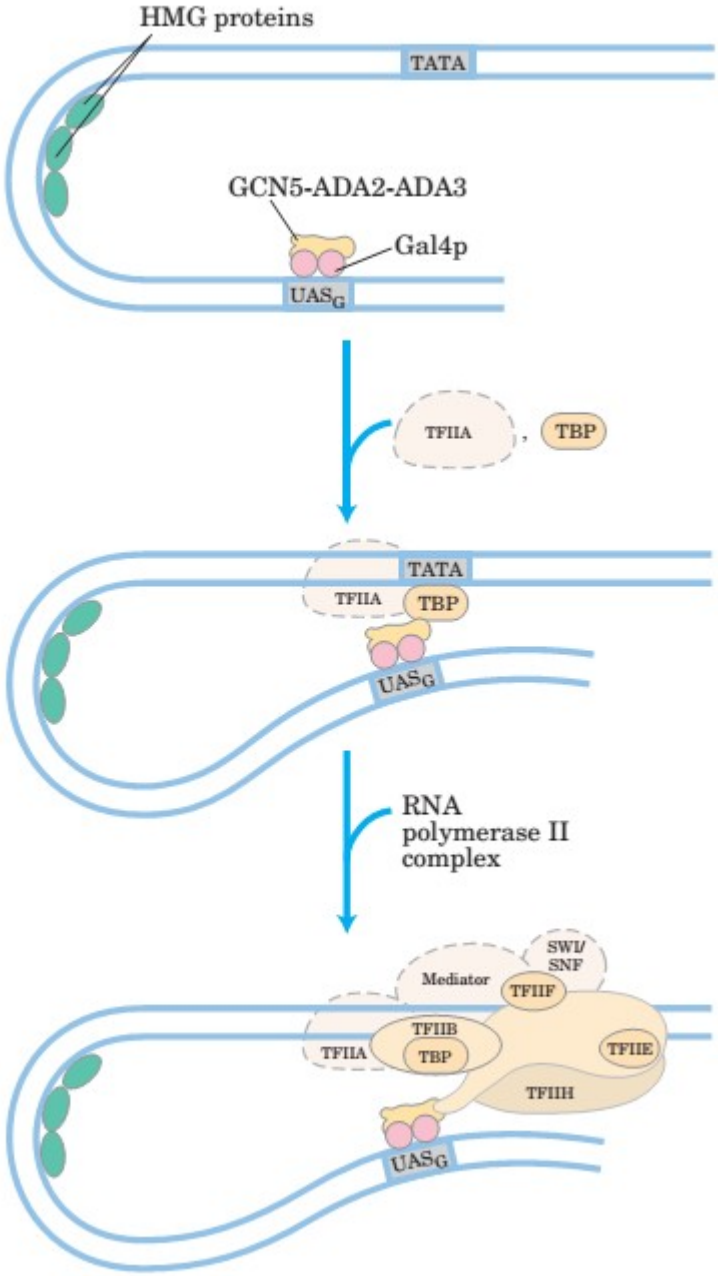




**TBP: TATA-Binding Protein;**

# Regulation of transcription at genes of galactose metabolism in yeast.



# Protein complexes involved in transcription activation of a group of related eukaryotic genes



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